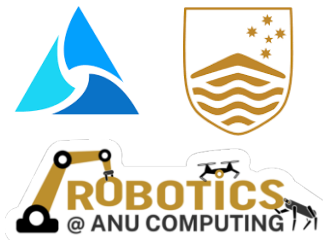




AuSRoS
Australian School of Robotic Systems

B3 – Task and Motion Planning

Rahul Shome
Lecturer, School of Computing
Australian National University

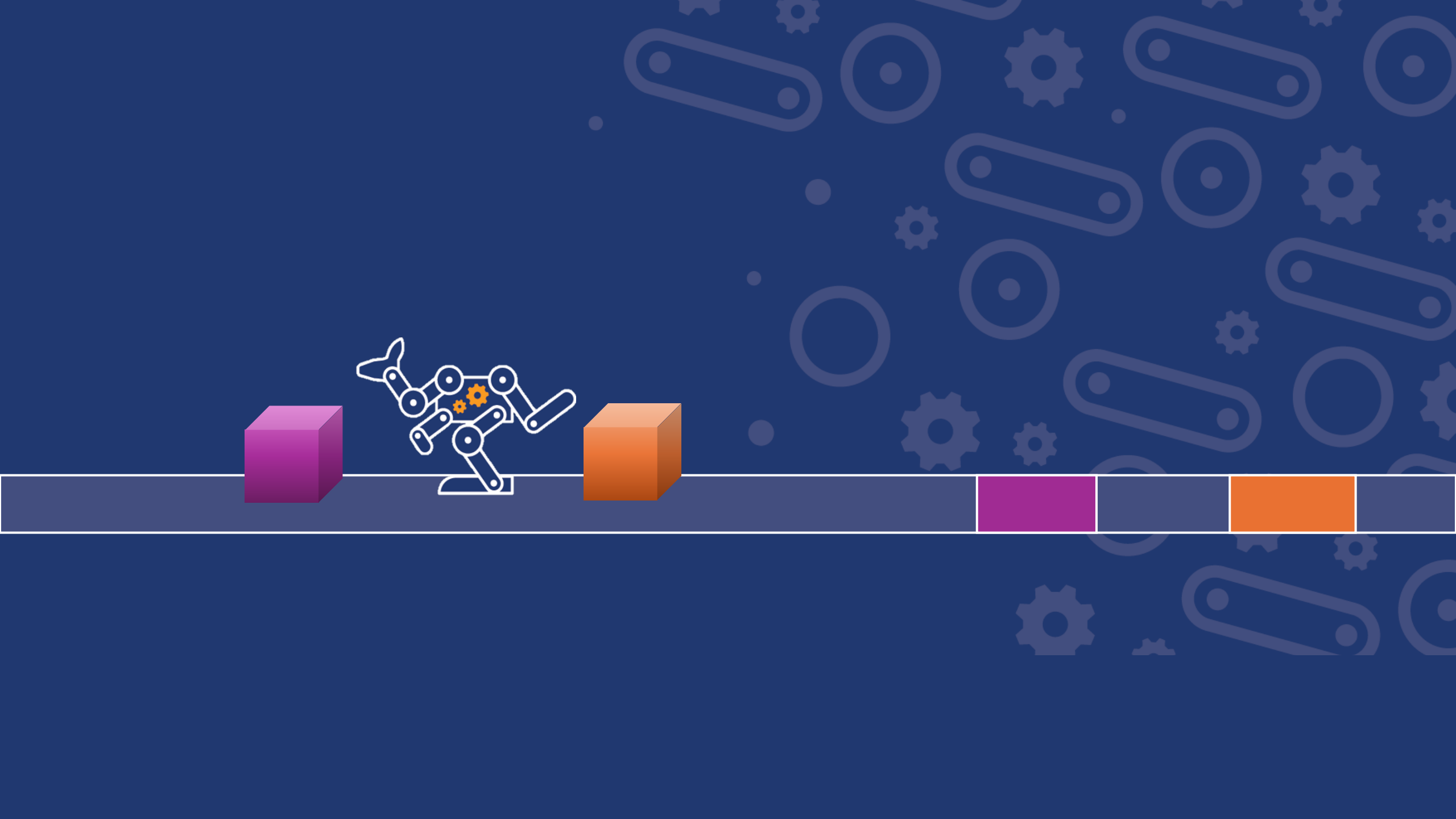


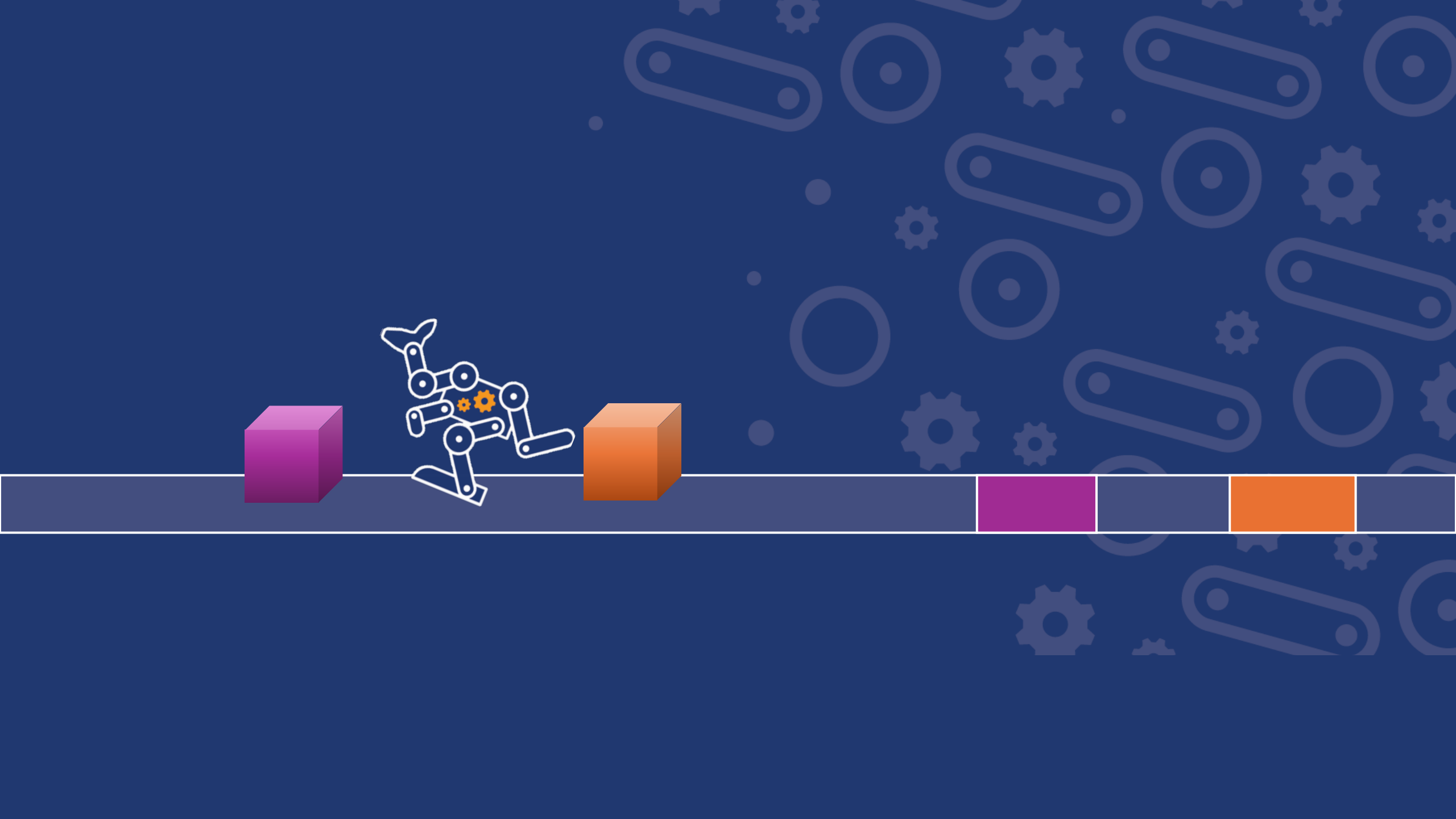


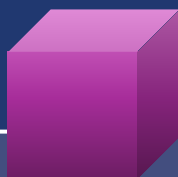
How to interact with and move the blocks?

Which block to move first?



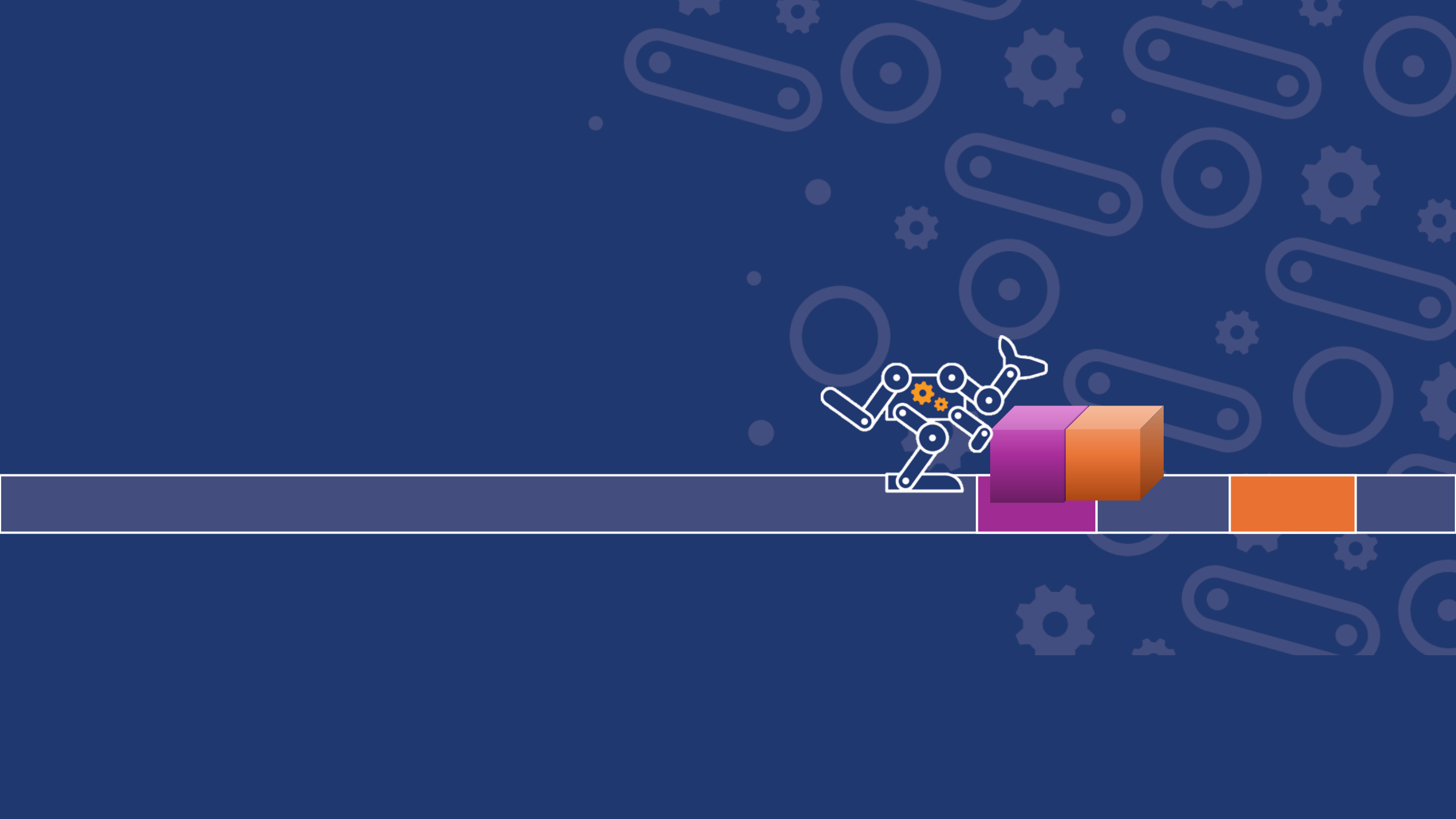


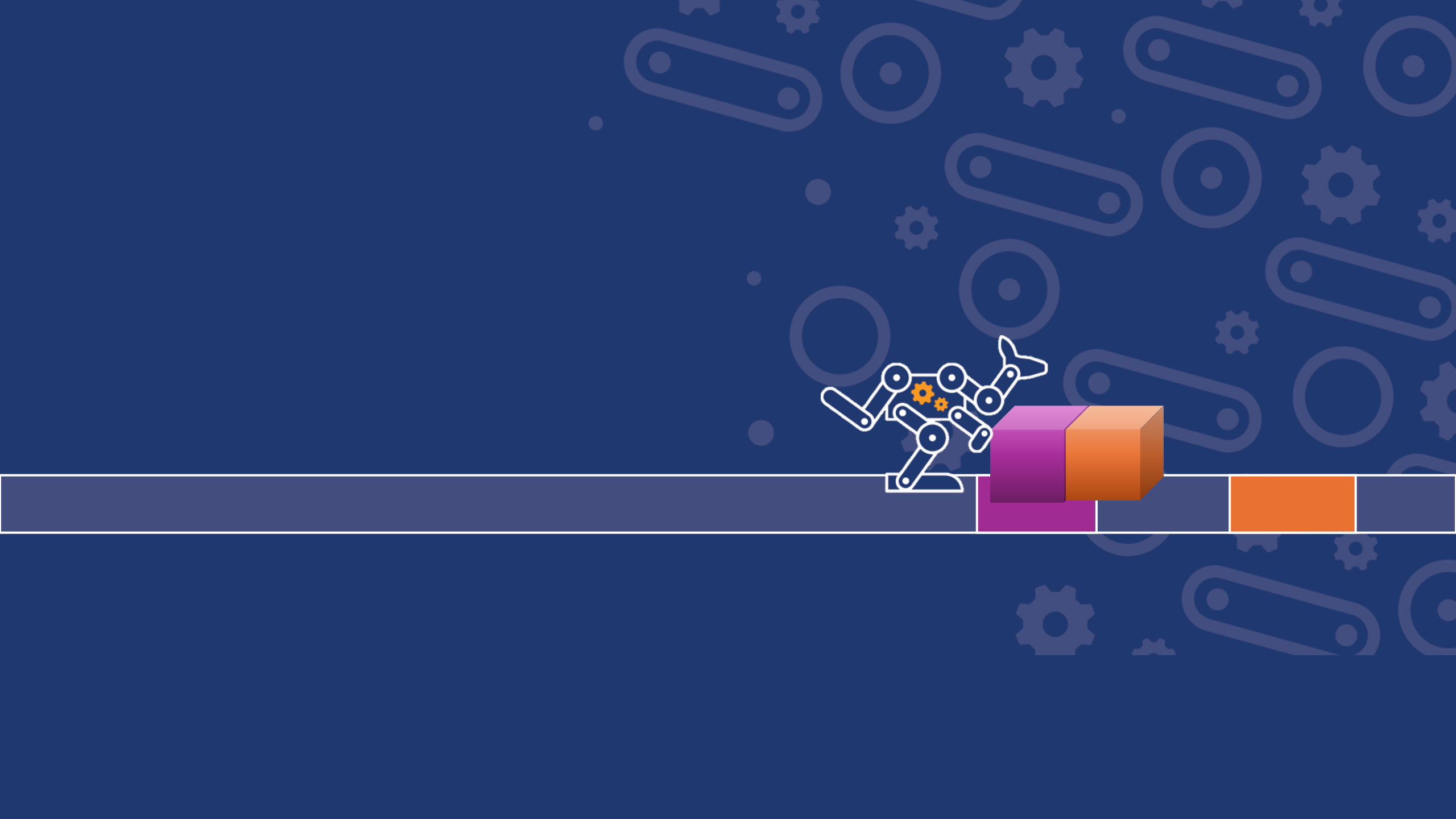








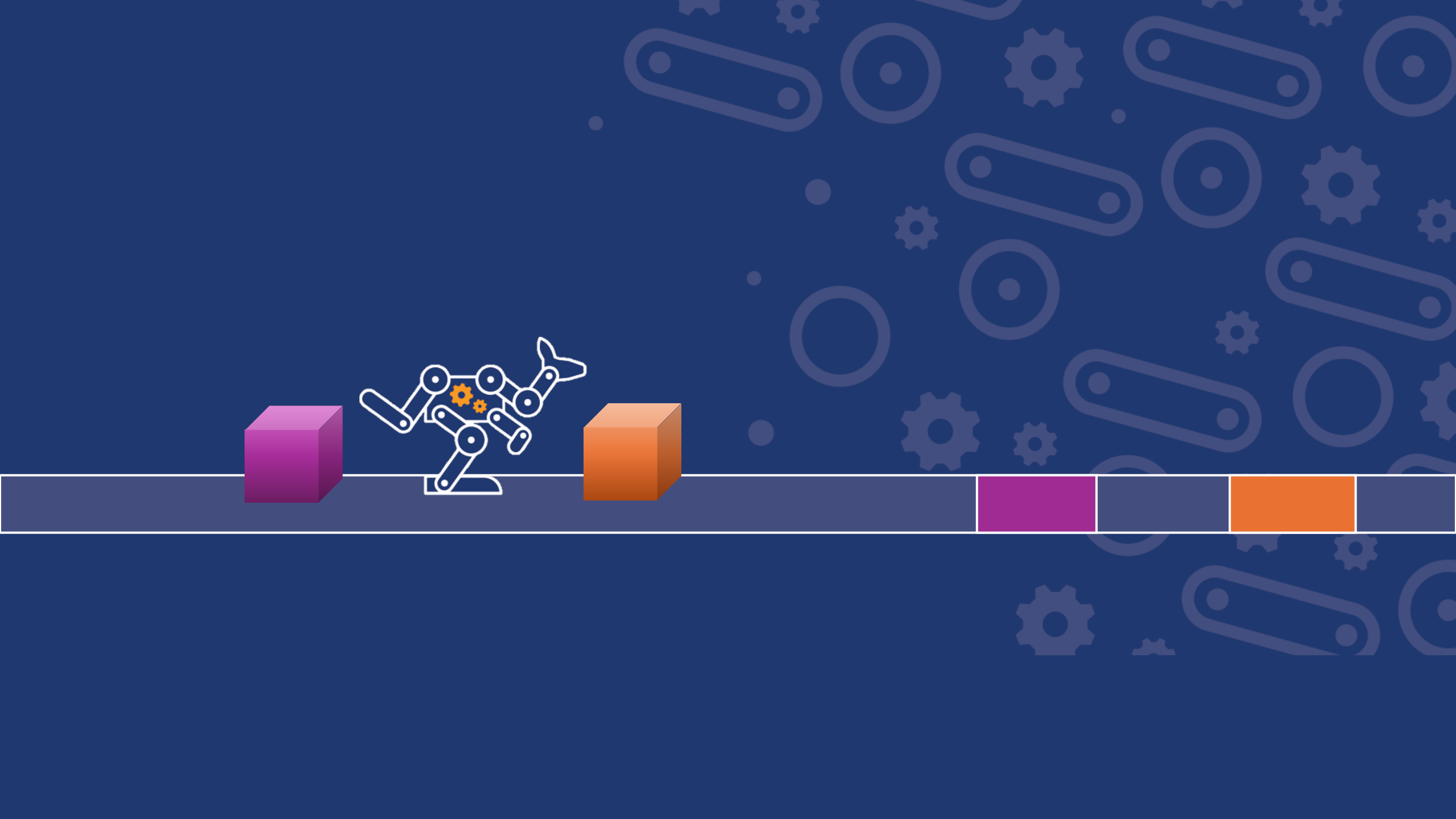


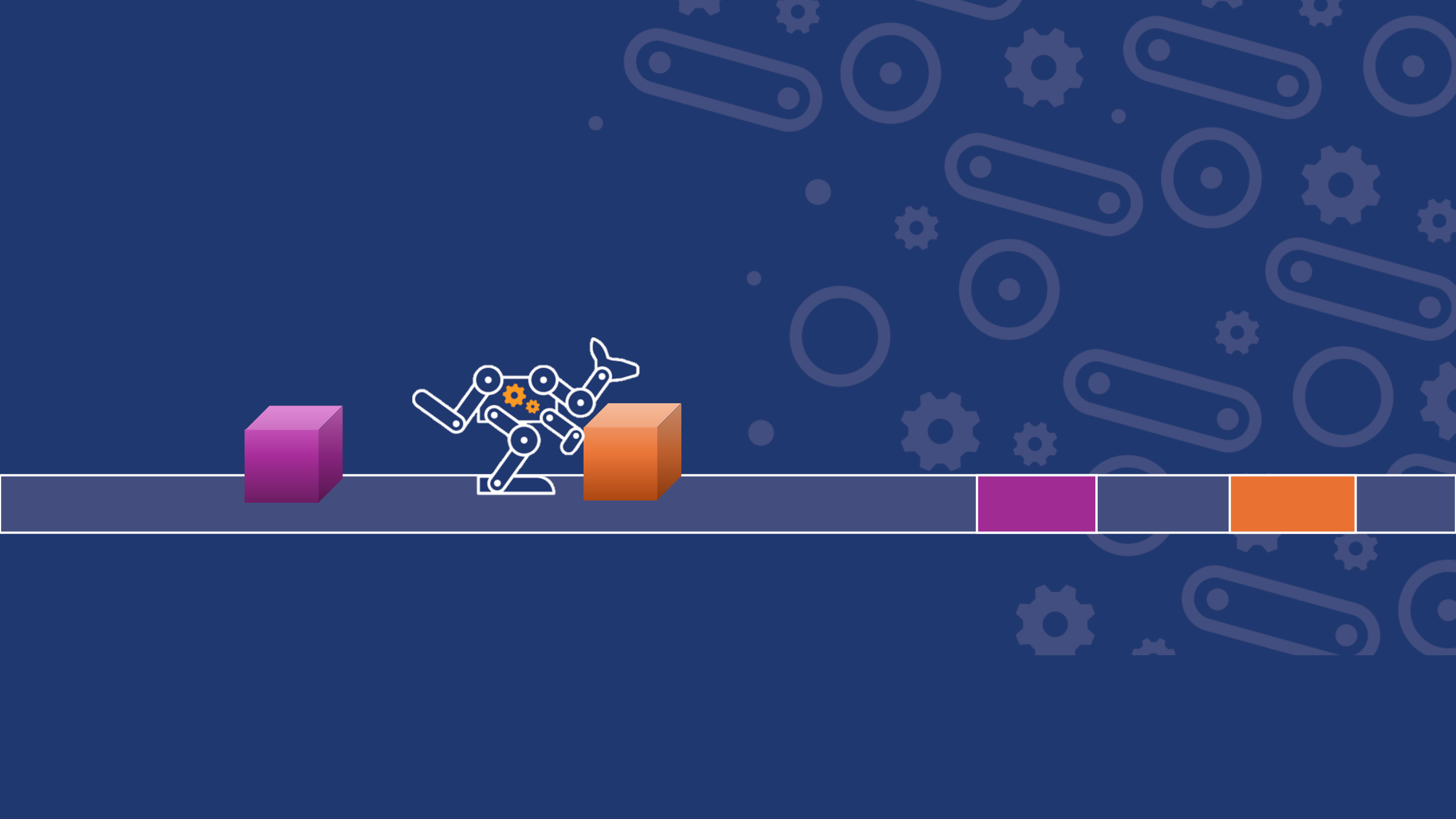


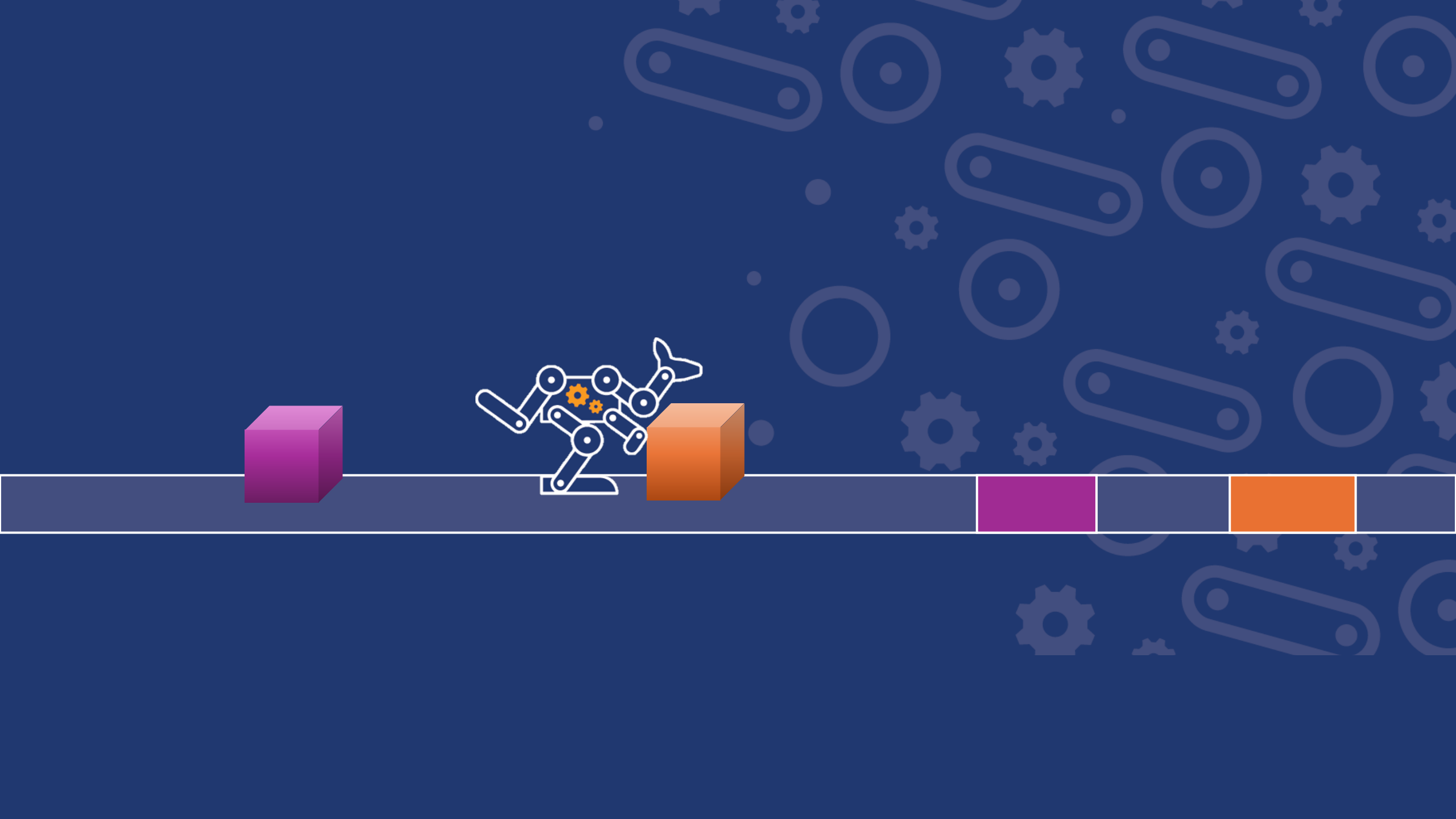
How to interact with and move the blocks?

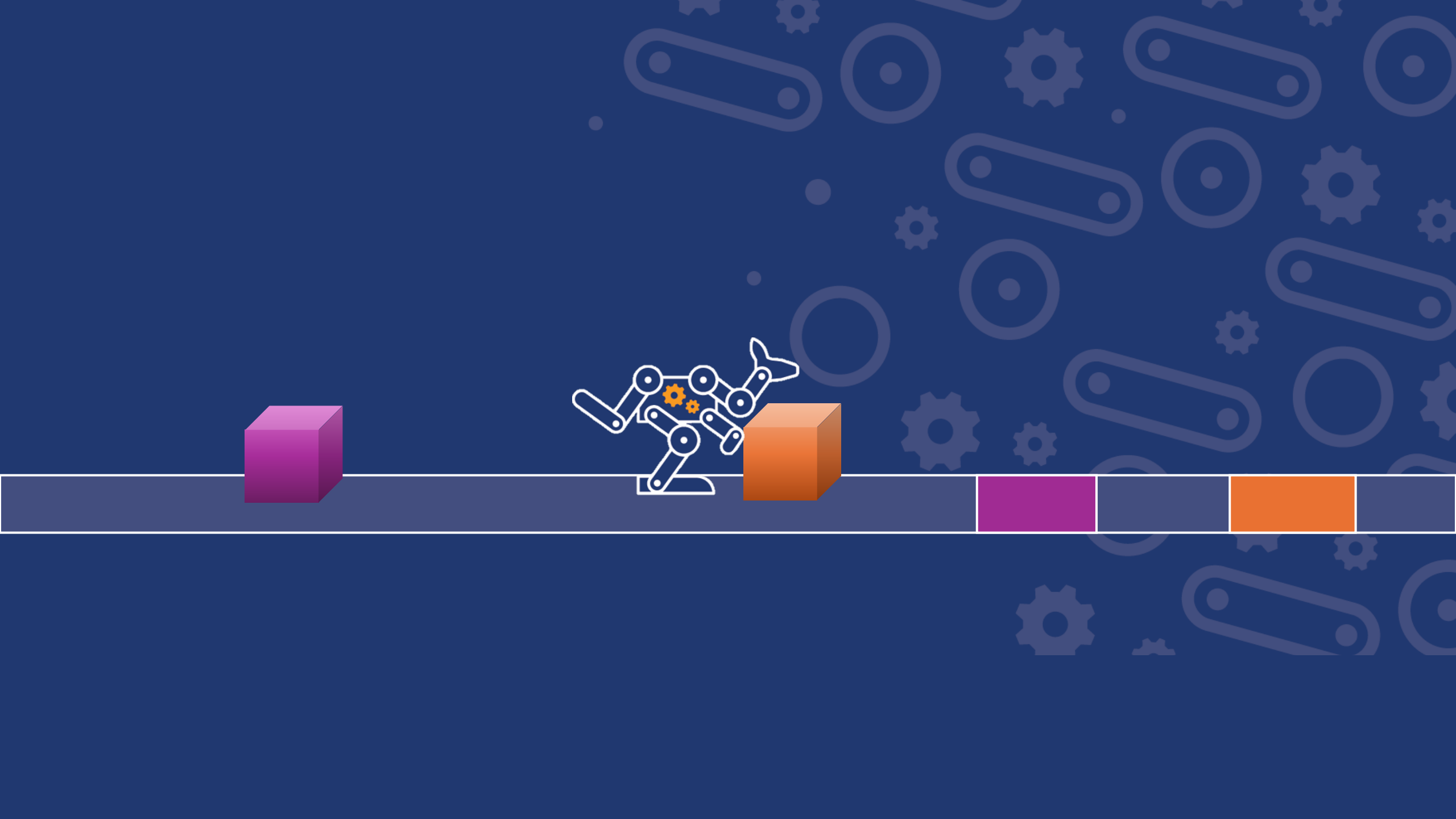
Which block to move first?

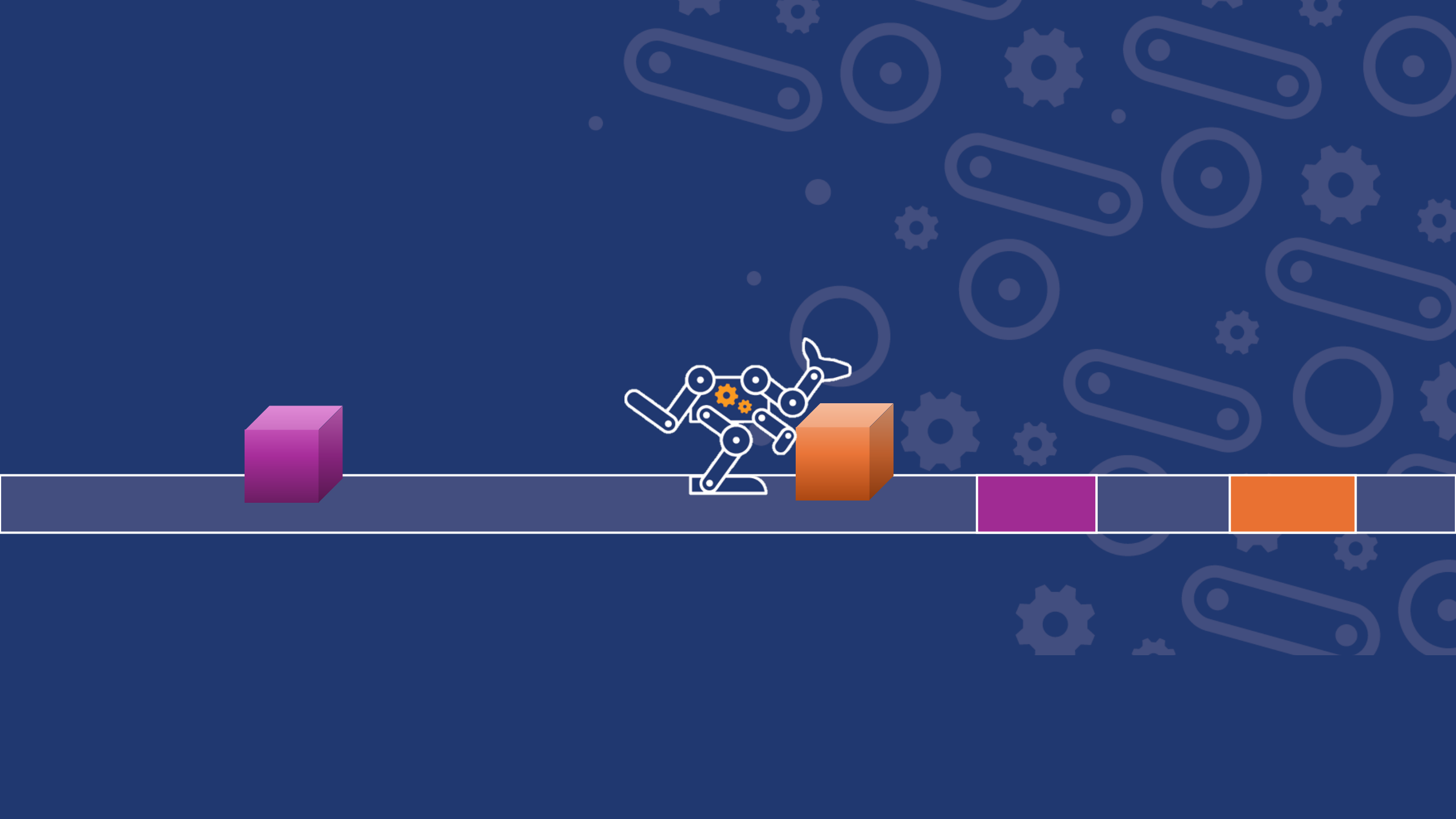


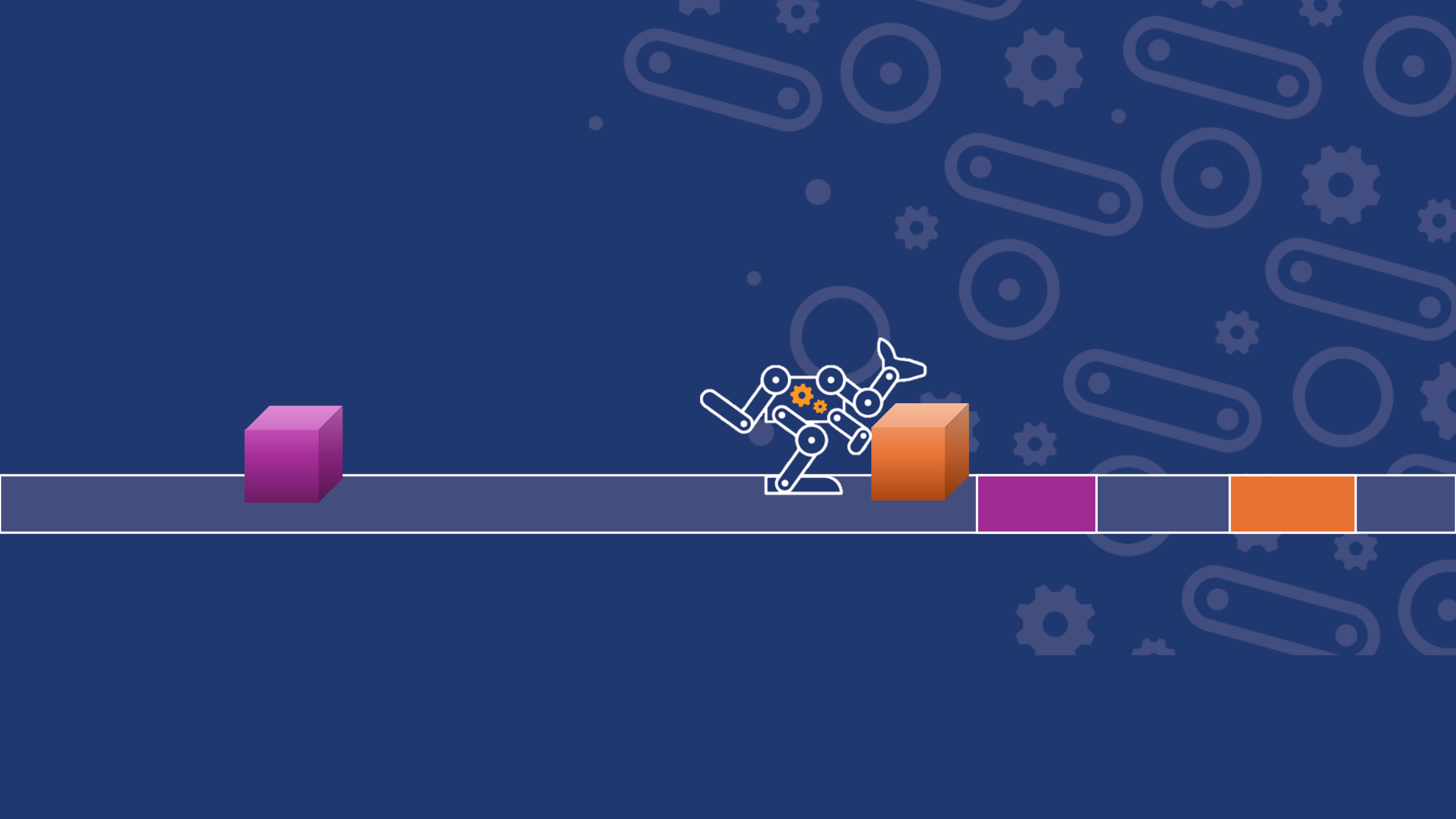


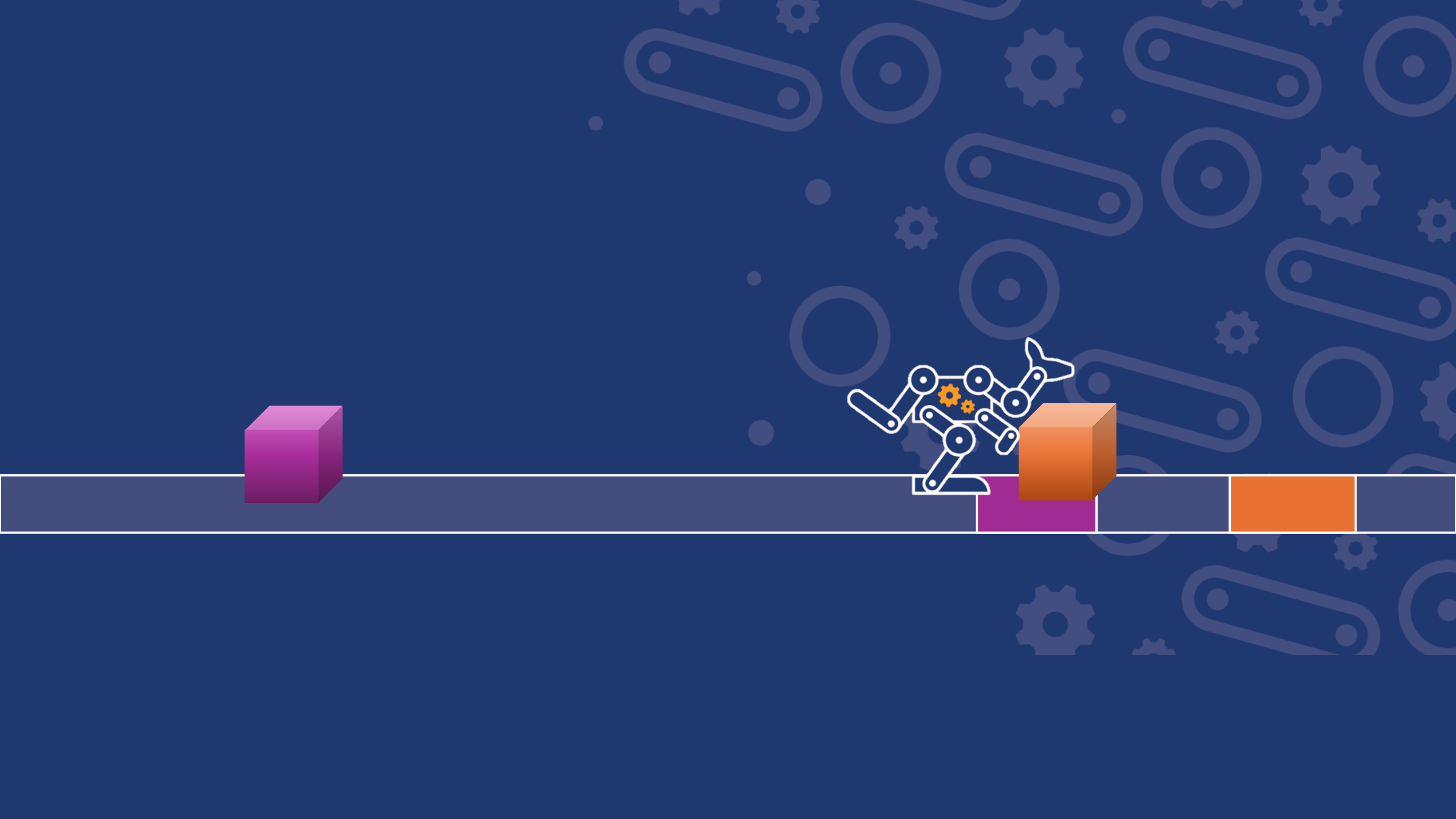


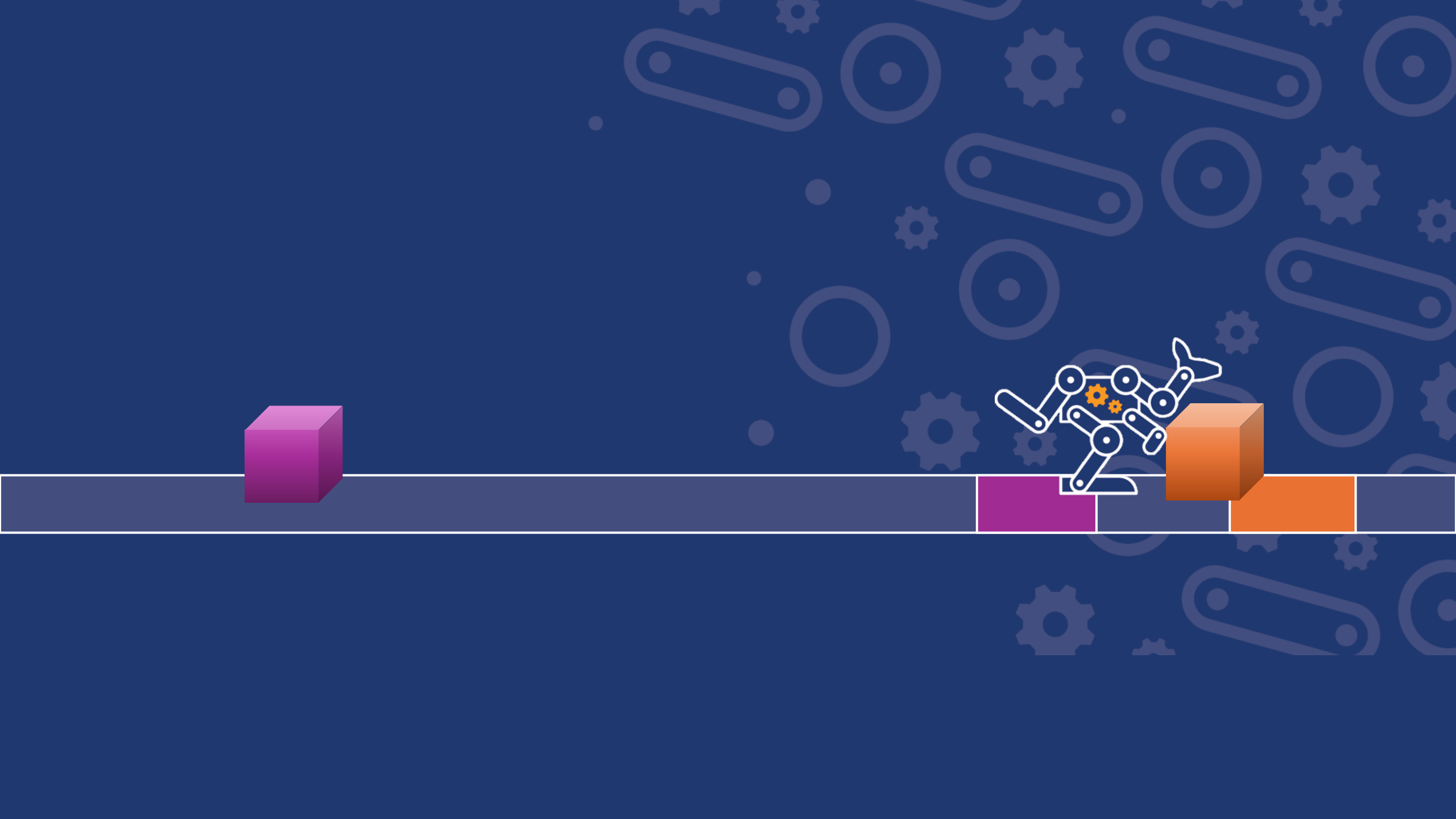


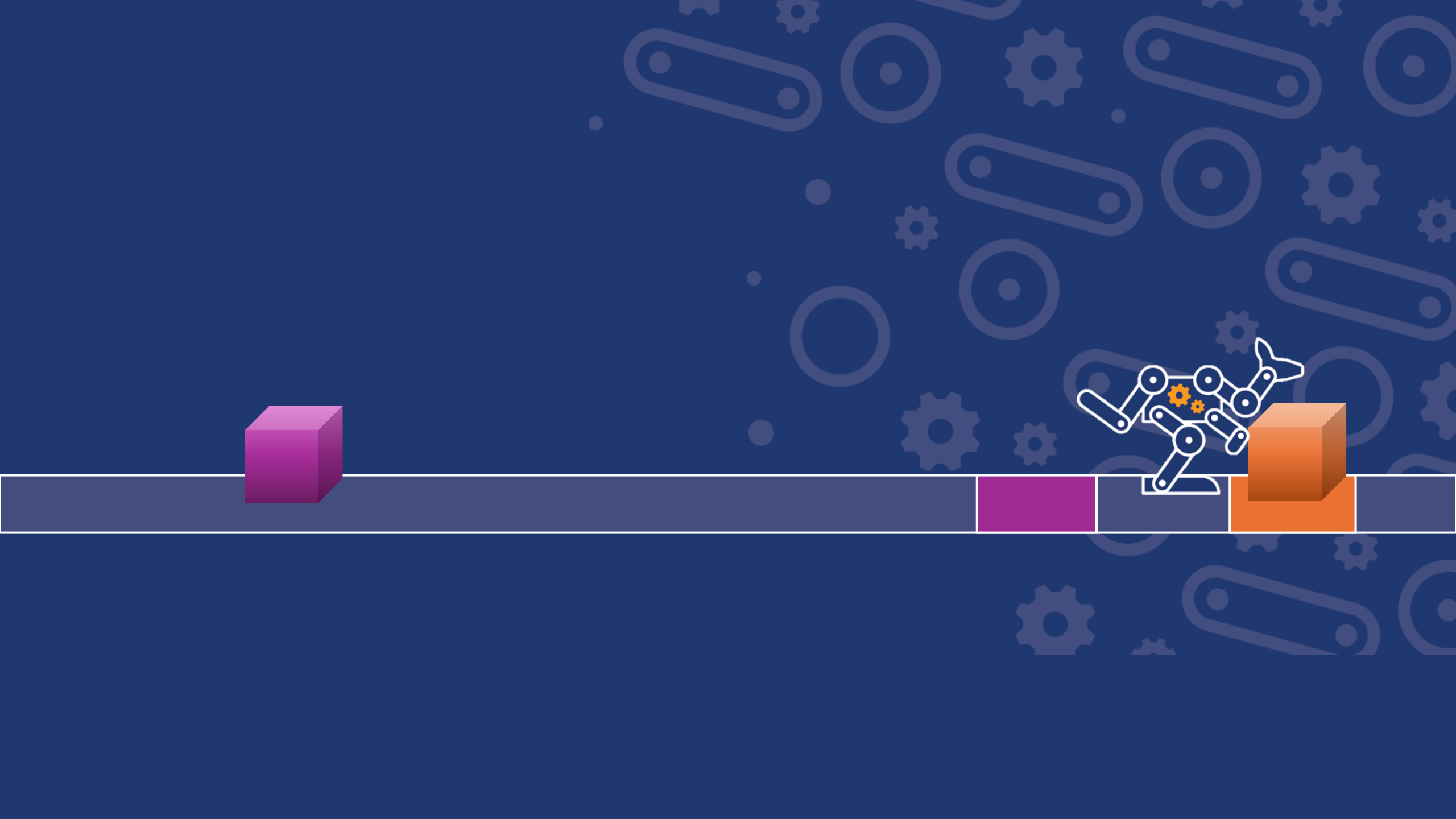


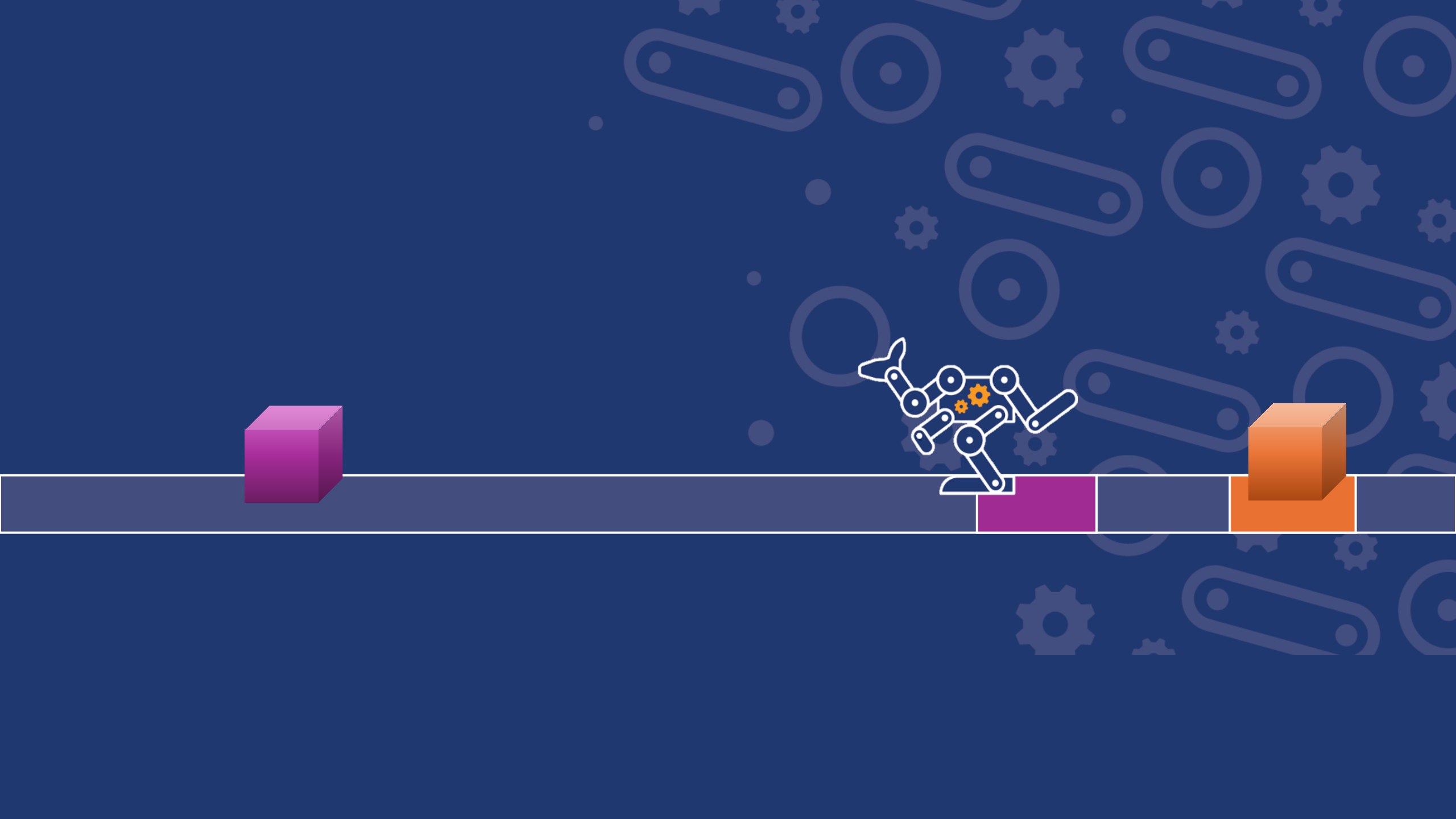


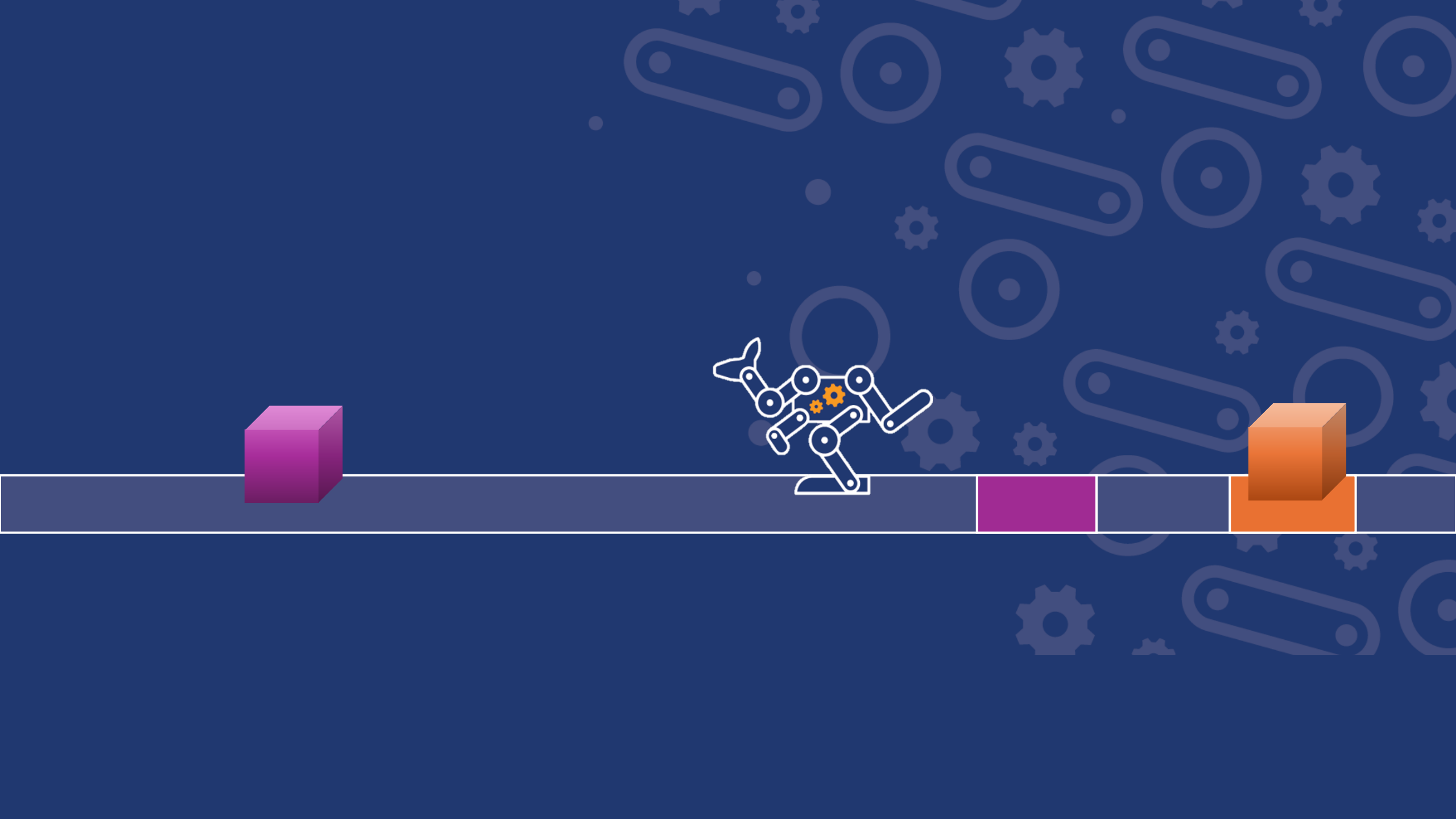


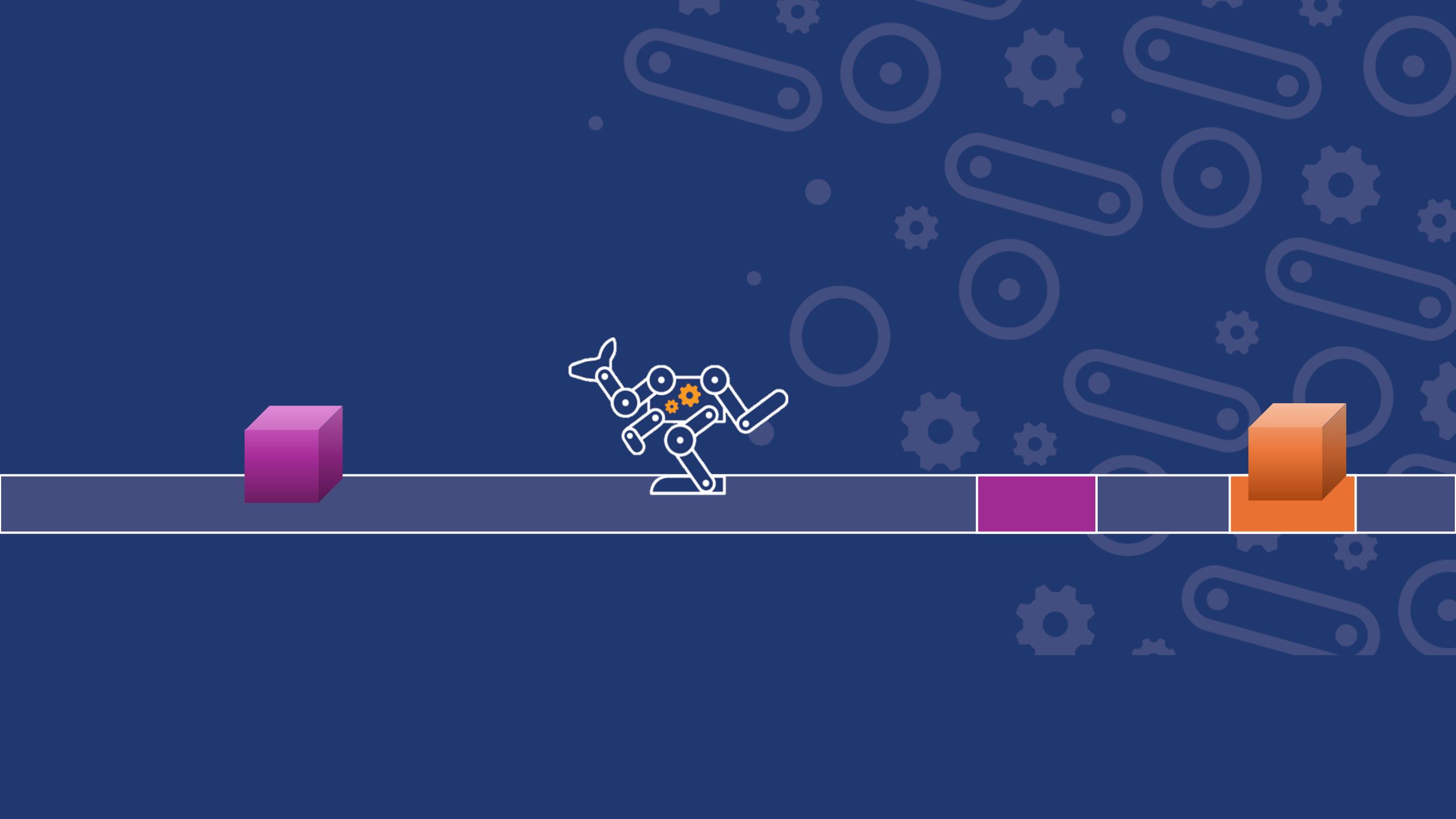


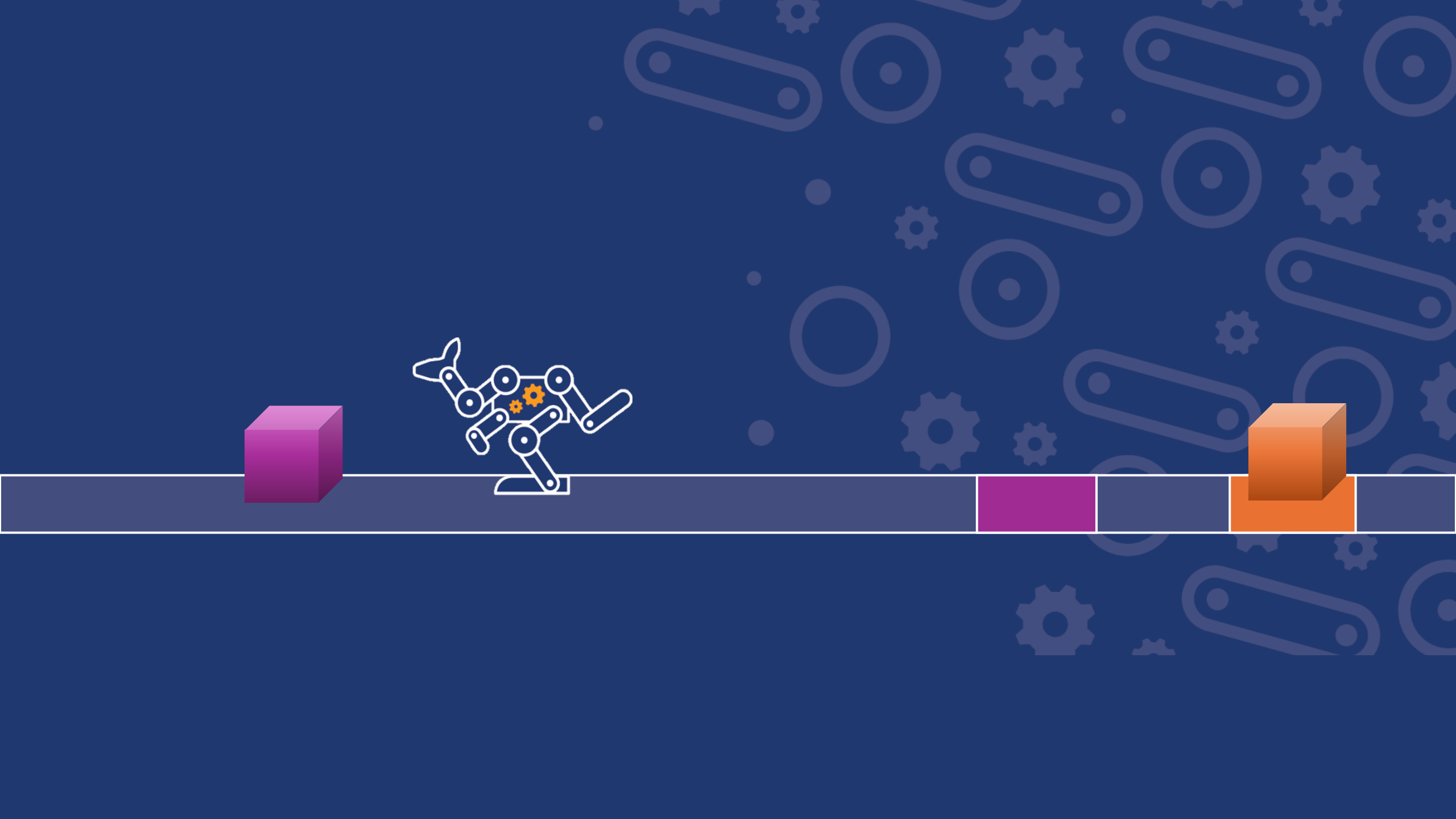


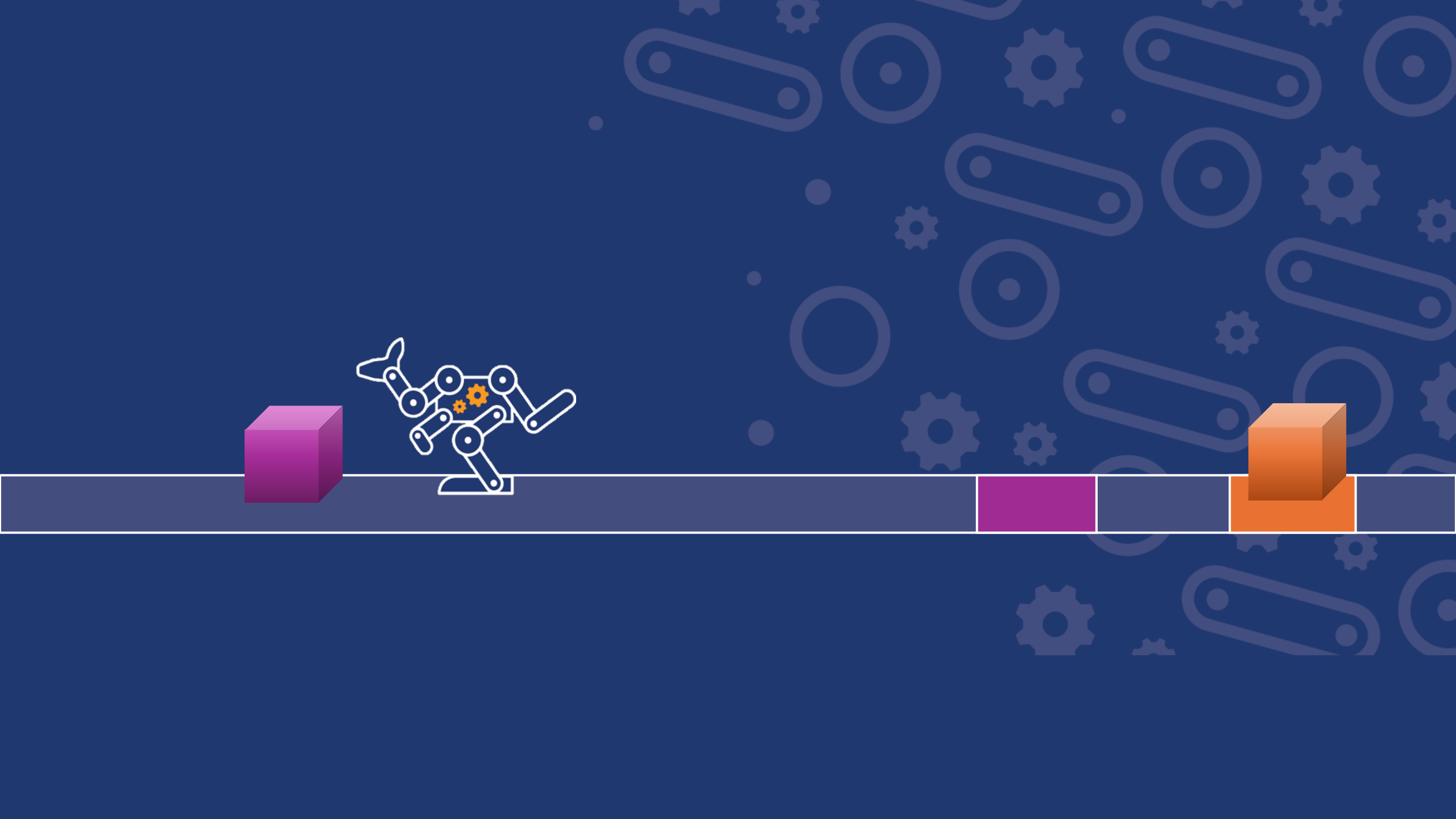


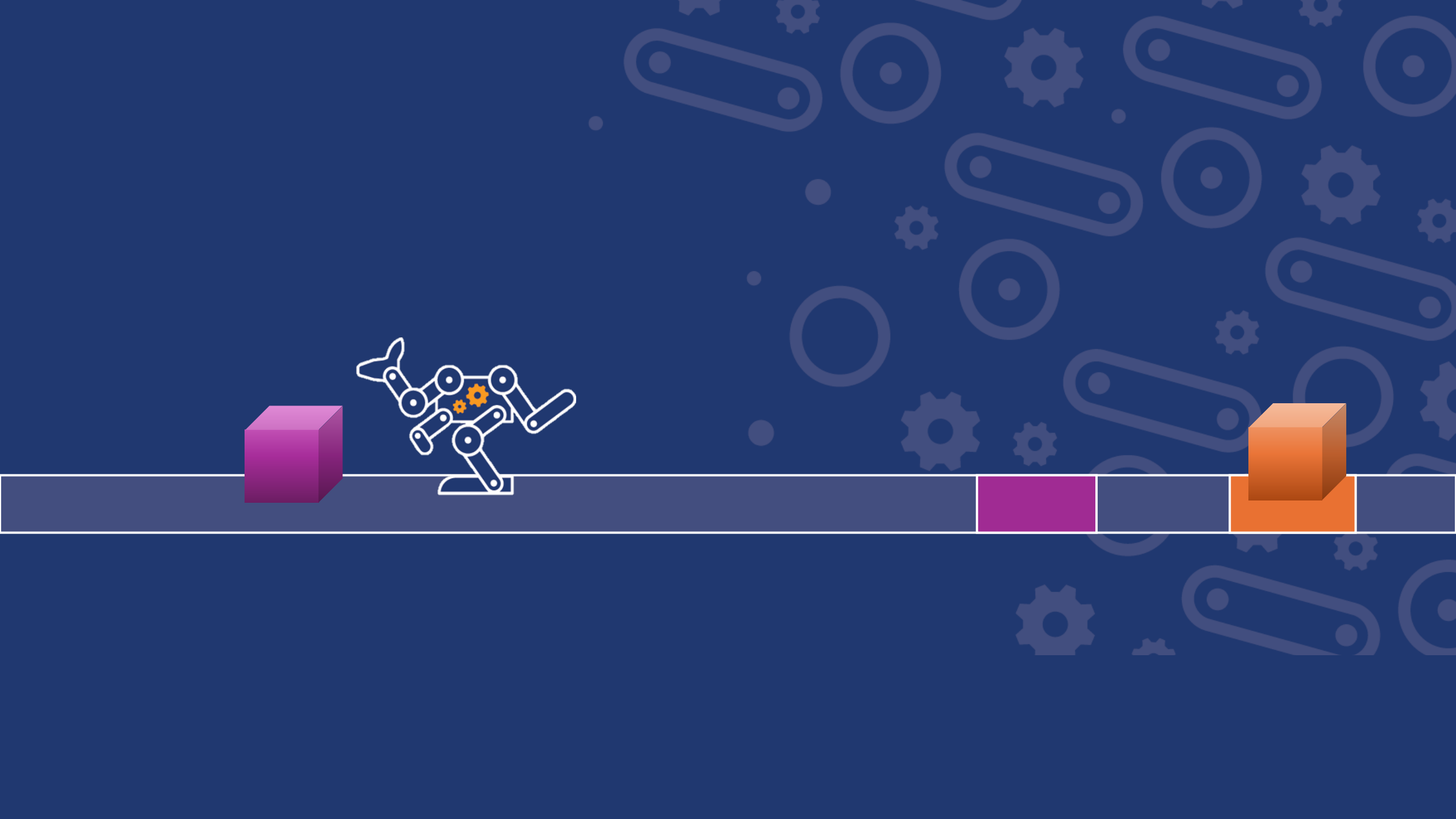


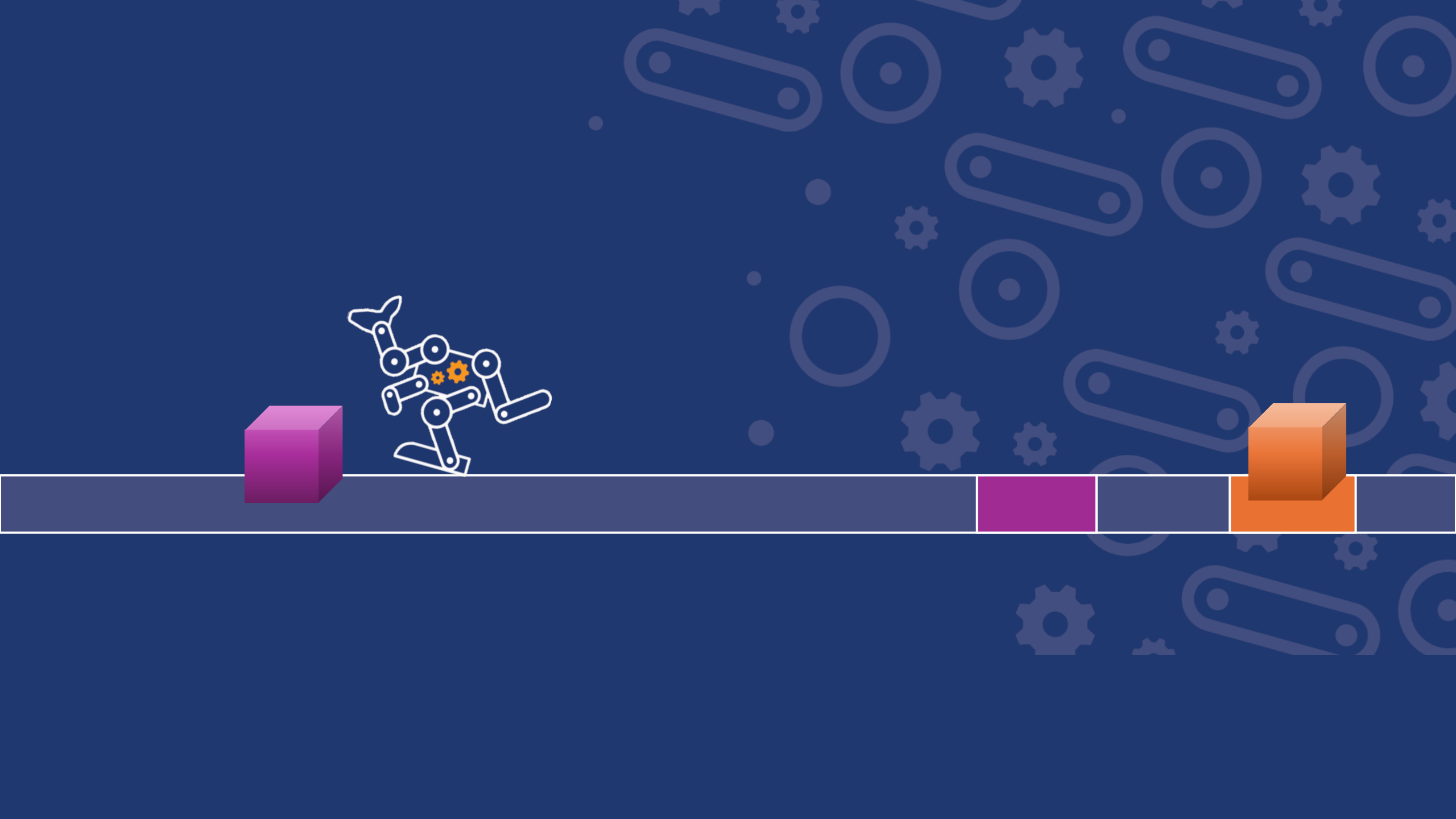


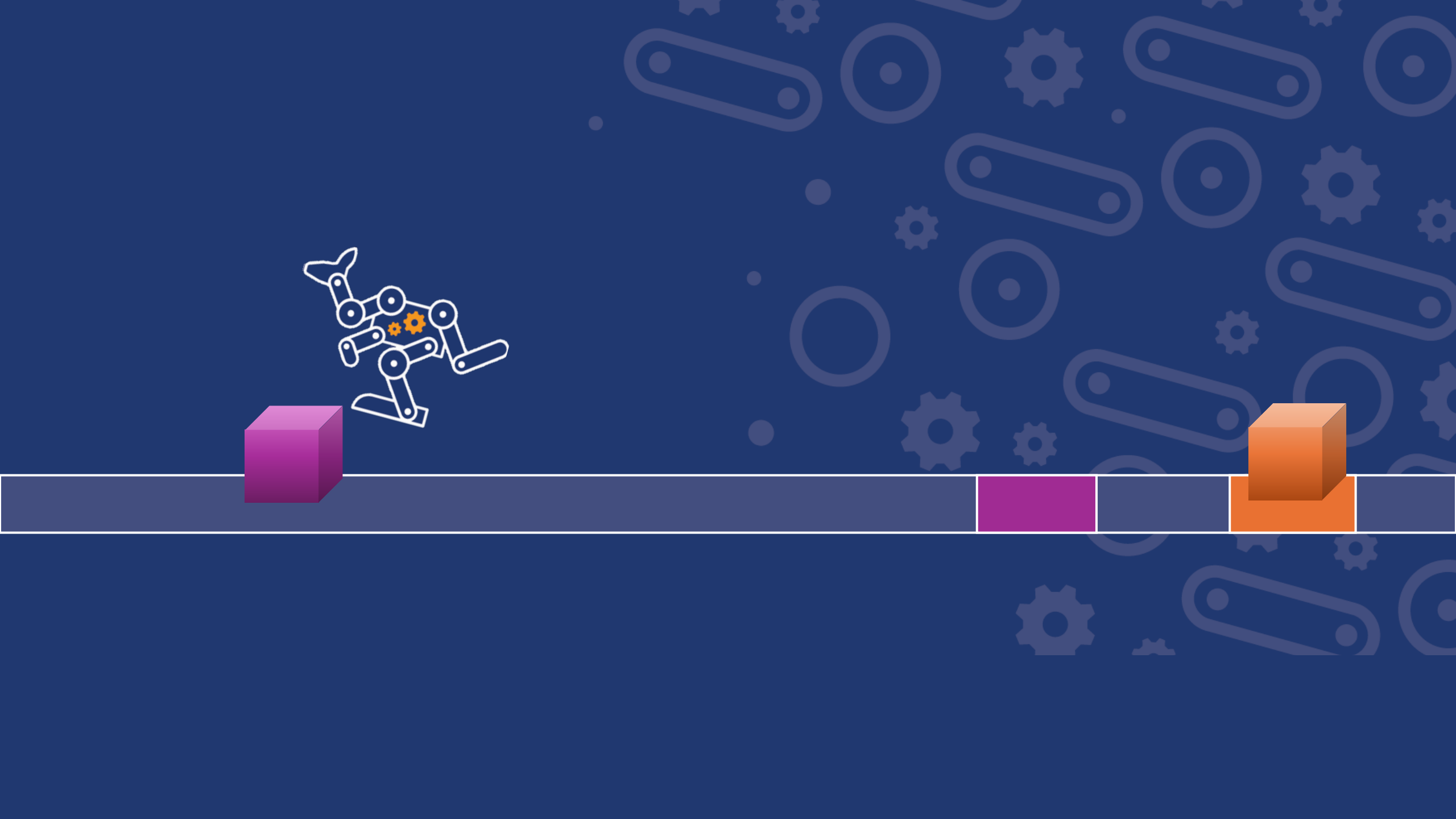


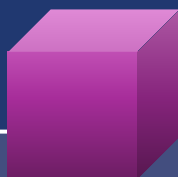


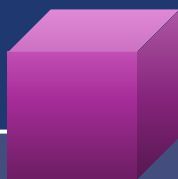


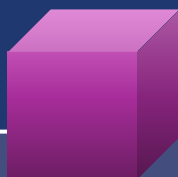


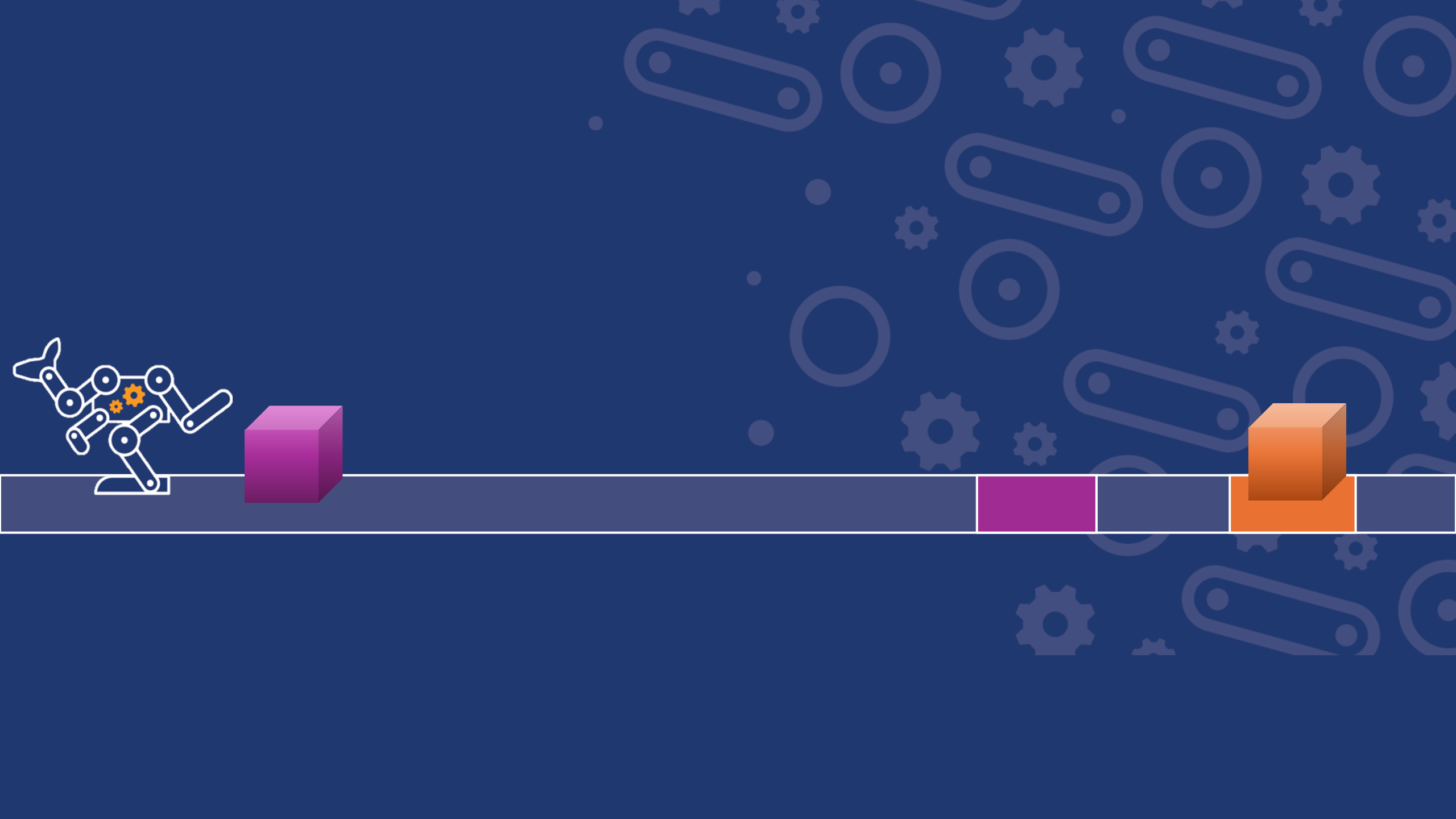






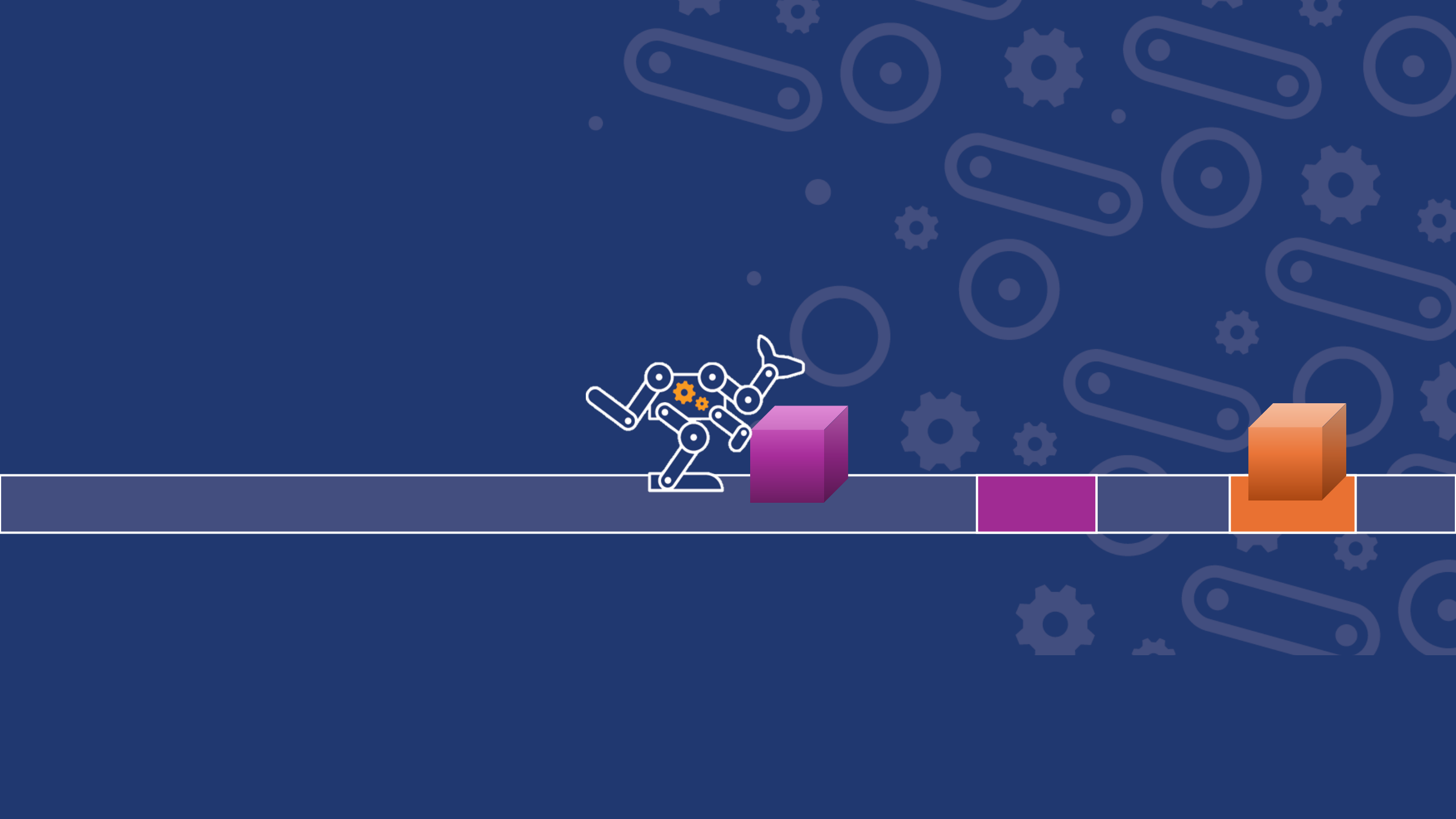


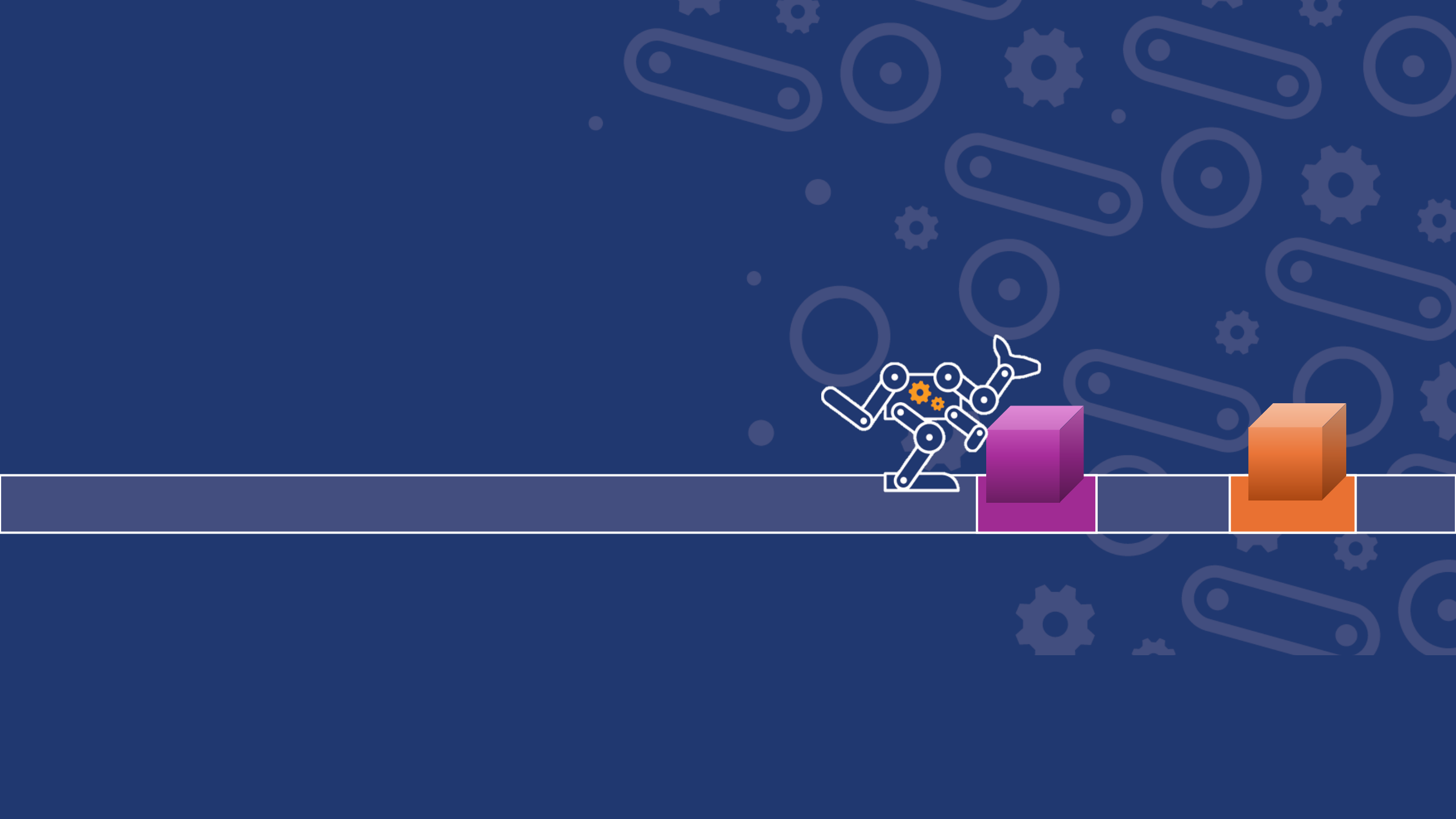


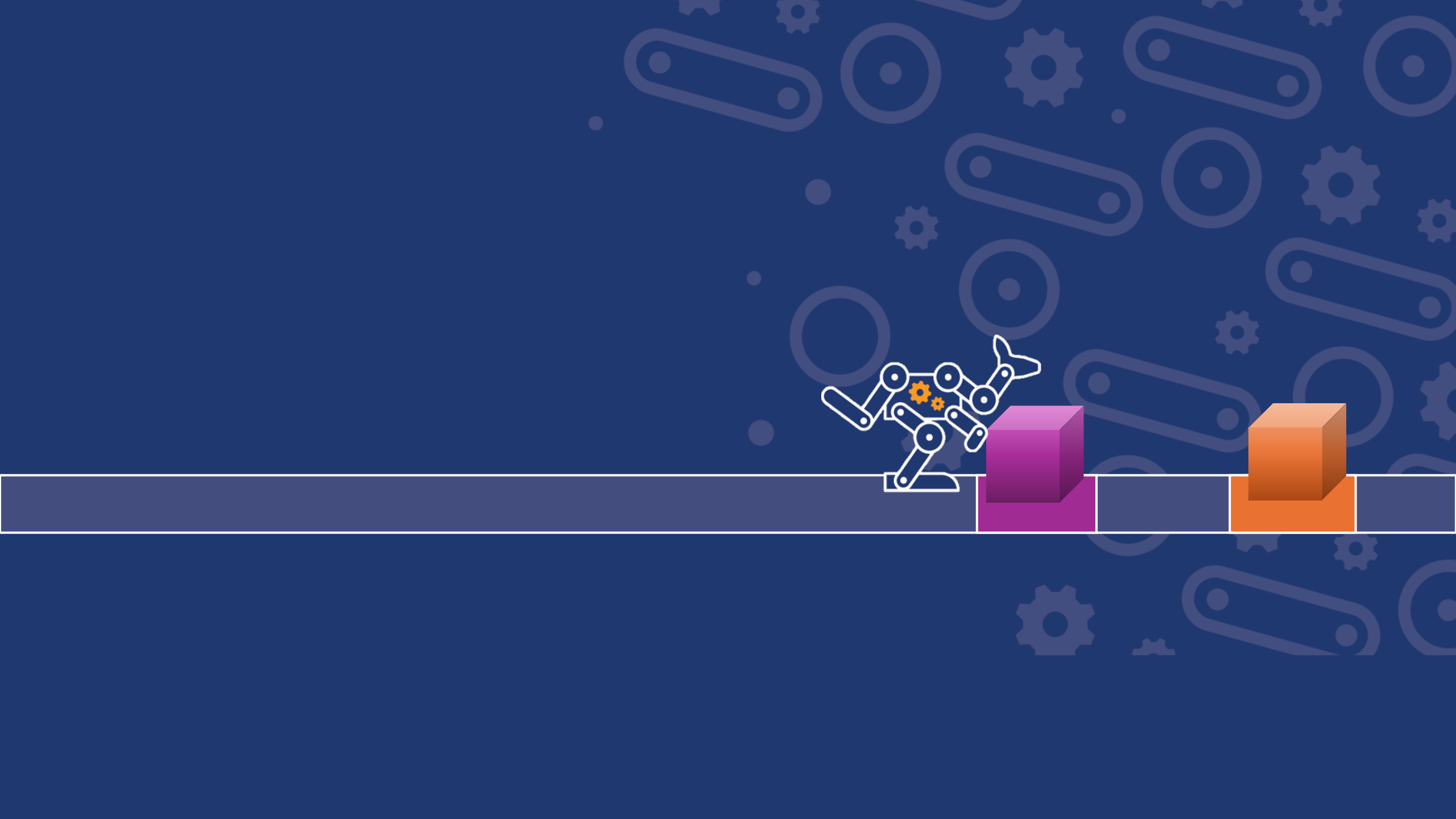






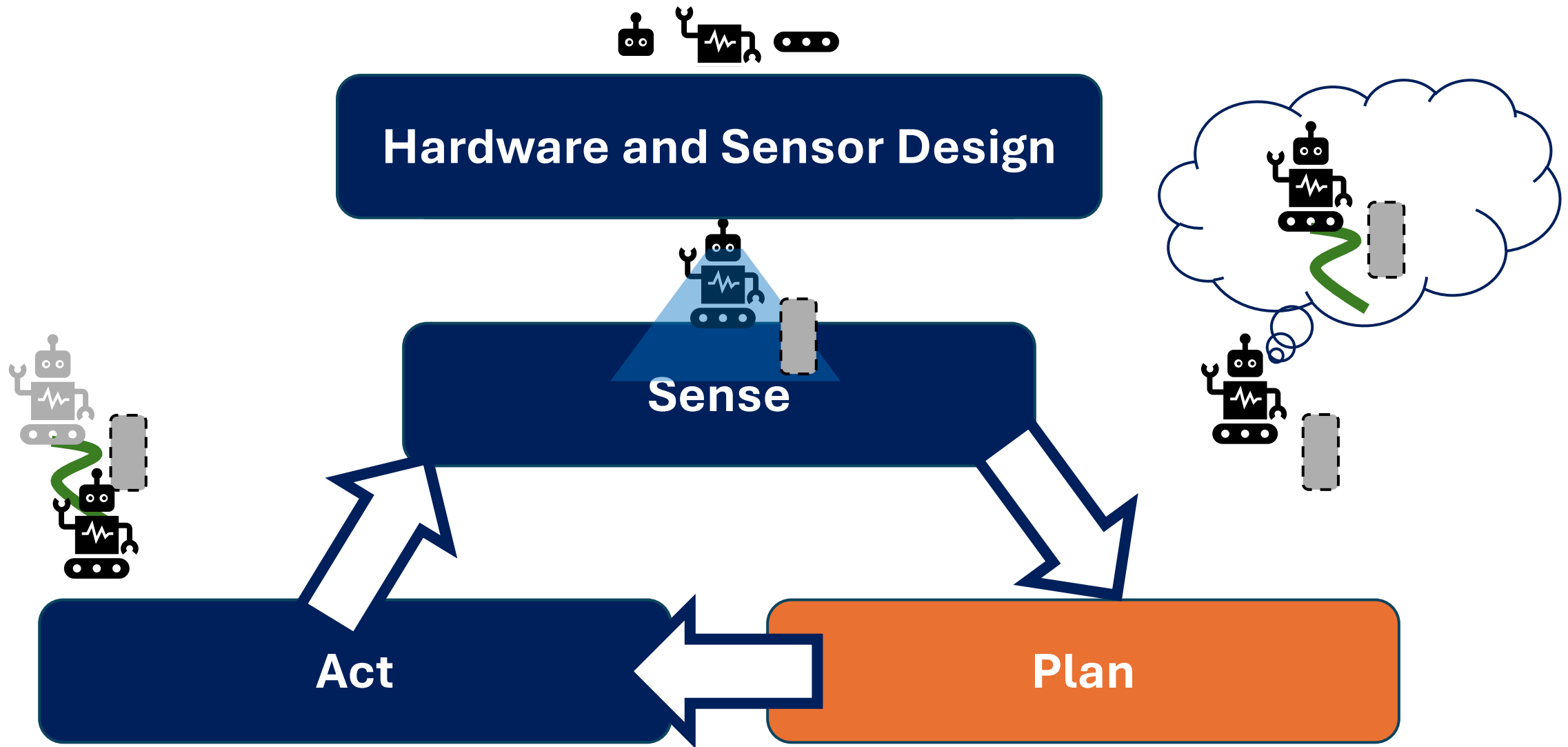










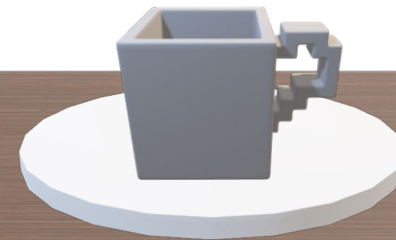


Task and Motion Planning

Logic Beyond Motion Planning



Logic Beyond Motion Planning



Logic Beyond Motion Planning

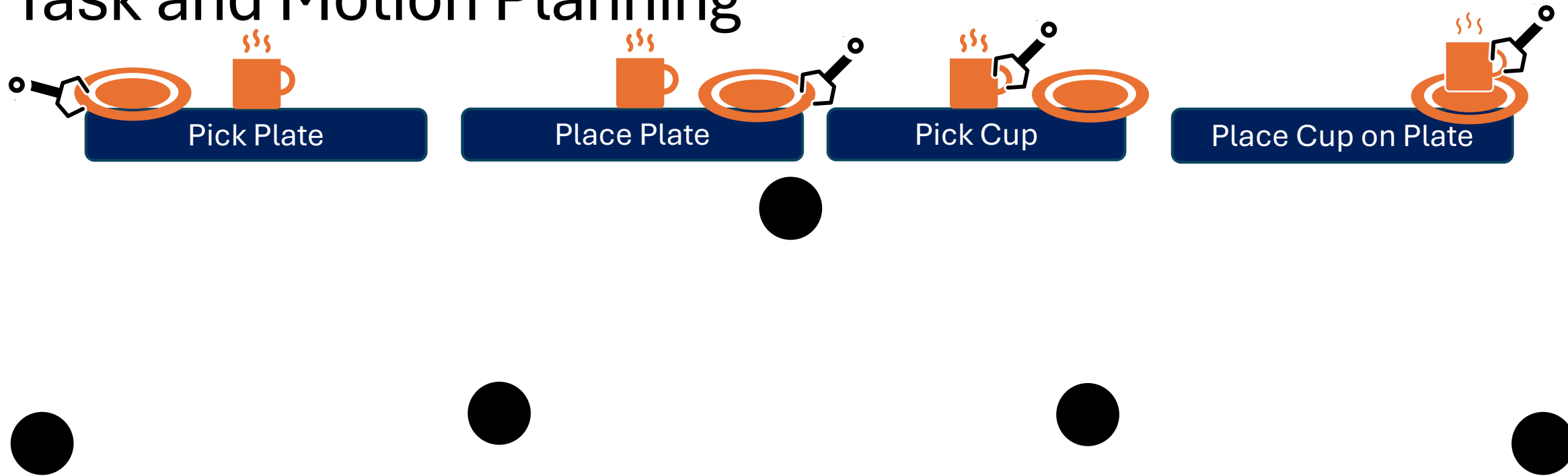


**start
configuration**



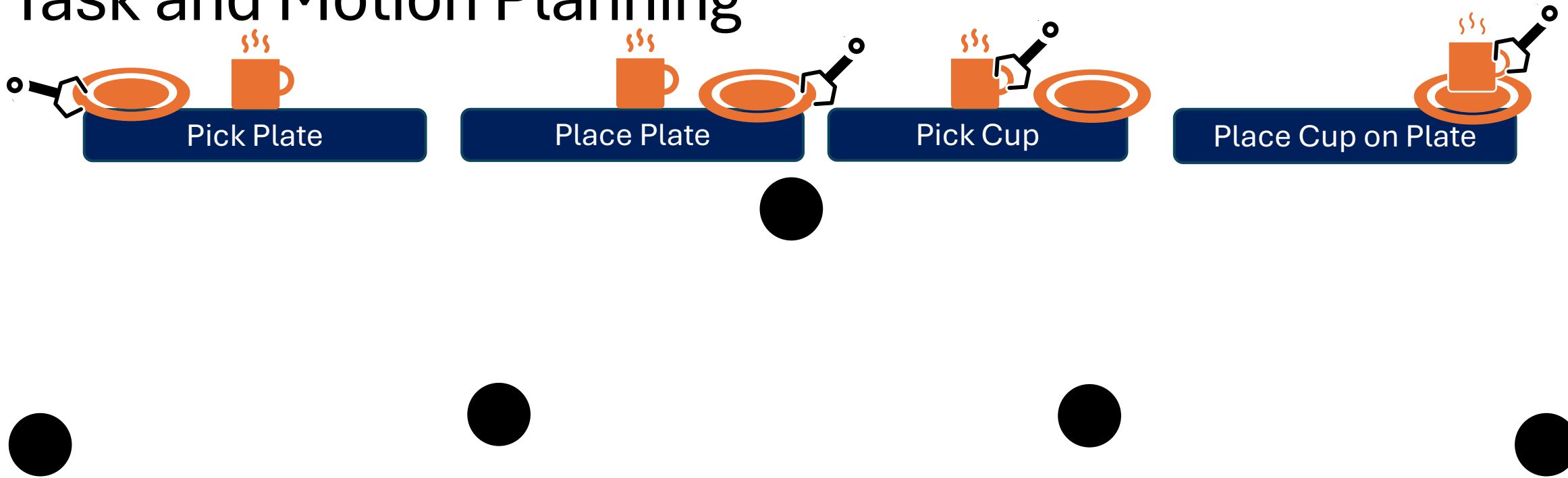
task goal

Task and Motion Planning



Sequence of **deliberate motions or actions** that achieve task goal

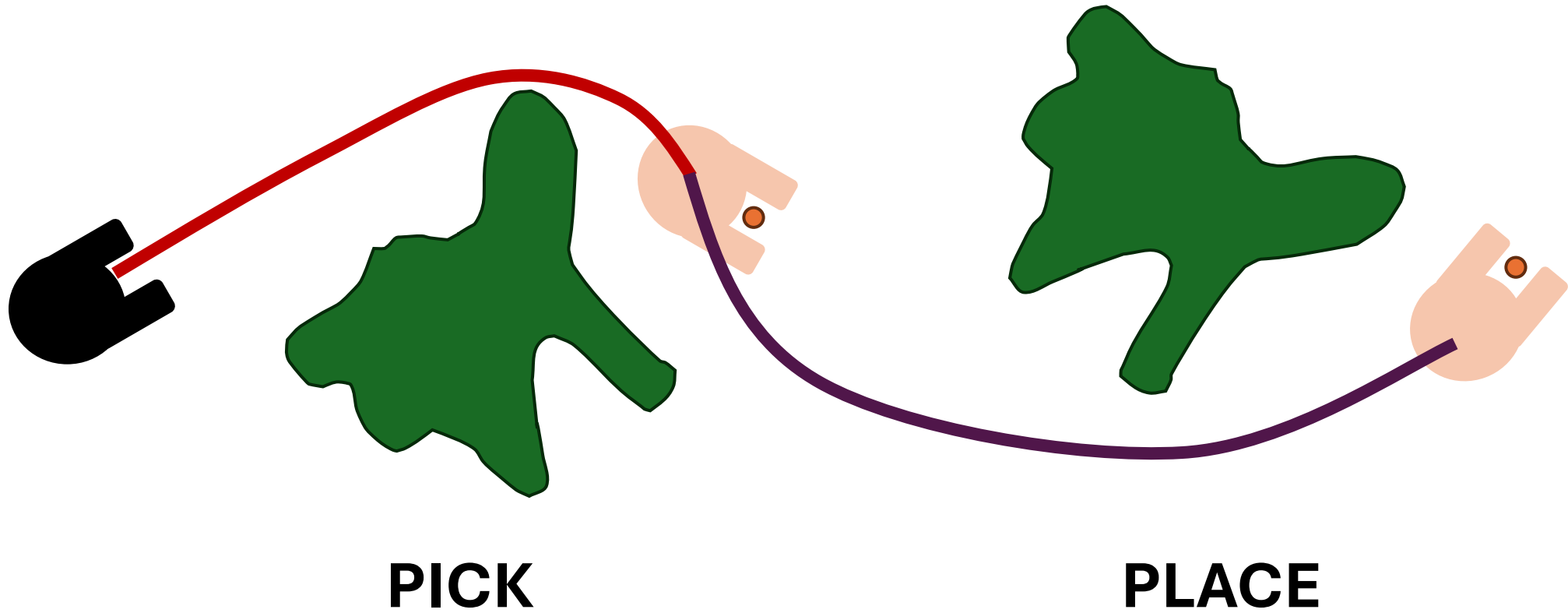
Task and Motion Planning



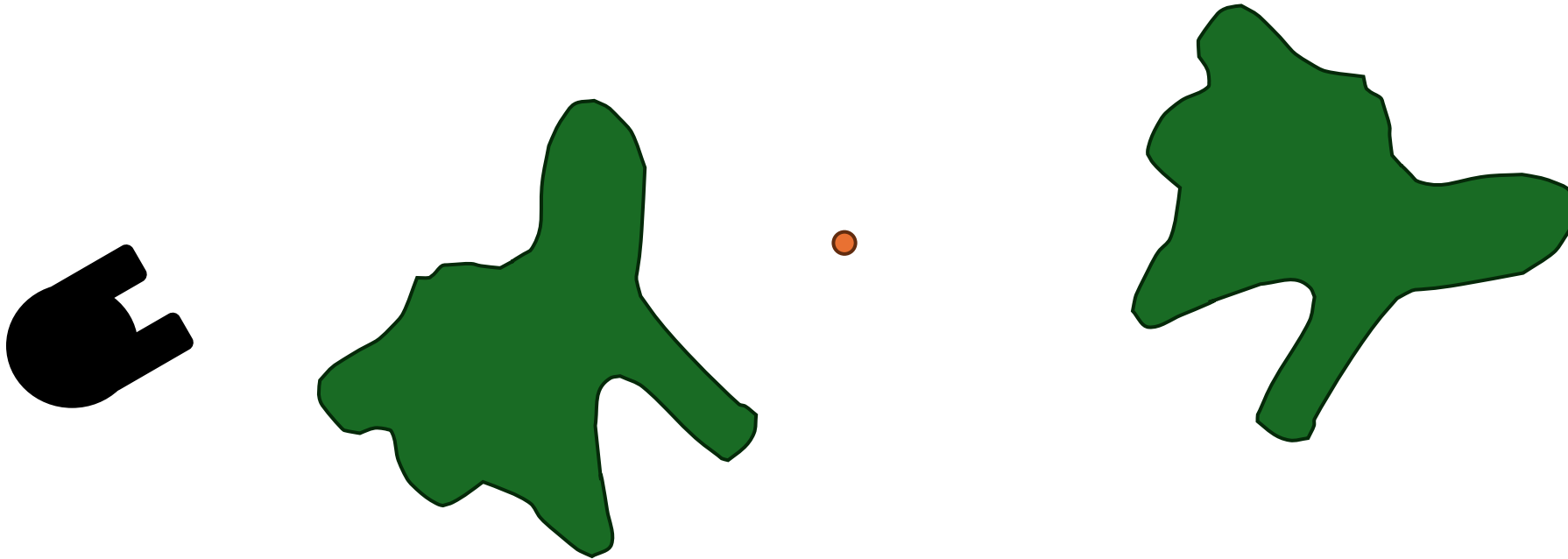
Each of these is a motion planning problem

Sequence of **deliberate motions or actions** that achieve task goal

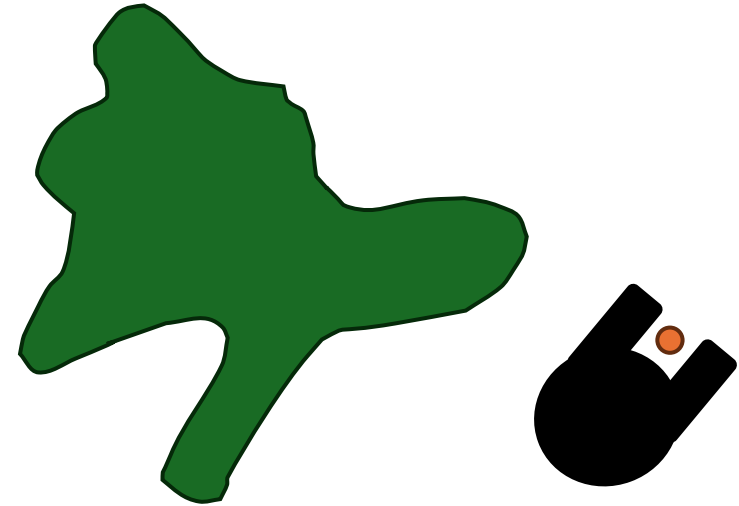
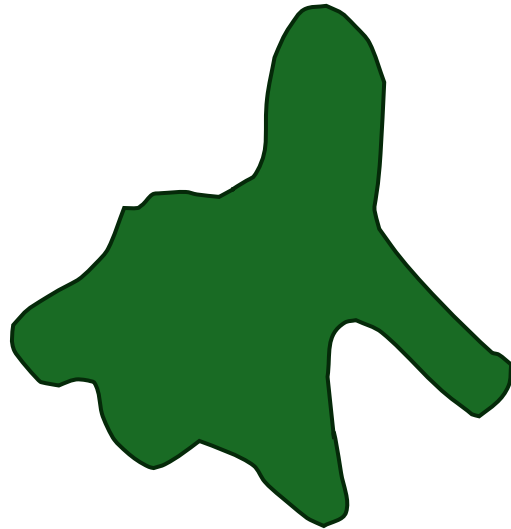
Motion Plans for Actions



Start state and configuration



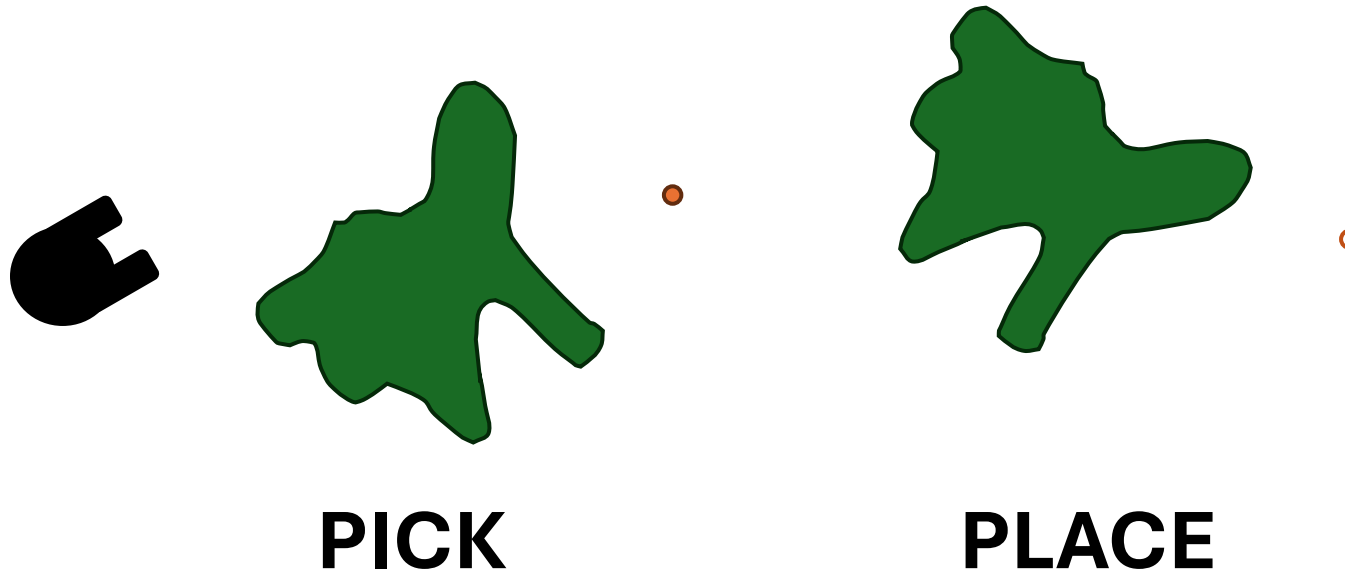
Goal state and configuration



Elements of the problem

🔧 Compute discrete actions to goal state

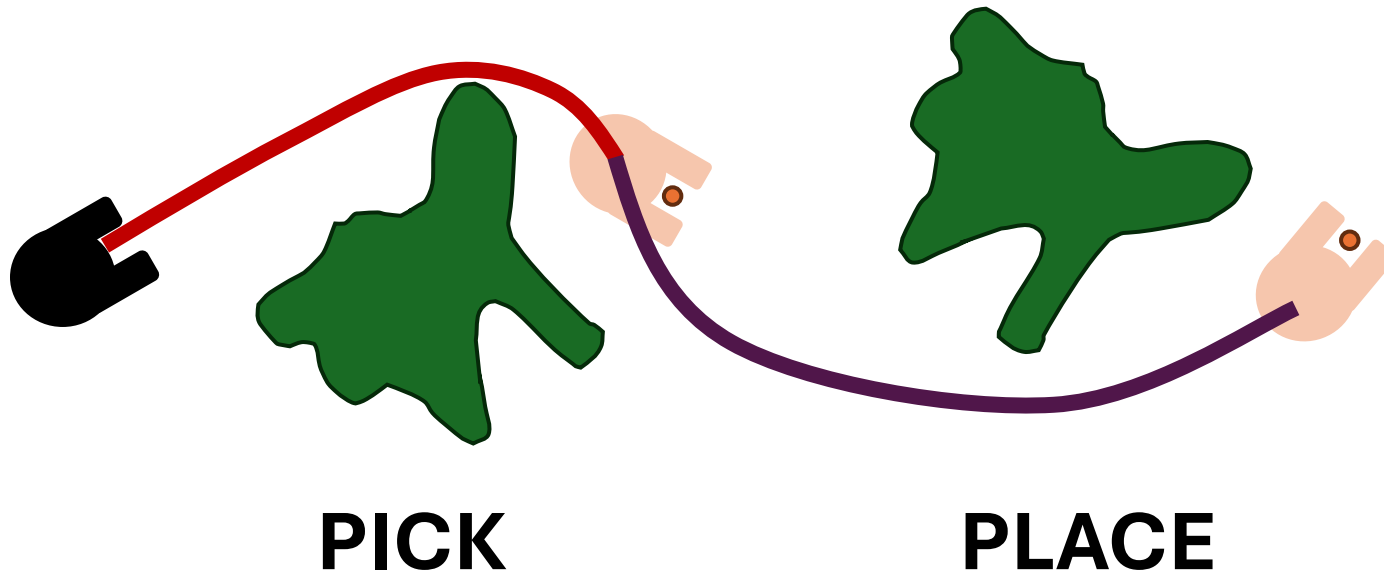
🔧 Compute feasible motions for actions



Elements of the problem

🔧 Compute discrete actions to goal state

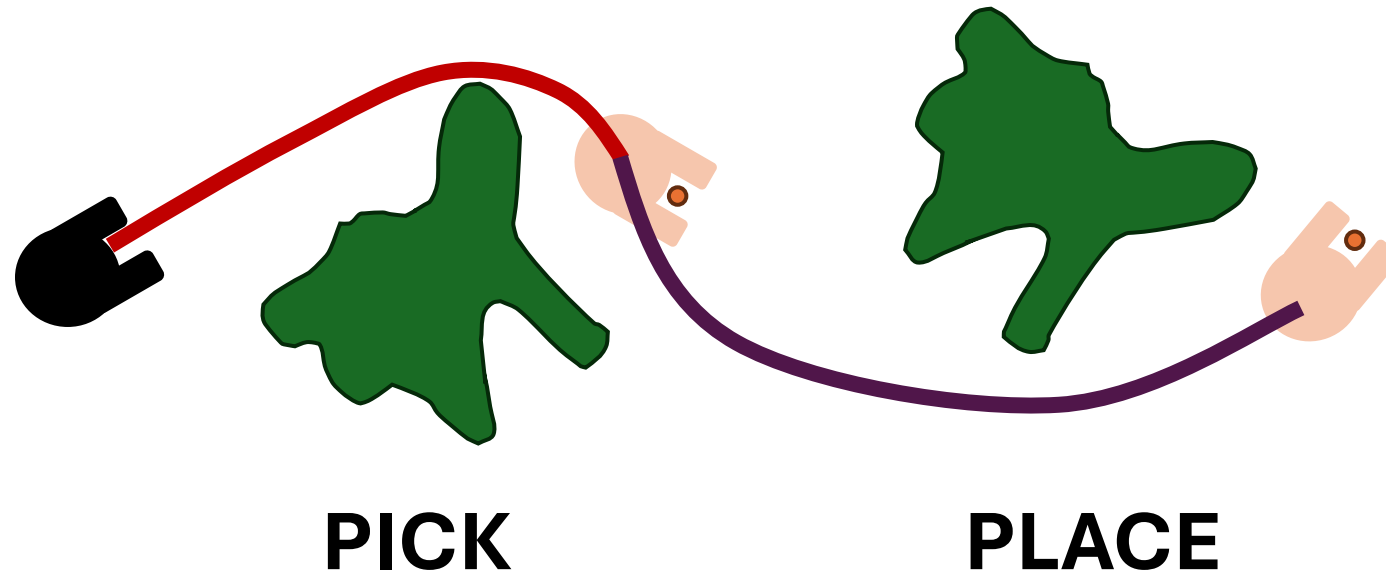
🔧 Compute feasible motions for actions



Elements of the problem

🔧 **Task Planning:** Compute discrete actions to goal state

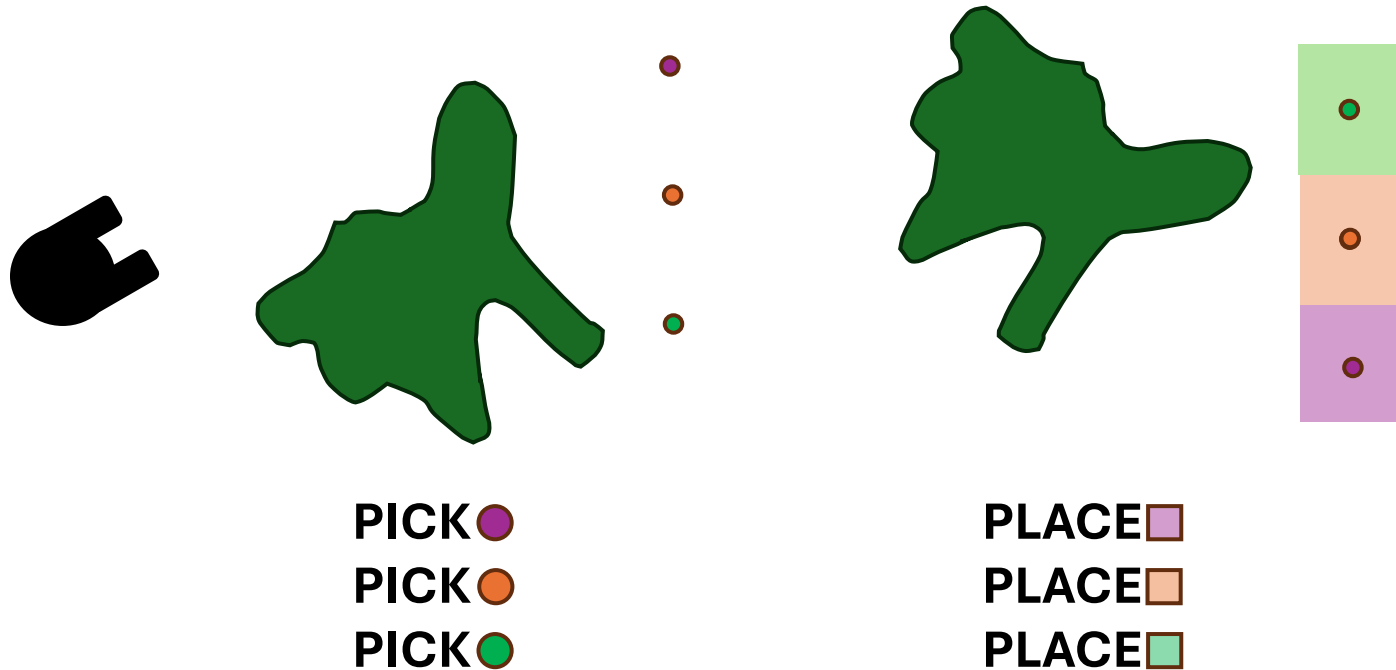
🔧 **Motion Planning:** Compute feasible motions for actions



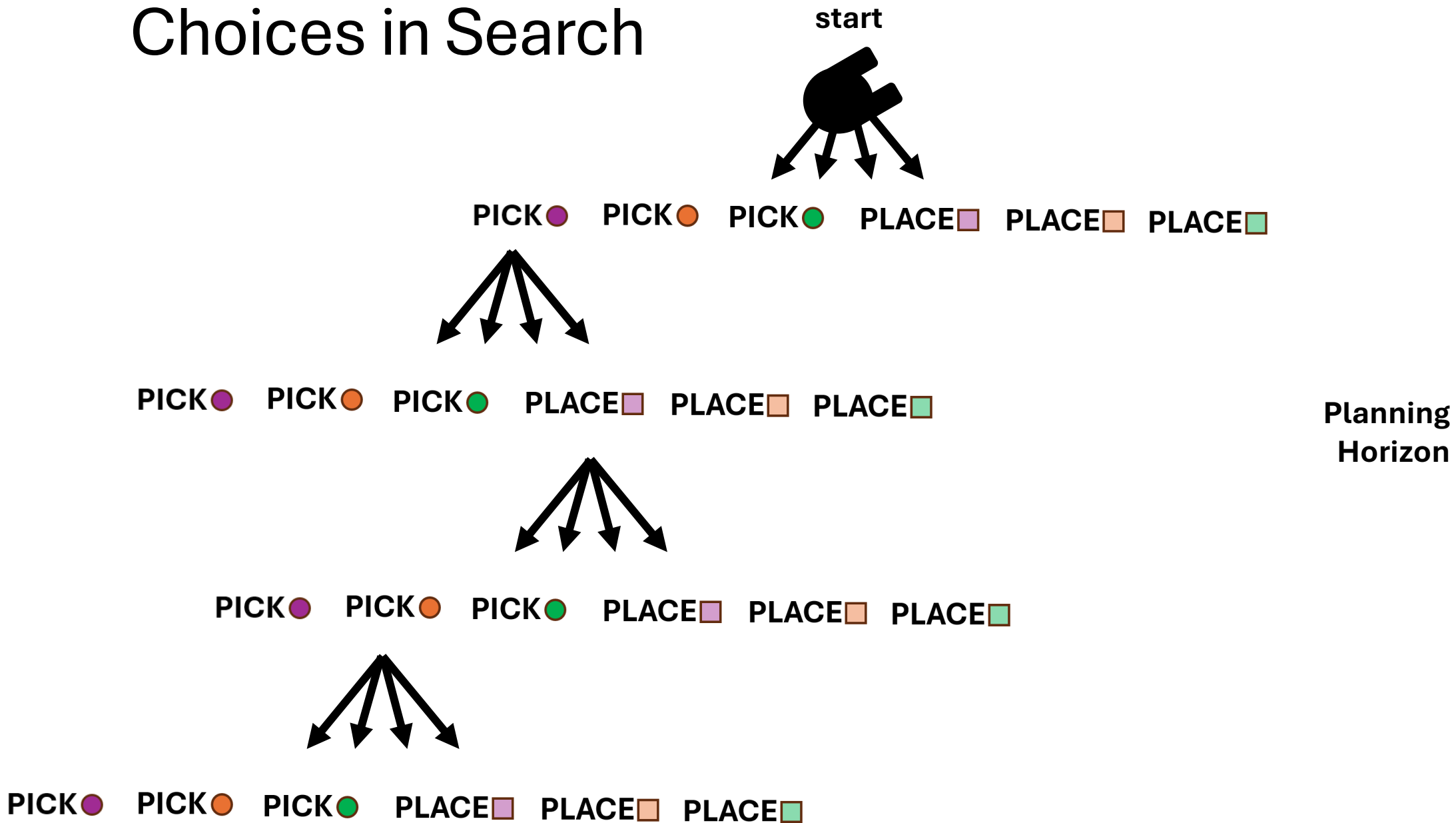
Choices in Search

🤖 **Task Planning:** Compute discrete actions to goal state

🤖 **Motion Planning:** Compute feasible motions for actions



Choices in Search

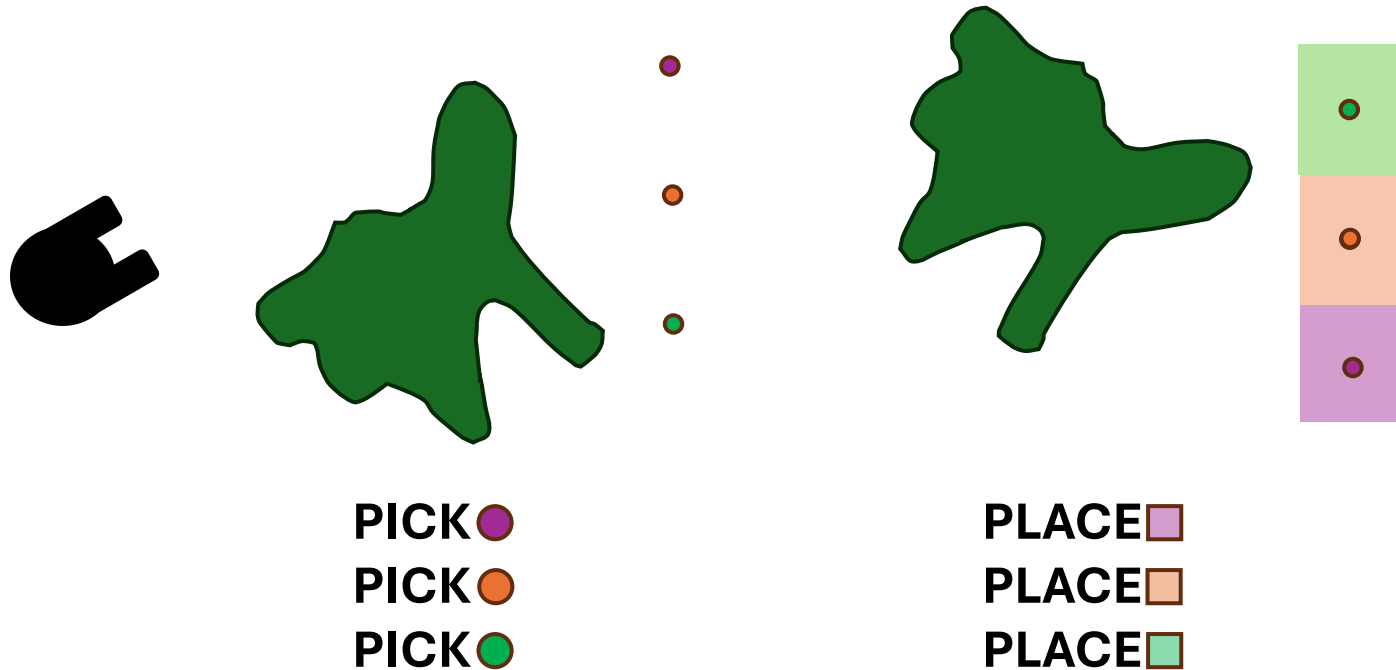


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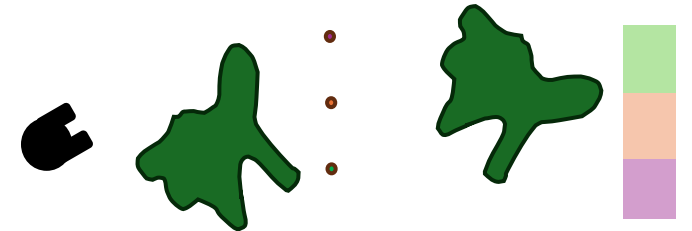
Choices in Search

🤖 **Task Planning:** Compute discrete actions to goal state

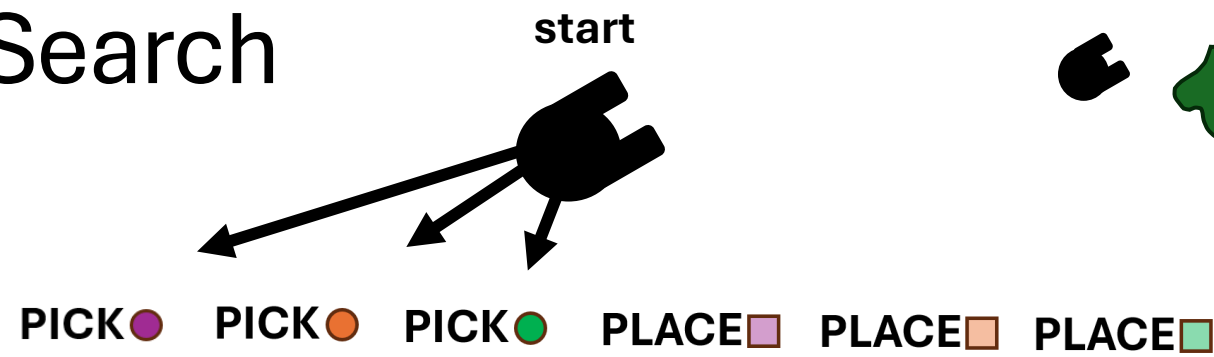
🤖 **Motion Planning:** Compute feasible motions for actions



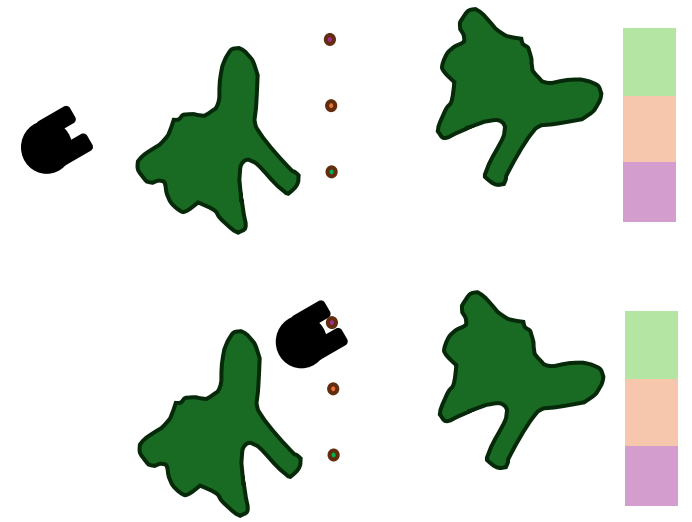
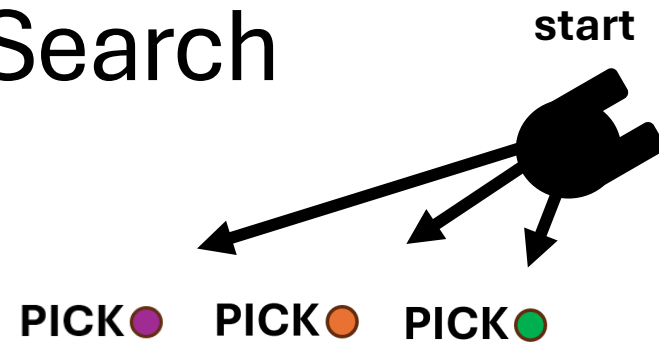
Choices in Search



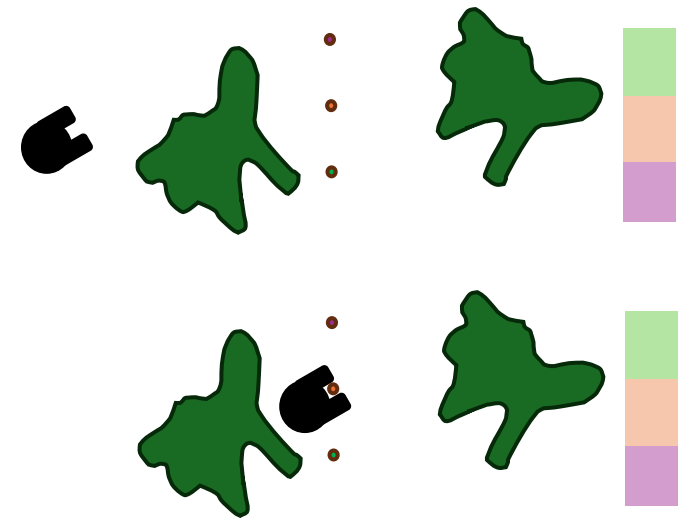
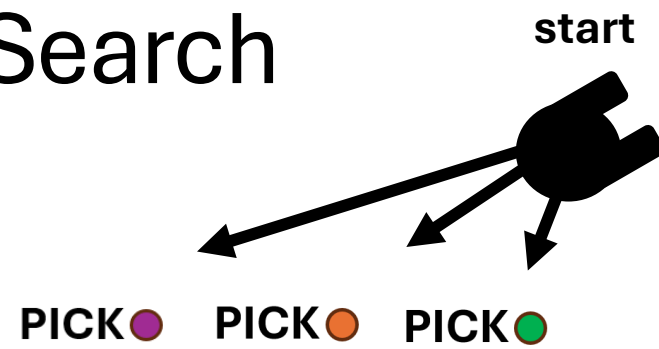
Choices in Search



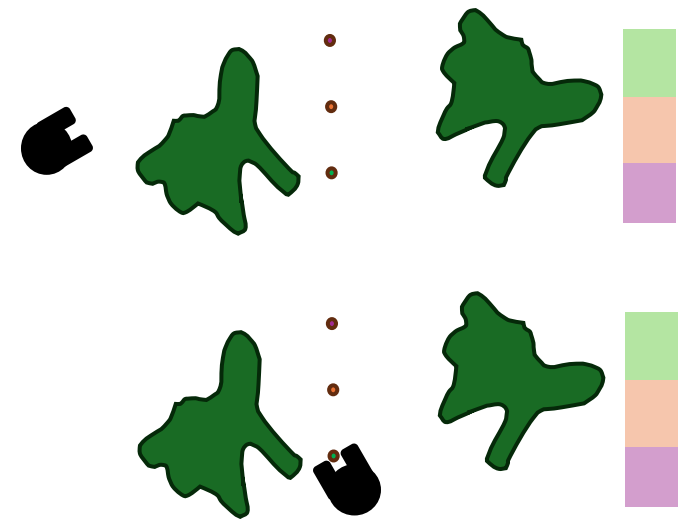
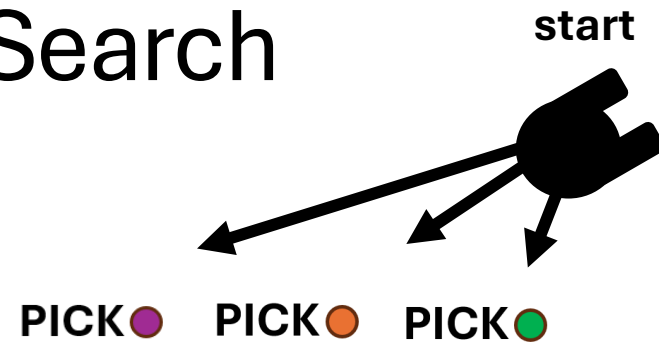
Choices in Search



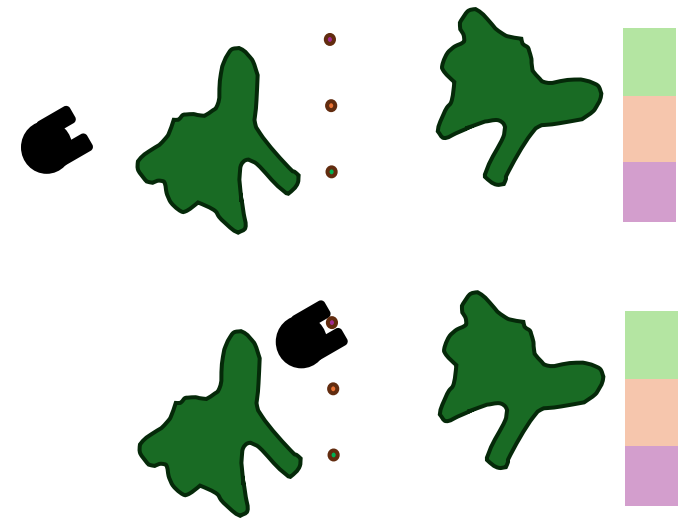
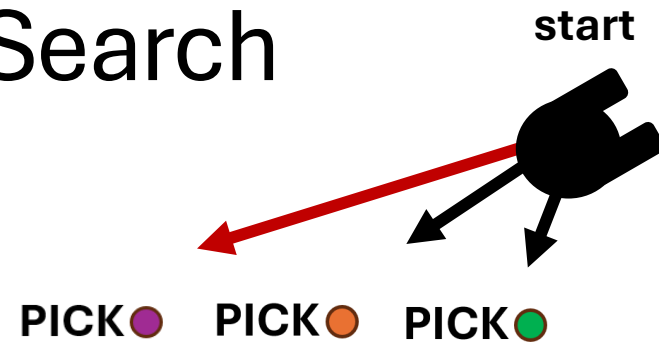
Choices in Search



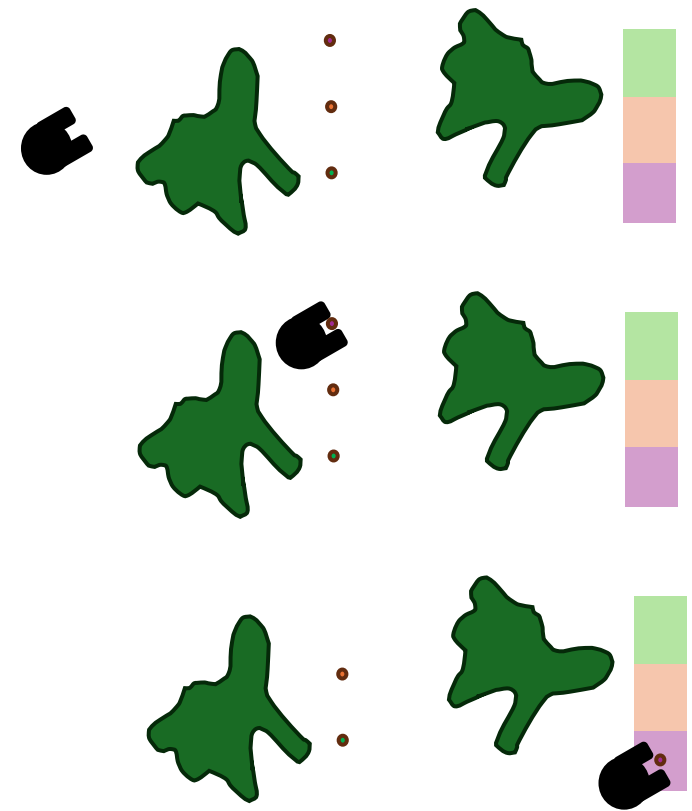
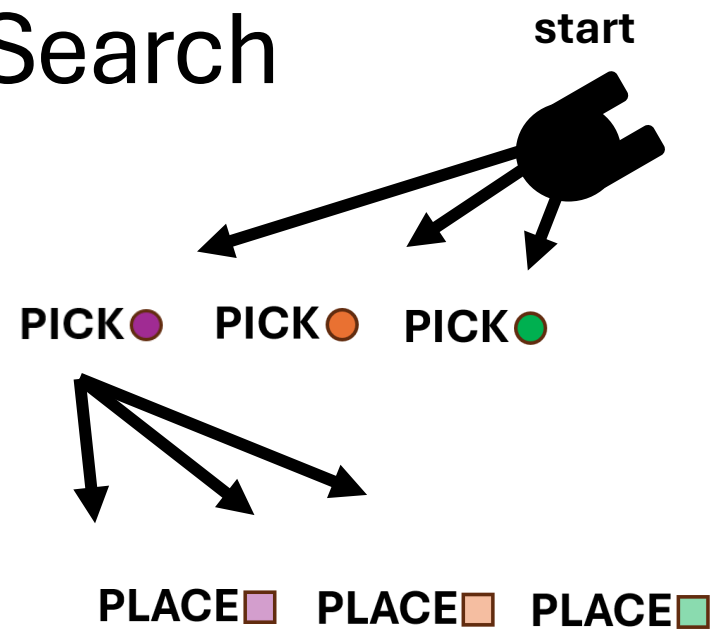
Choices in Search



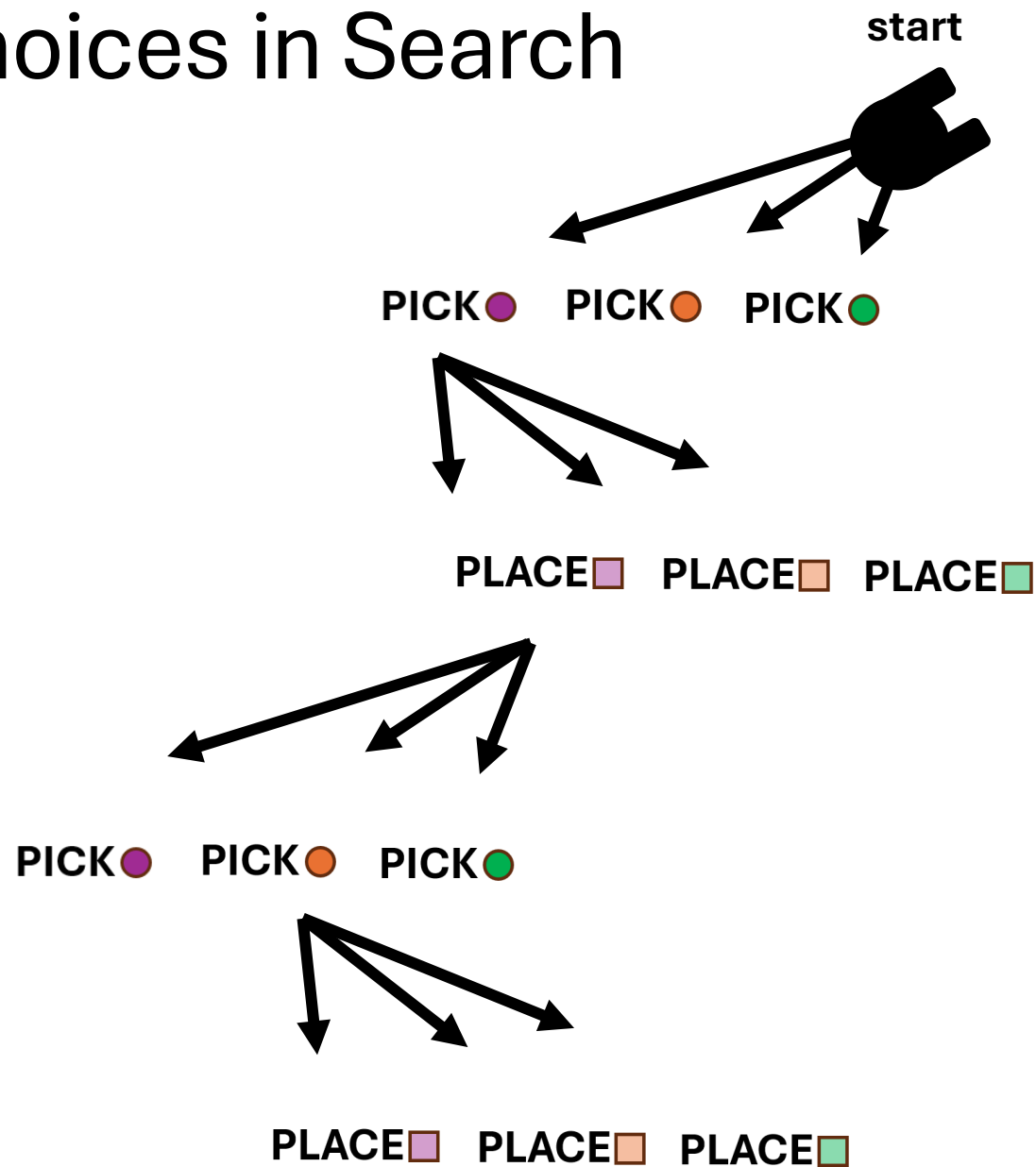
Choices in Search



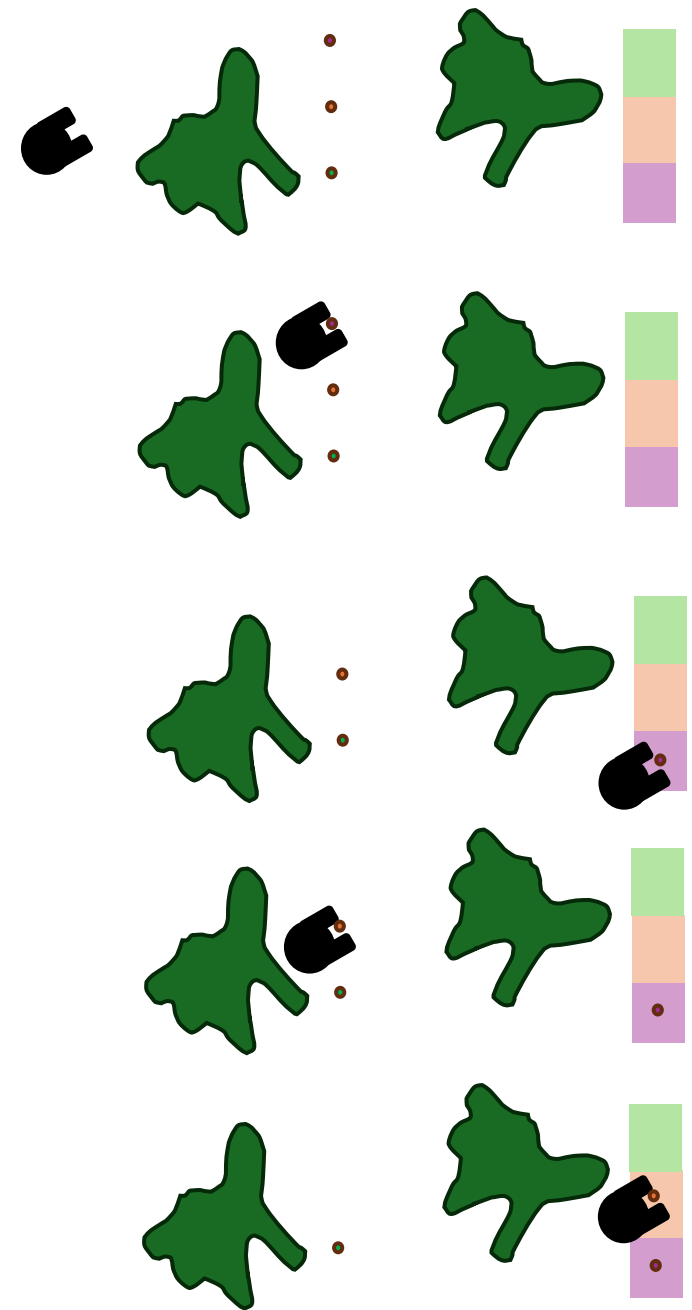
Choices in Search



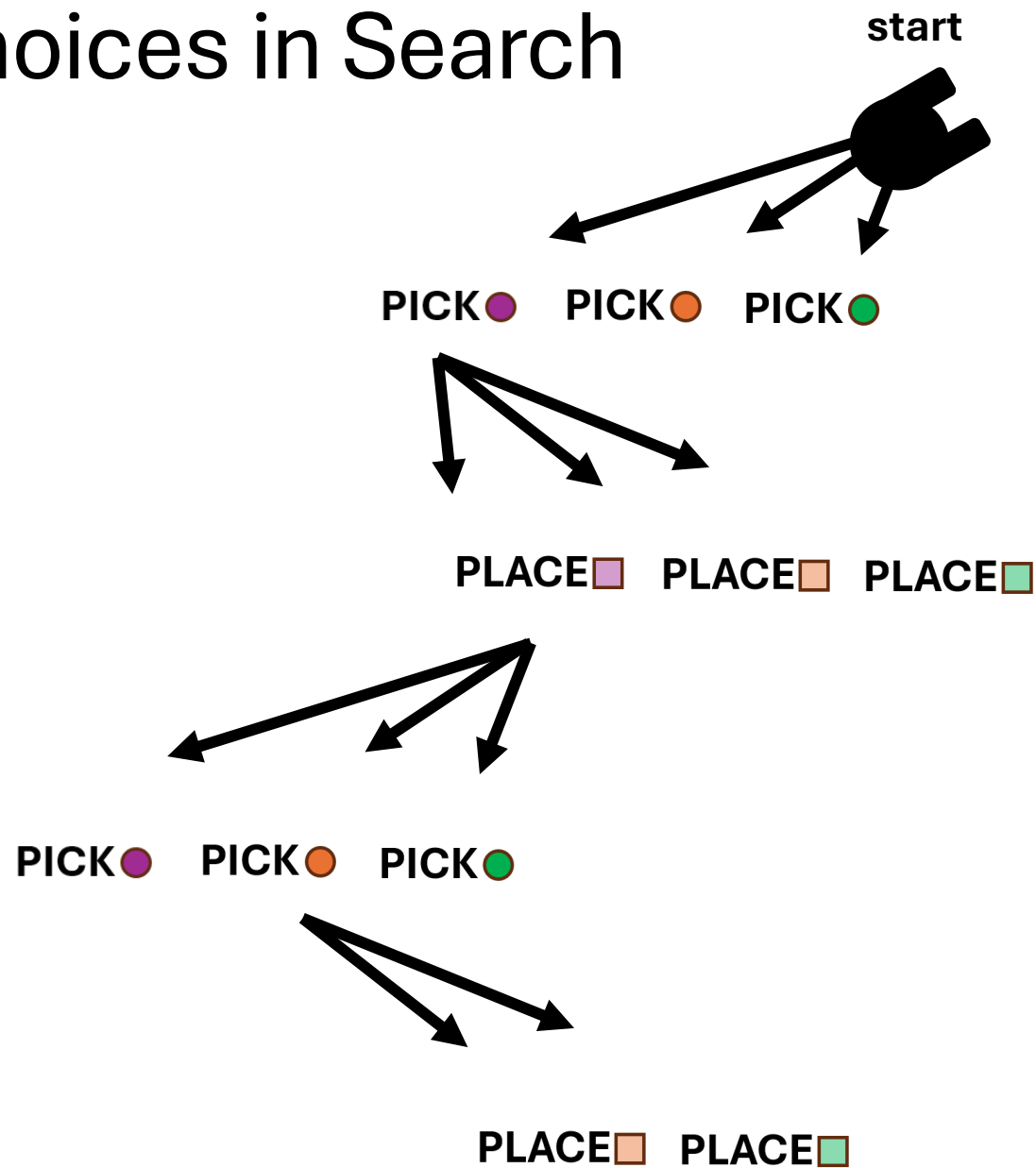
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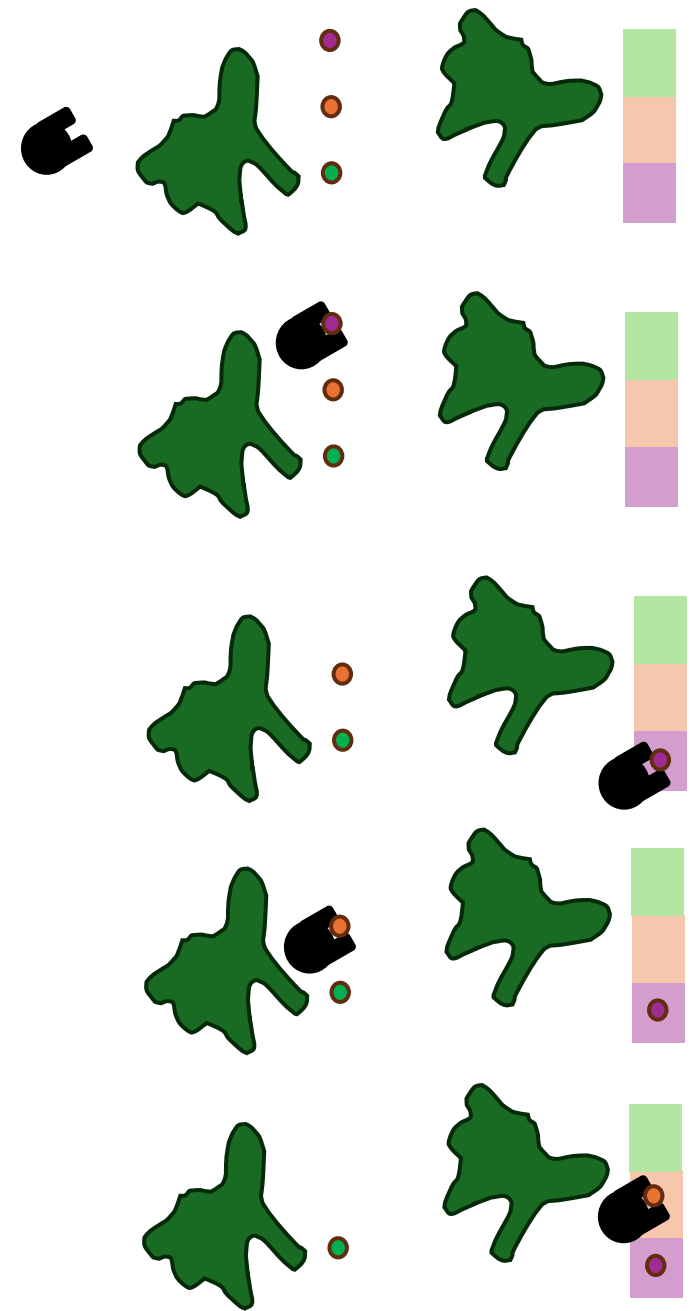
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Choices in Search



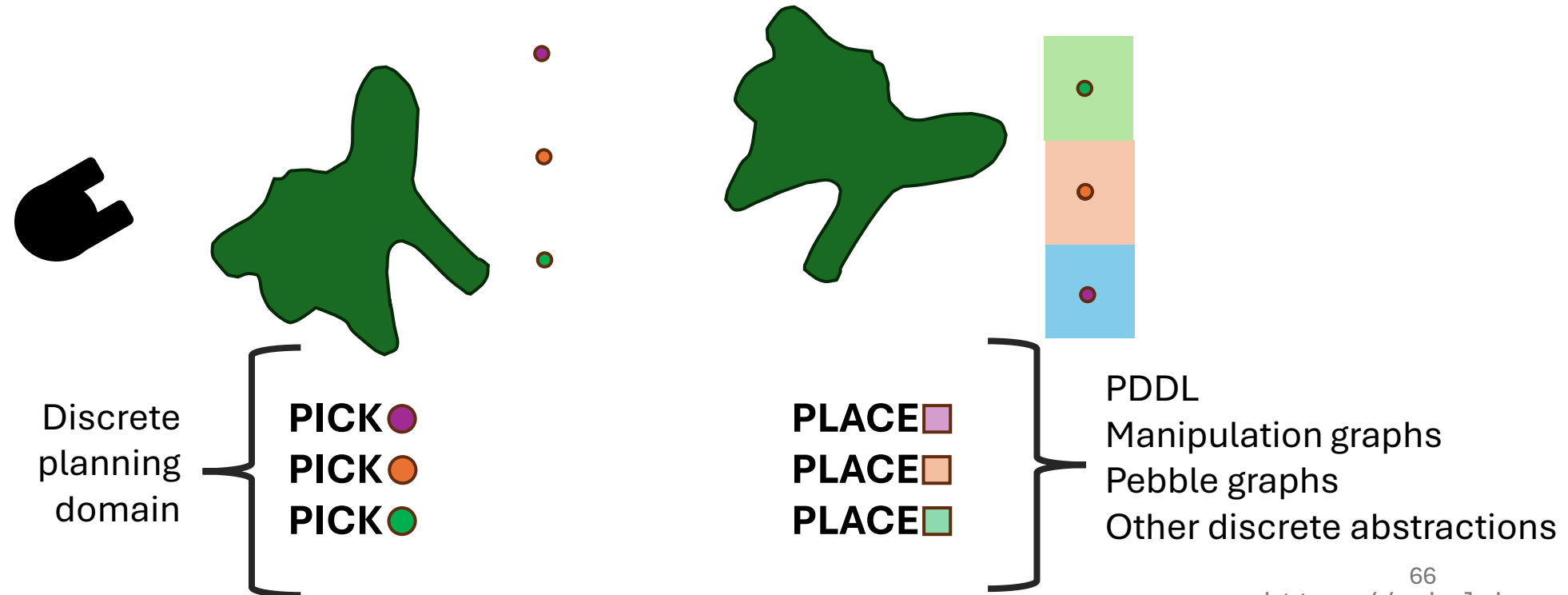
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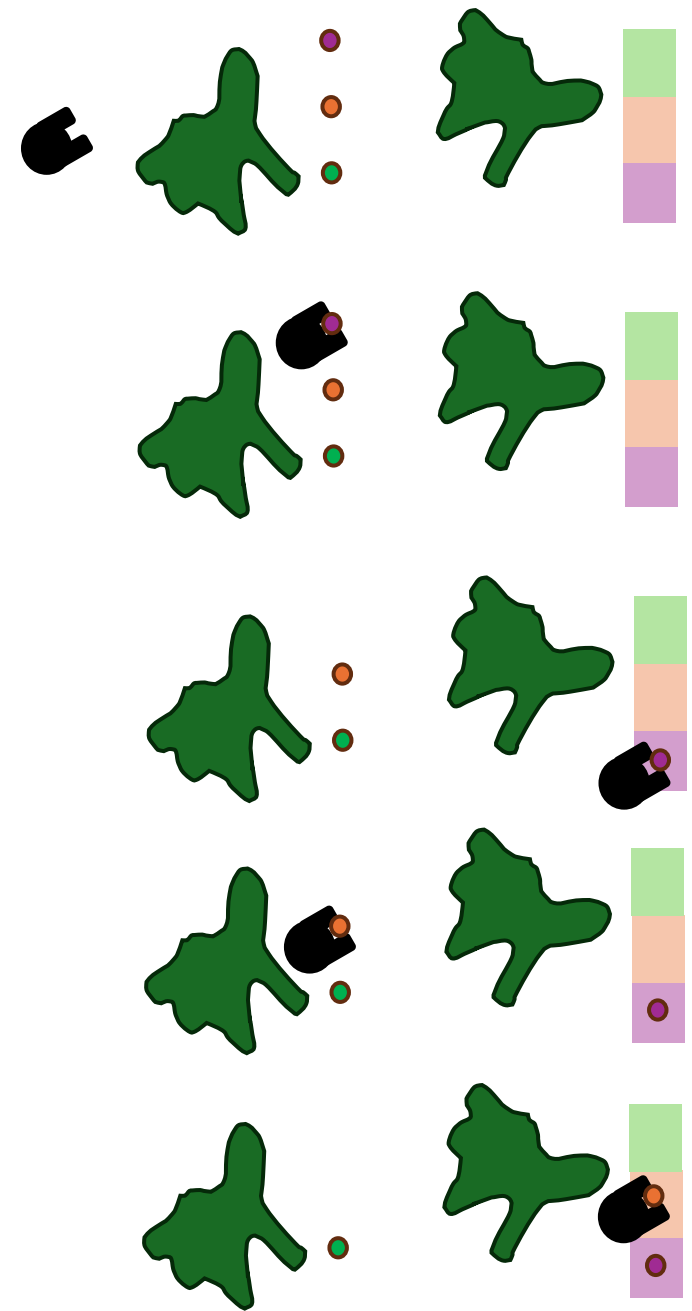
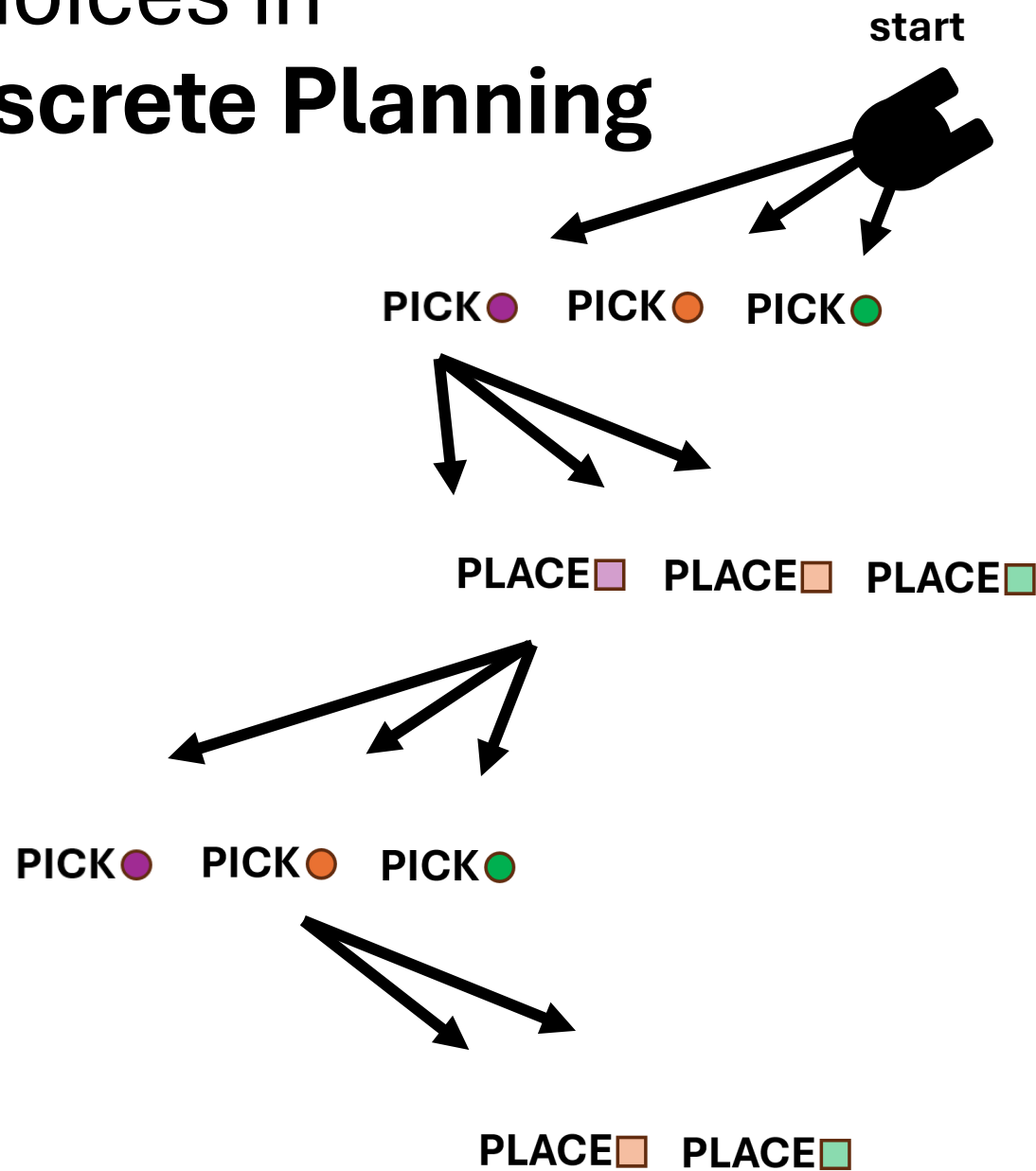
Planning Domain Representations

🔧 **Preconditions:** When can an action be performed

🔧 **Effects:** What is the change of state an action results in



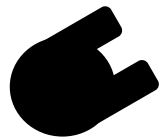
Choices in Discrete Planning



Task Planning, Motion Planning

🤖 Find task plan (task planner query)

🤖 Find motion plans (sequence of motion planner queries)



goal

all preconditions
satisfied

PICK



effect is now
object grasped

all preconditions
satisfied

PLACE

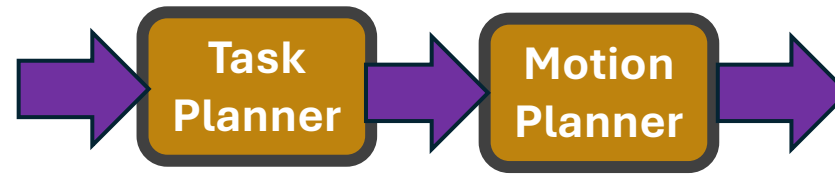


effect is object
location
changed and
object
ungrasped

Task Planning, Motion Planning

🤖 Find task plan (task planner query)

🤖 Find motion plans (sequence of motion planner queries)



goal

all preconditions
satisfied

PICK



effect is now
object grasped

all preconditions
satisfied

PLACE

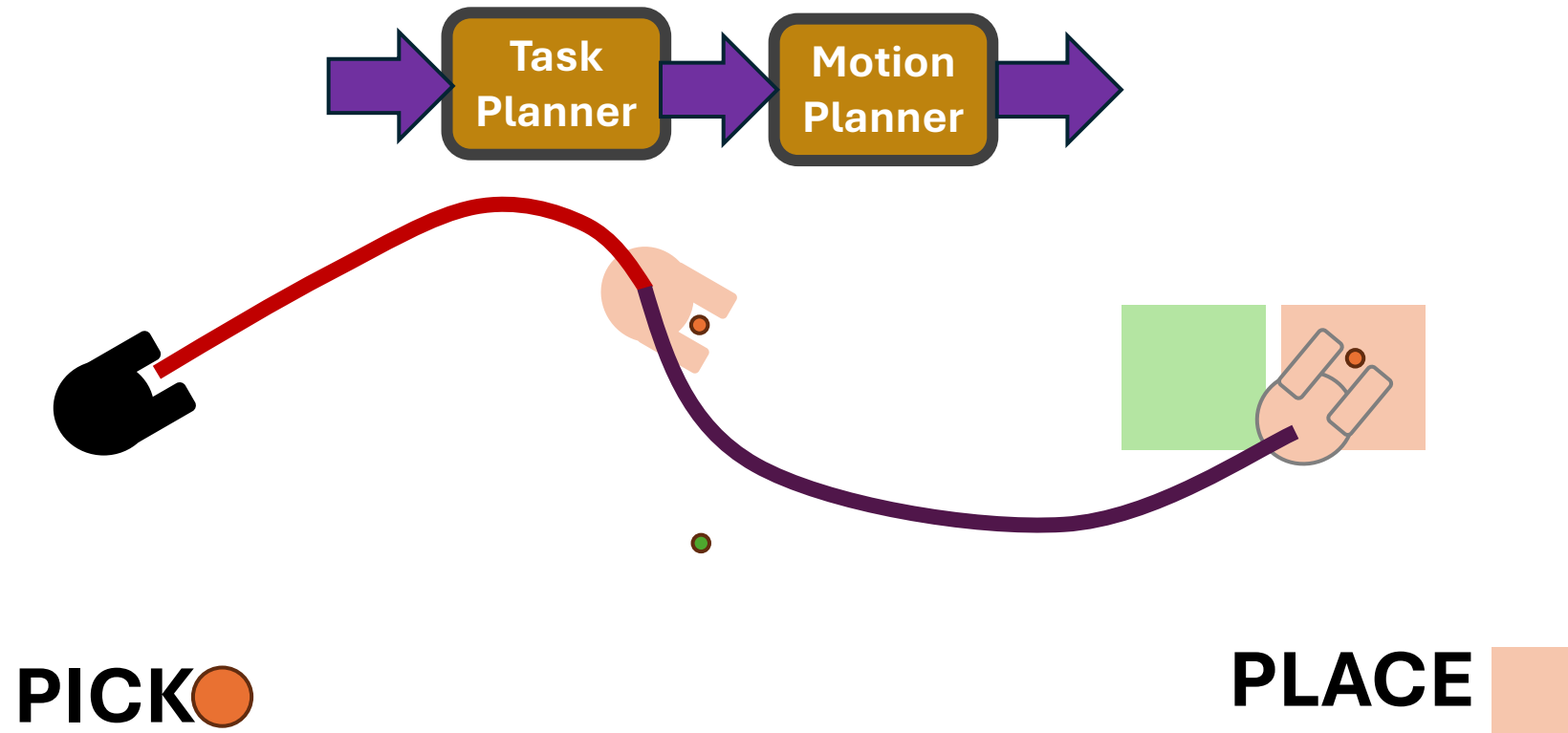


effect is object
location
changed and
object
ungrasped

Task Planning THEN Motion Planning

🔍 Find task plan (task planner query)

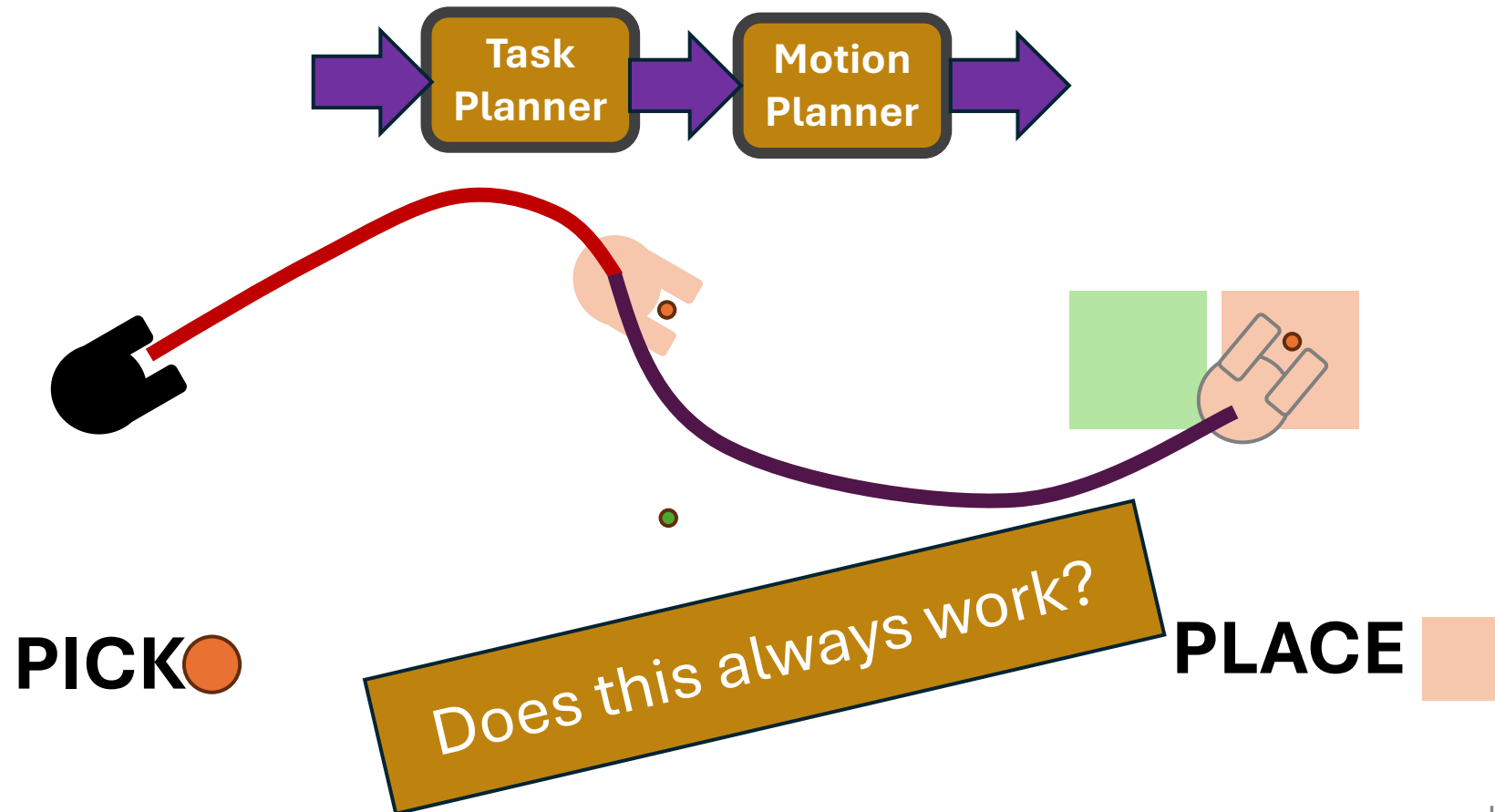
🔍 **THEN** Find motion plans (sequence of motion planner queries)



Task Planning THEN Motion Planning

🔍 Find task plan (task planner query)

🔍 **THEN** Find motion plans (sequence of motion planner queries)

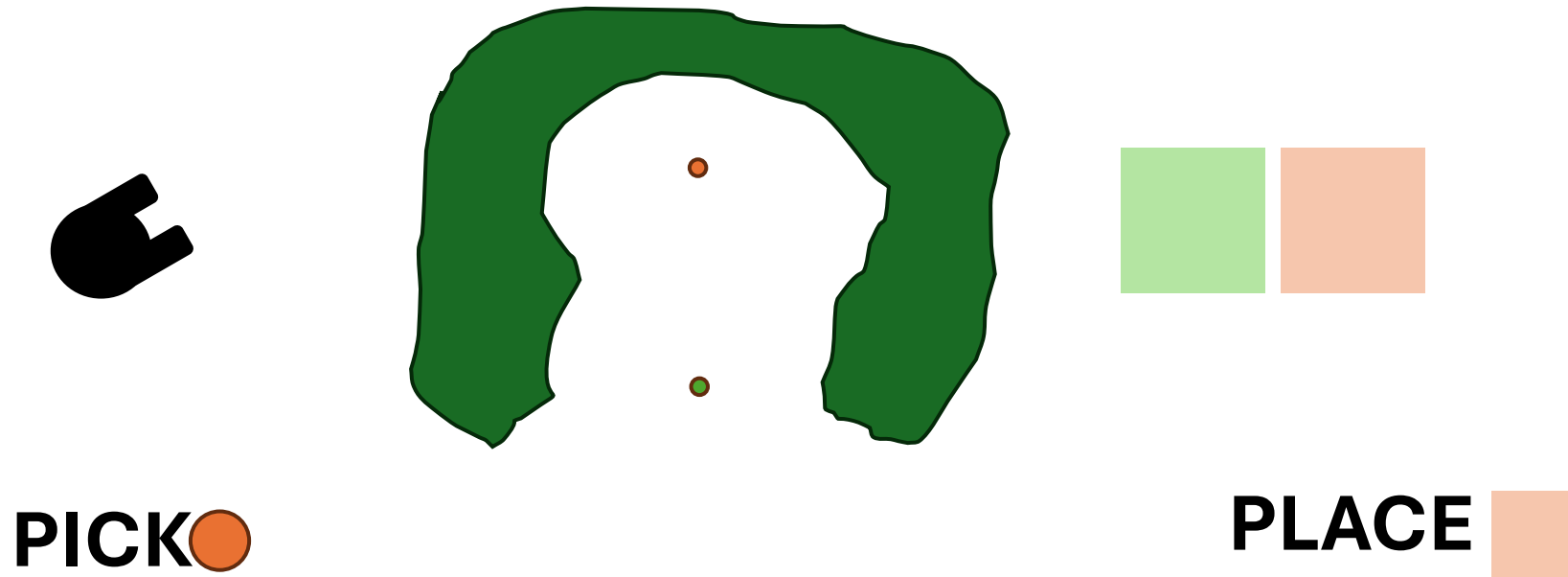


Task Planning THEN Motion Planning

🤖 Find task plan (task planner query)

🤖 Then find motion plans (sequence of motion planner queries)

Motion infeasibility in high-dimensional configuration space is not known beforehand

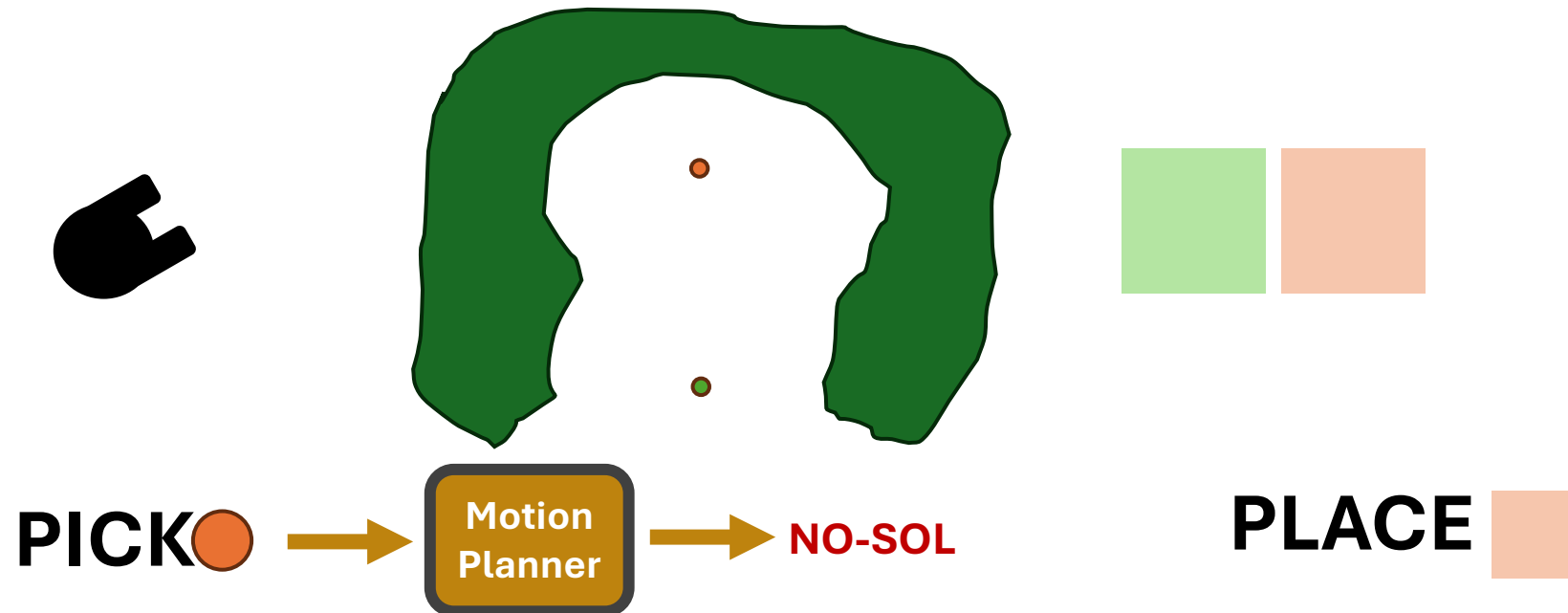


Task Planning **AND** Motion Planning

🔍 Find task plan (task planner query)

🔍 Find motion plans (sequence of motion planner queries)

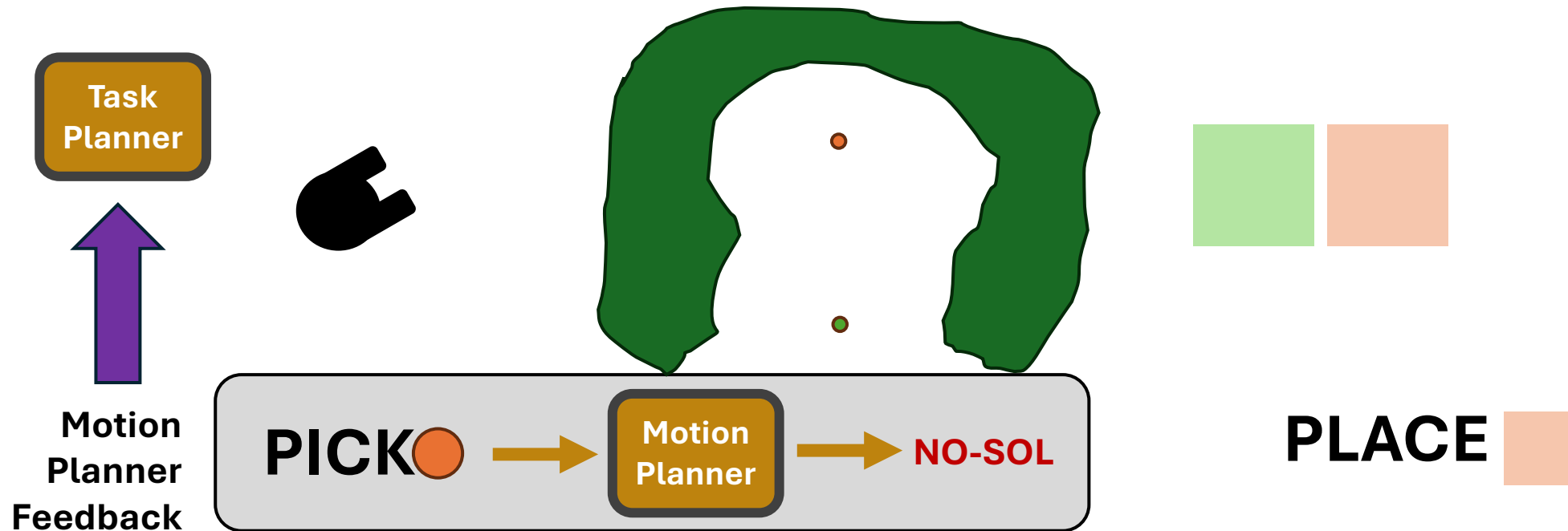
Motion infeasibility in high-dimensional configuration space is not known beforehand



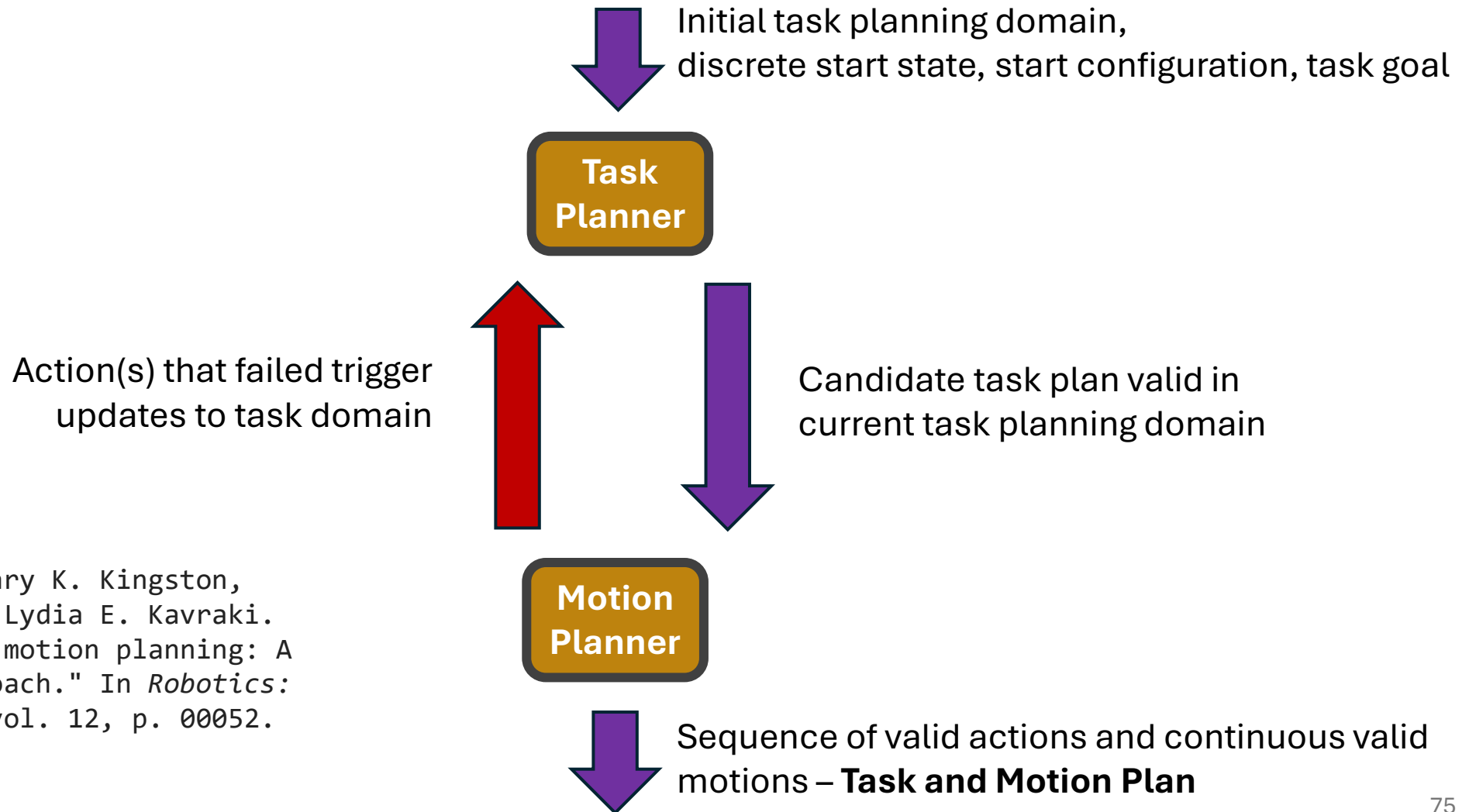
Task Planning **AND** Motion Planning

🔍 Find task plan (task planner query)

🔍 Find motion plans (sequence of motion planner queries)

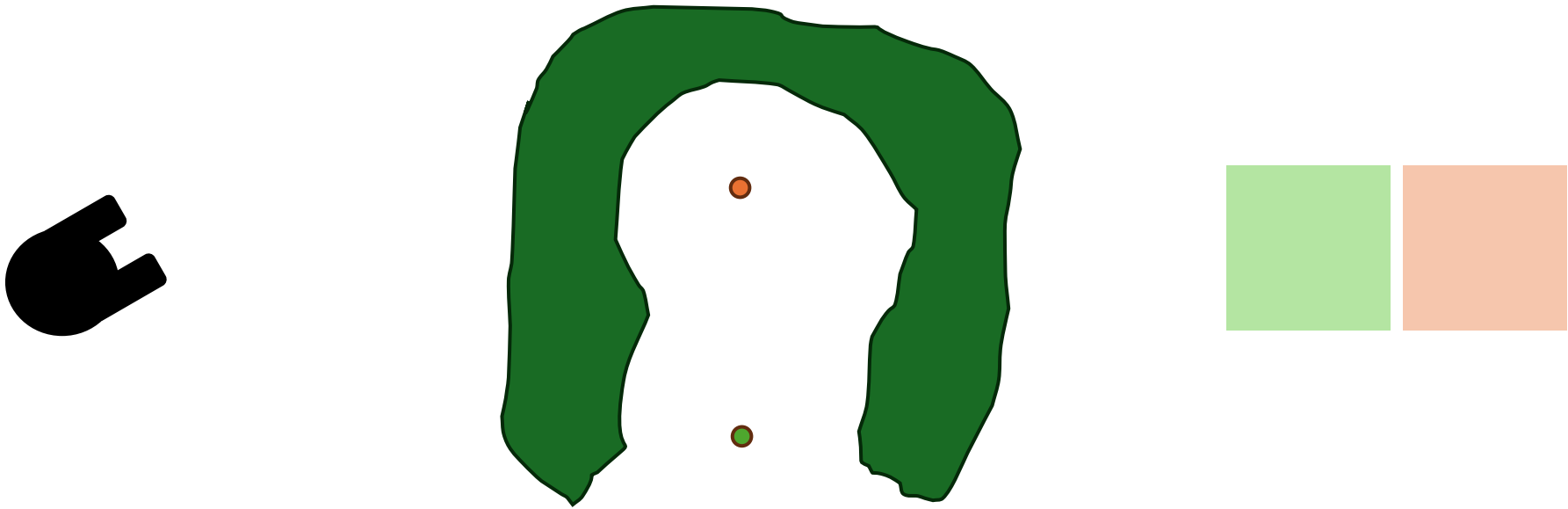


Task and Motion Planning



Dantam, Neil T., Zachary K. Kingston, Swarat Chaudhuri, and Lydia E. Kavraki. "Incremental task and motion planning: A constraint-based approach." In *Robotics: Science and systems*, vol. 12, p. 00052. 2016.

Task and Motion Planning



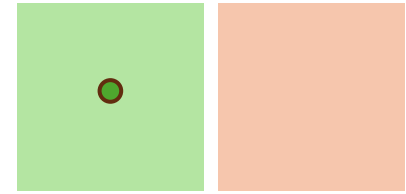
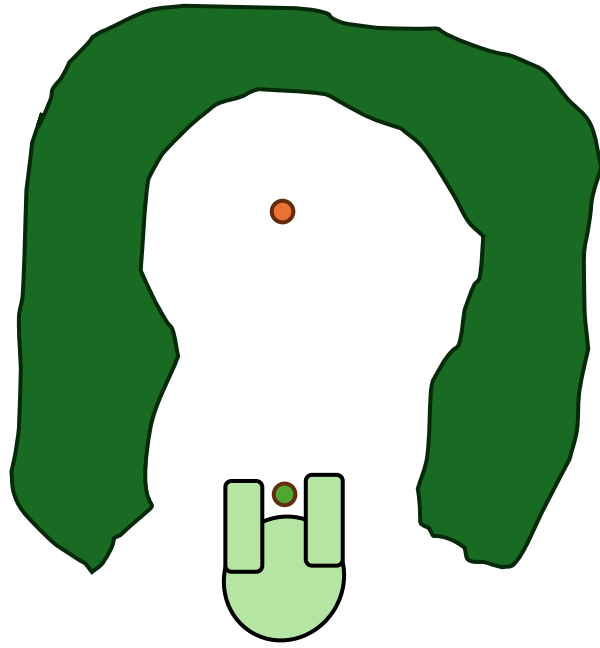
PICK ●

PLACE ■

PICK ●

PLACE ■

Task and Motion Planning



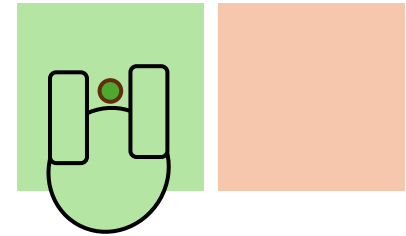
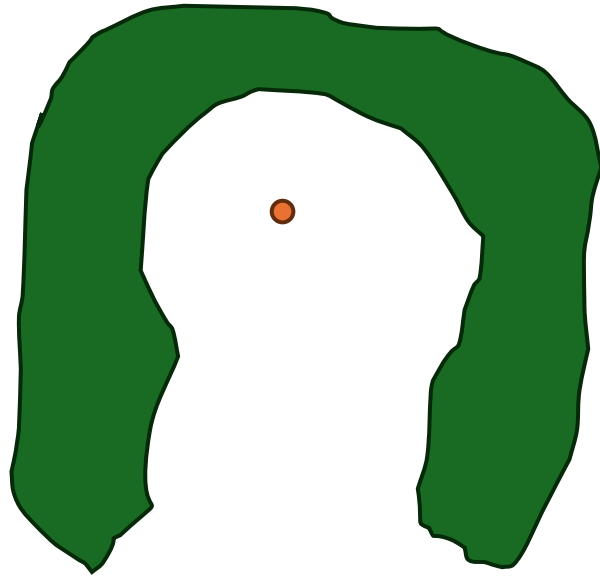
PICK ●

PLACE ■

PICK ●

PLACE ■

Task and Motion Planning



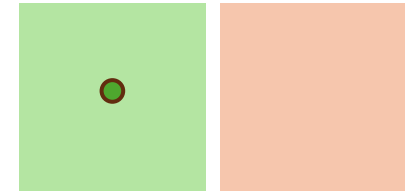
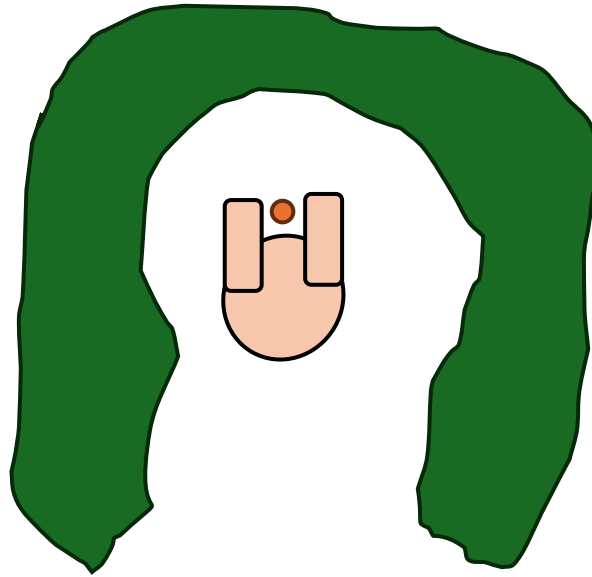
PICK●

PLACE■

PICK●

PLACE■

Task and Motion Planning



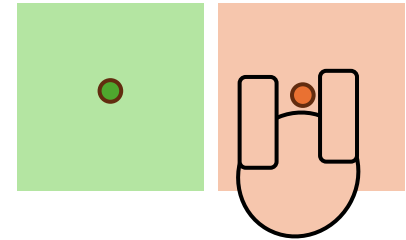
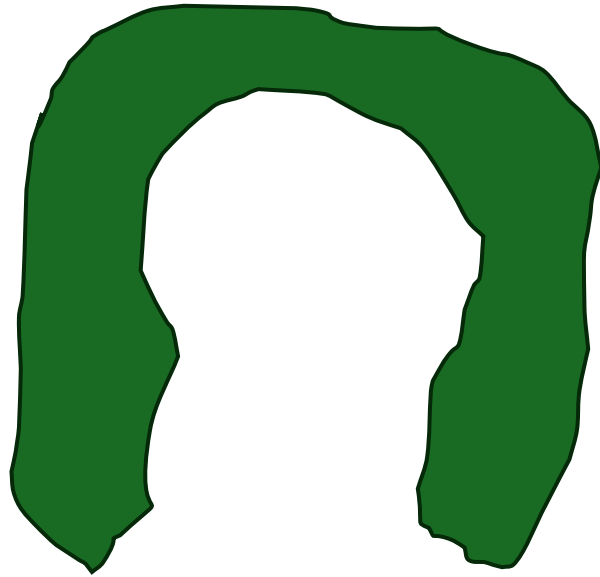
PICK ●

PLACE ■

PICK ●

PLACE ■

Task and Motion Planning



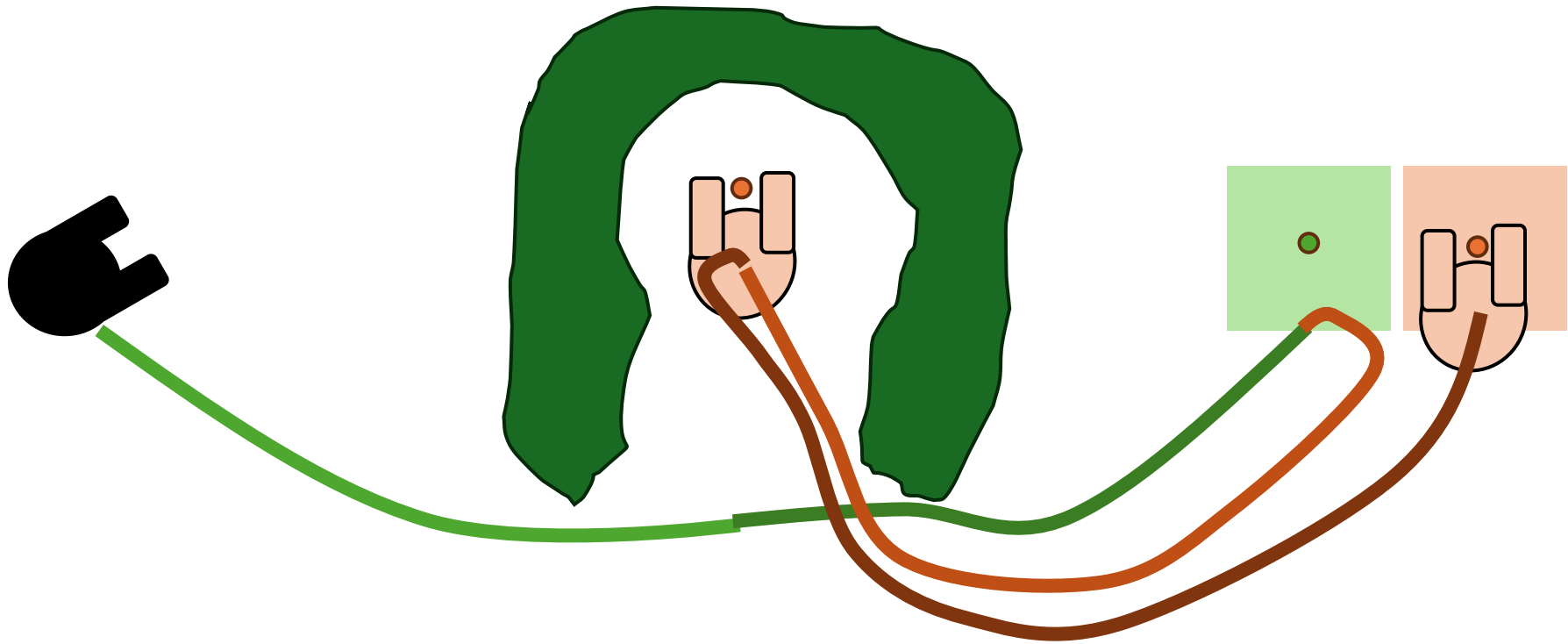
PICK ●

PLACE ■

PICK ●

PLACE ■

Task and Motion Plan



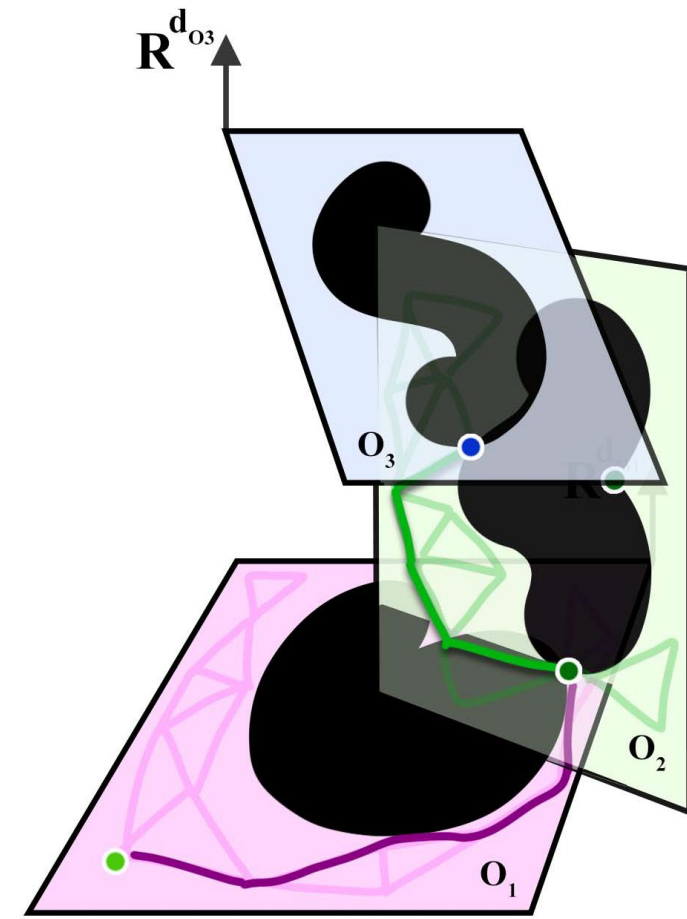
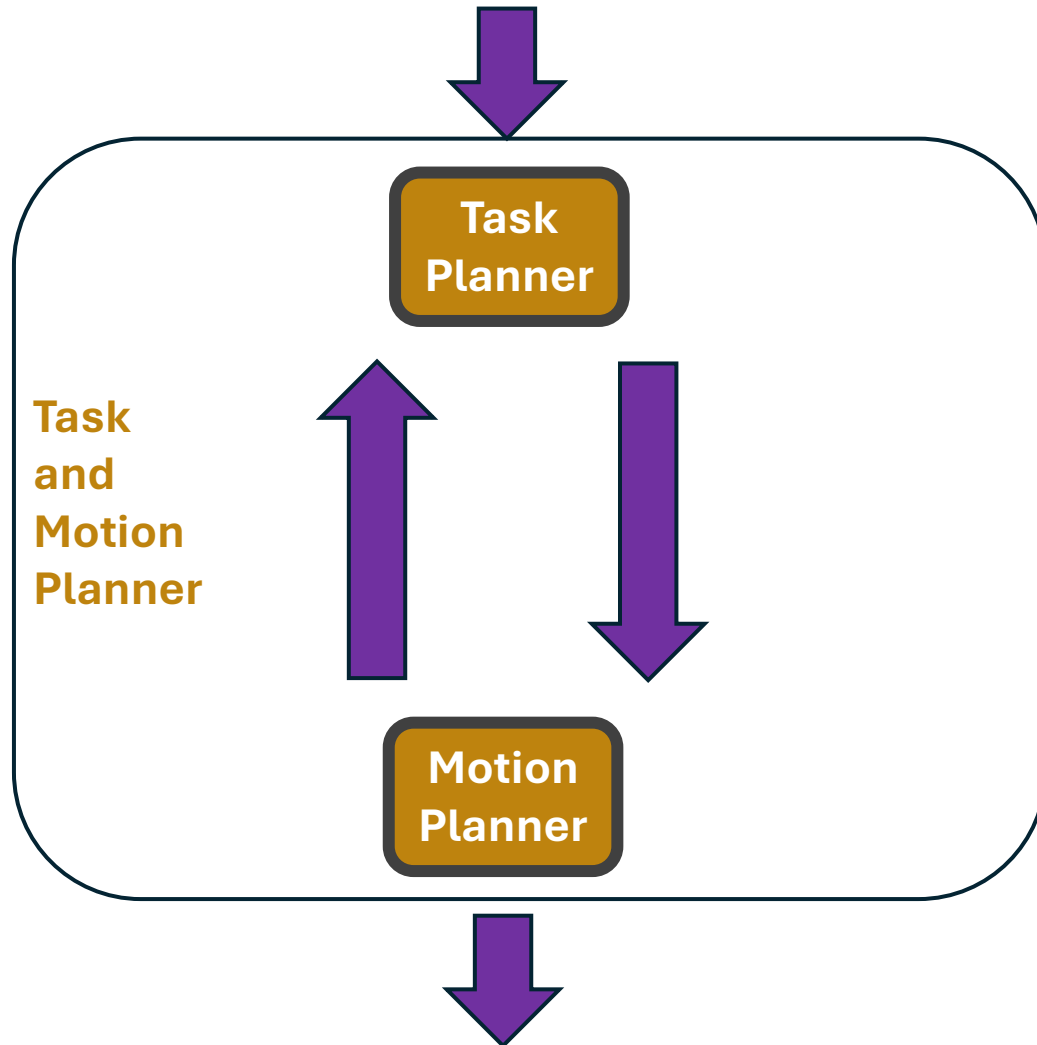
PICK ●

PLACE ■

PICK ●

PLACE ■

Task and Motion Planning





- 🔧 Resolve logical constraints
- 🔧 Resolve motion, geometric, physical constraints
- 🔧 Achieve completeness and optimality

Manipulation

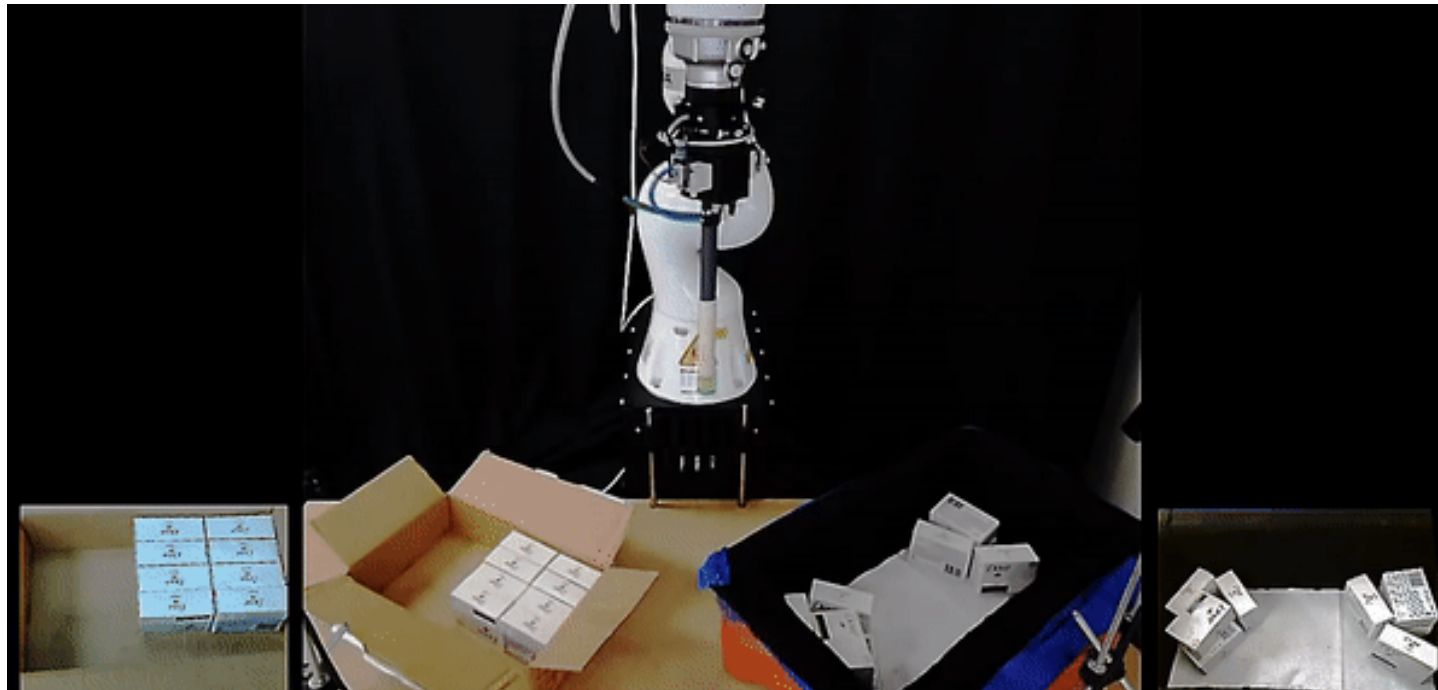
Manipulation

Manipulation is when an **agent moves things** other than itself through **selective contact**.

Mason, Matthew T. "Toward robotic manipulation." *Annual Review of Control, Robotics, and Autonomous Systems* 1, no. 1 (2018): 1-28.

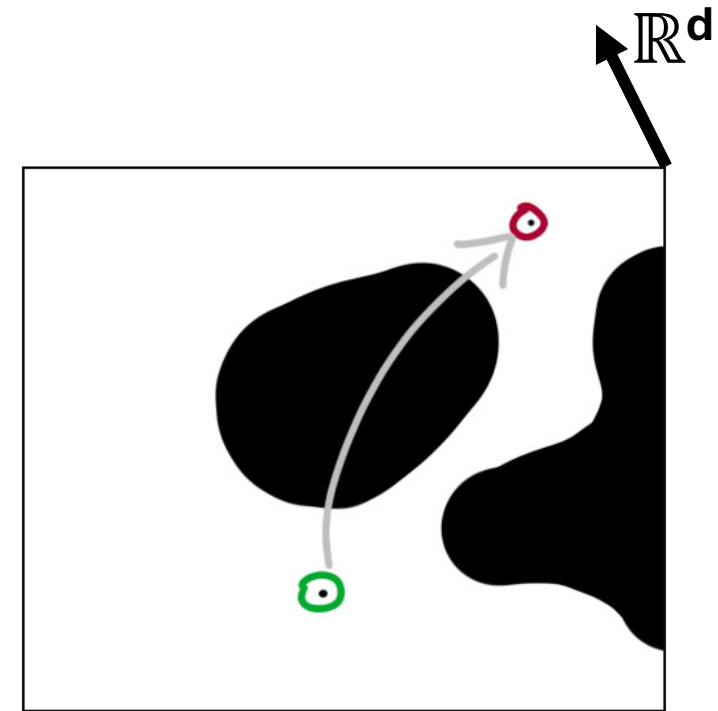
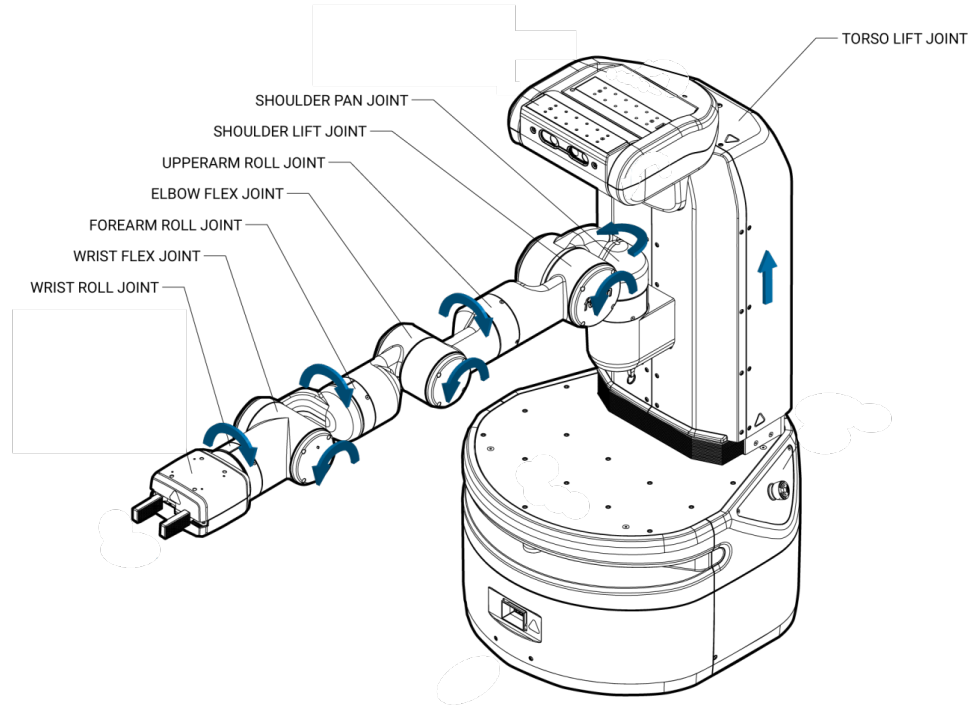
Manipulation

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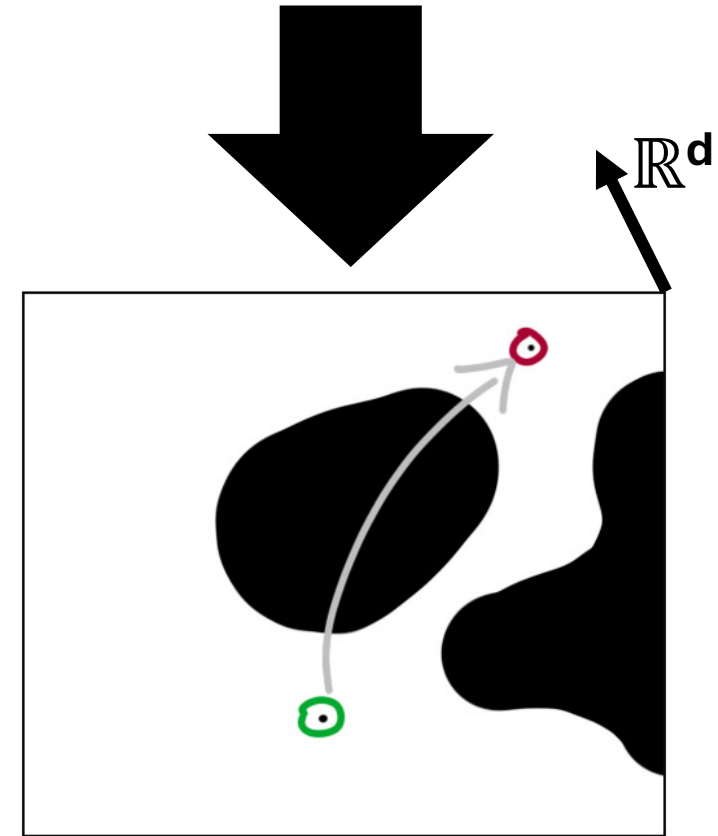
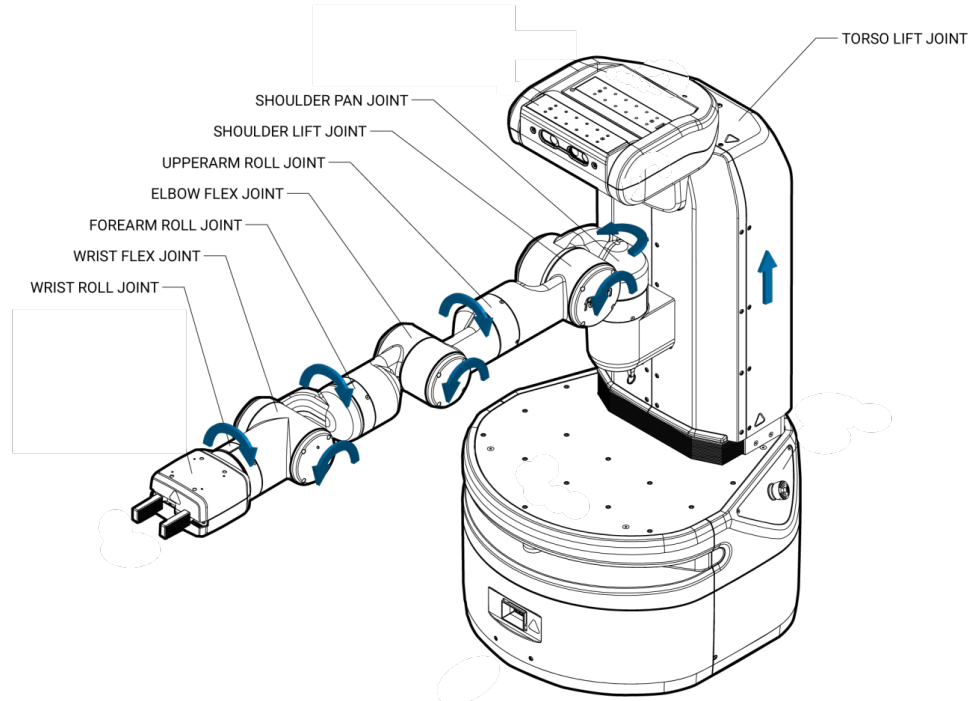


Shome, Rahul, Wei N. Tang, Changkyu Song, Chaitanya Mitash, Hristiyan Kourtev, Jingjin Yu, Abdeslam Boularias, and Kostas E. Bekris. "Towards robust product packing with a minimalistic end-effector." In *2019 International Conference on Robotics and Automation (ICRA)*, pp. 9007-9013. IEEE, 2019.

Spaces

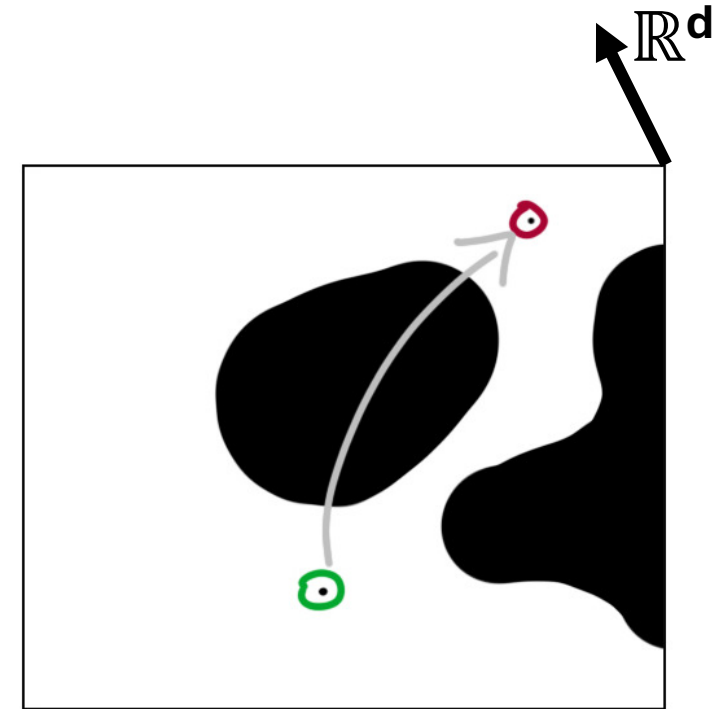
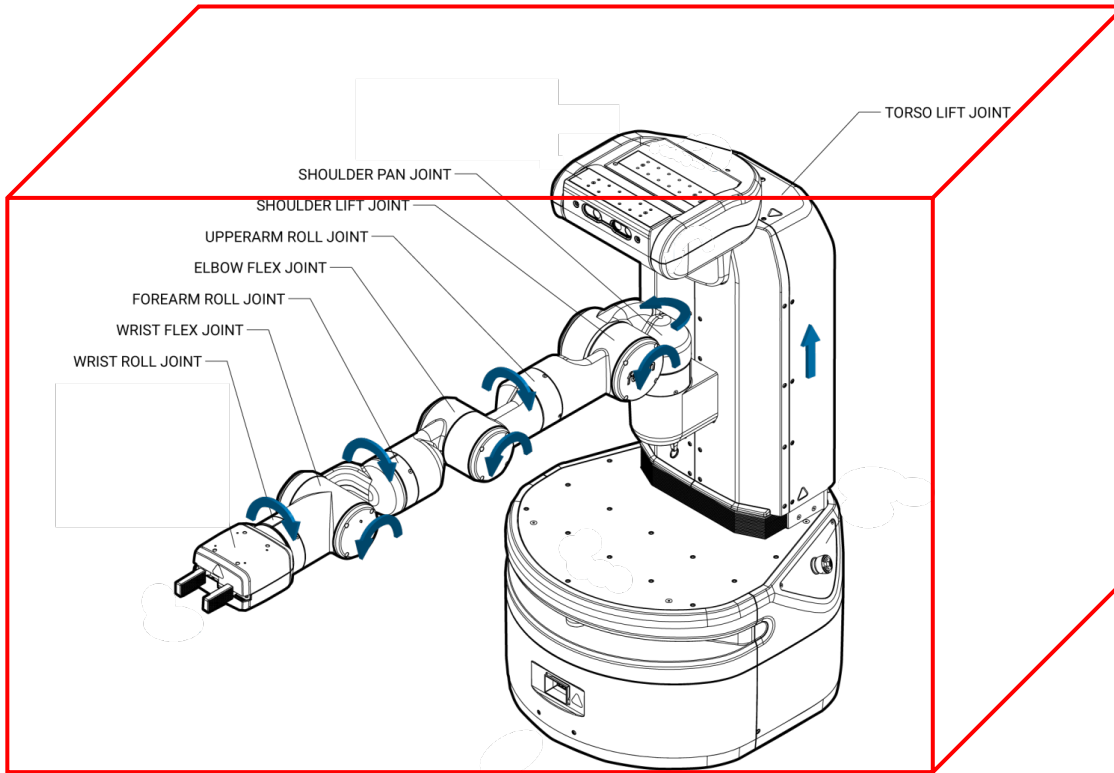


Configuration Space



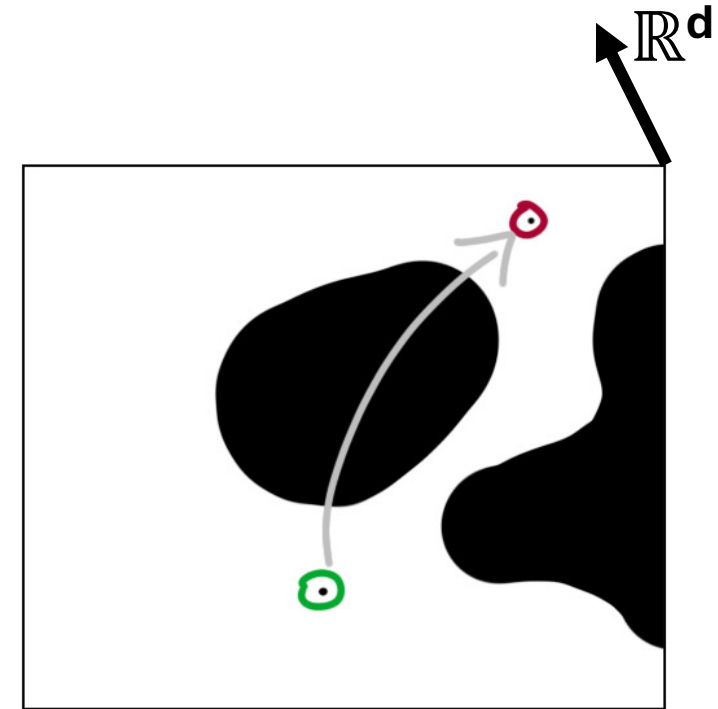
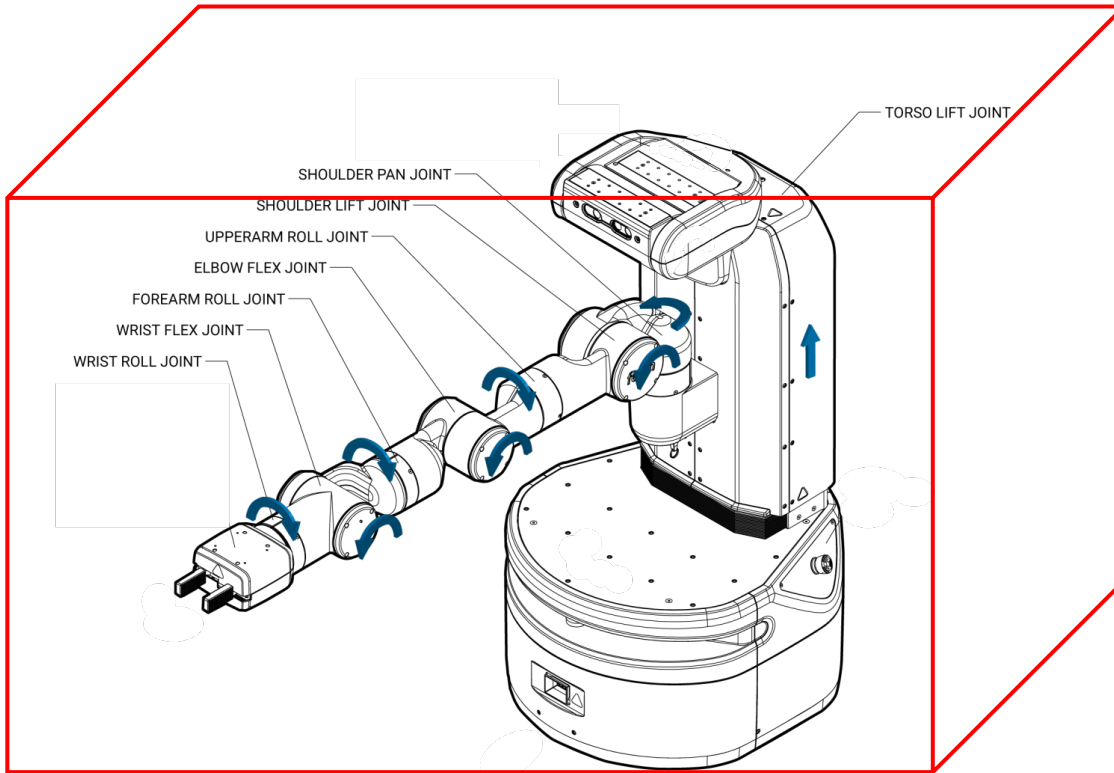
High-dimensional space where a point is a robot configuration.

Workspace



The space occupied by geometries of the robot and environment.
Here, it is 3D.

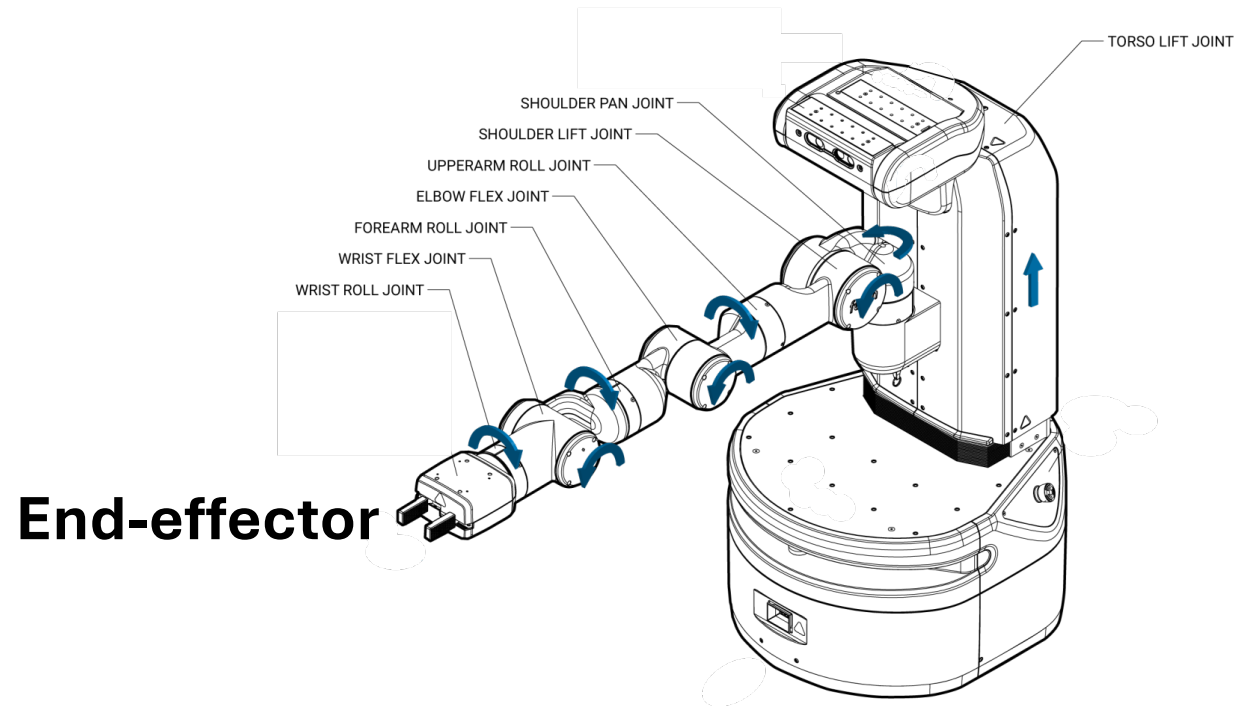
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Manipulation

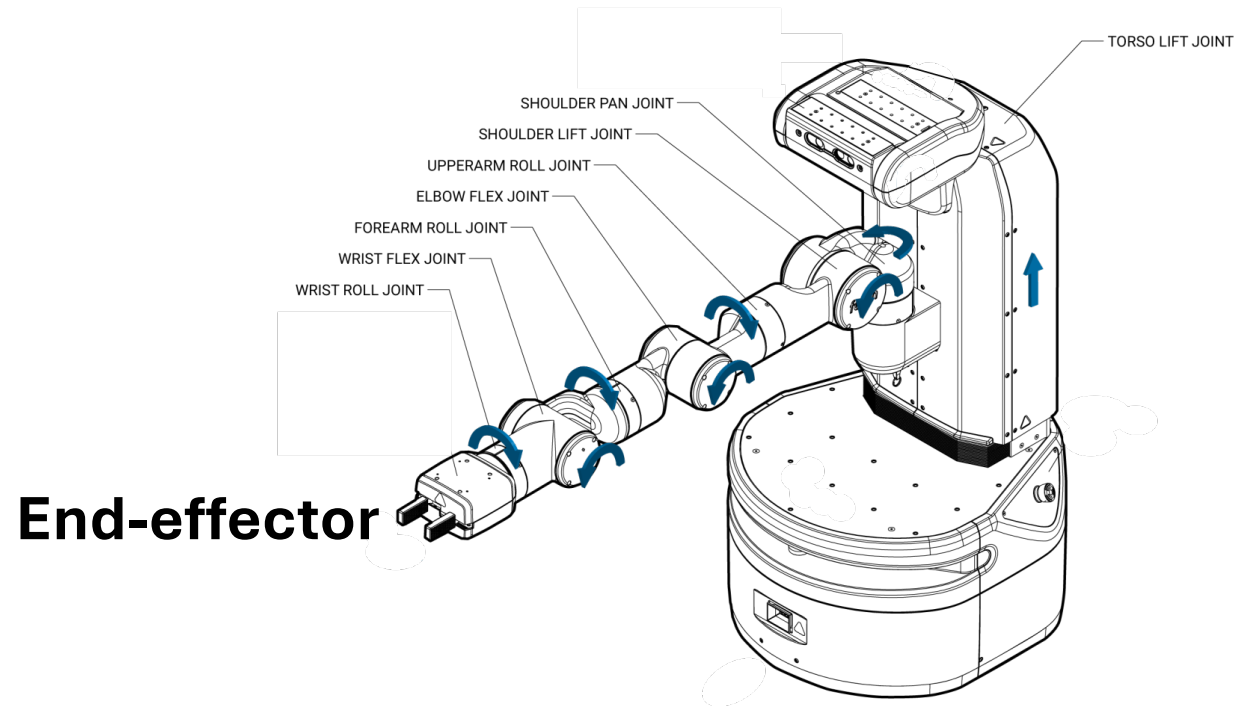
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Manipulation

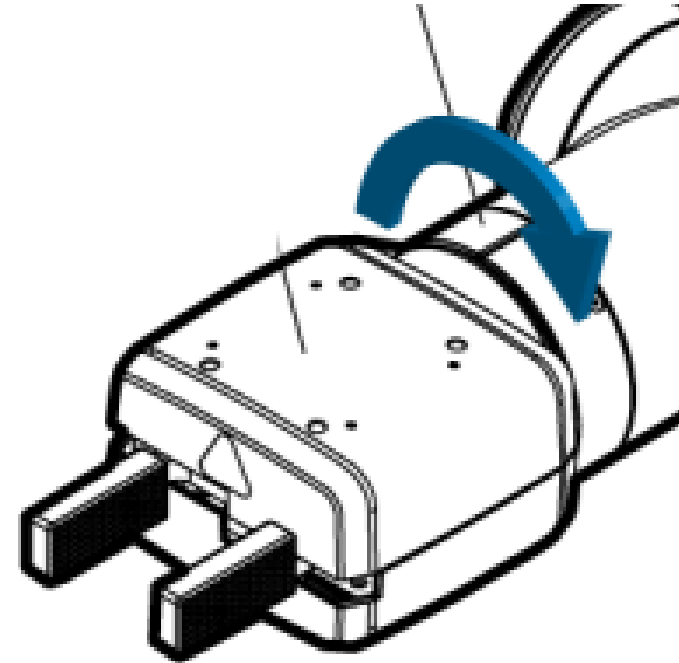
Manipulation is when an agent moves things other than itself through **selective contact**.



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Grasping

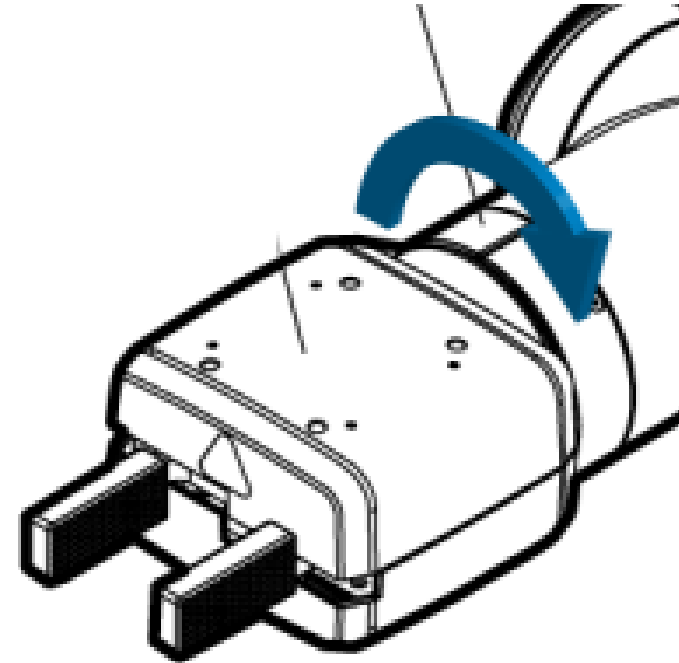
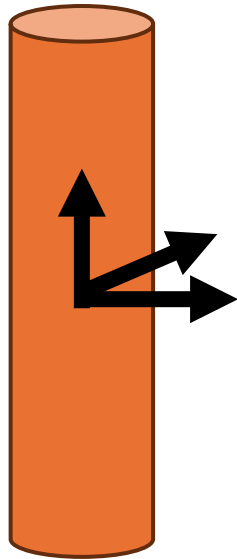
End-effector is the robot part making contact with an object.
The selective contact is a grasp.



Grasping

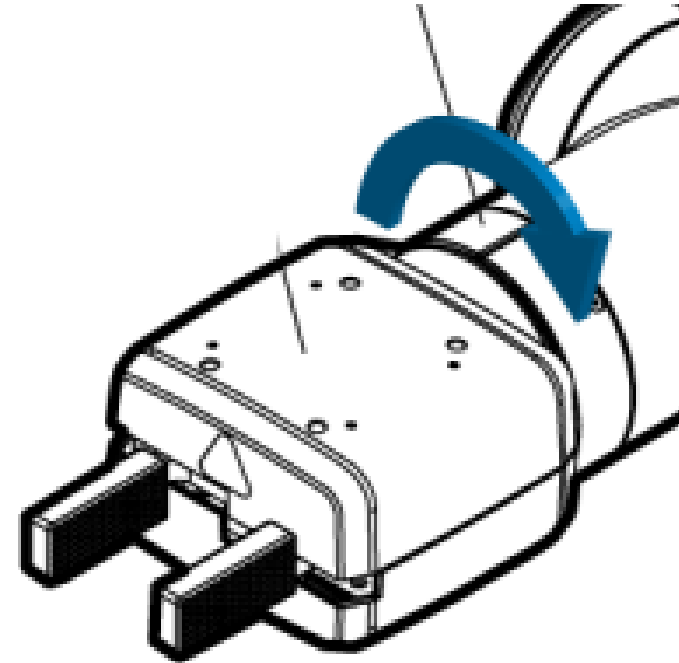
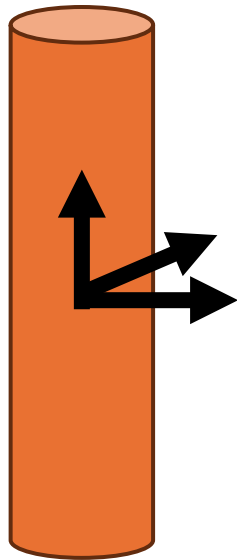
End-effector is the robot part making contact with a **rigid object**.
The selective contact is a grasp.

Pose of
object



Grasping

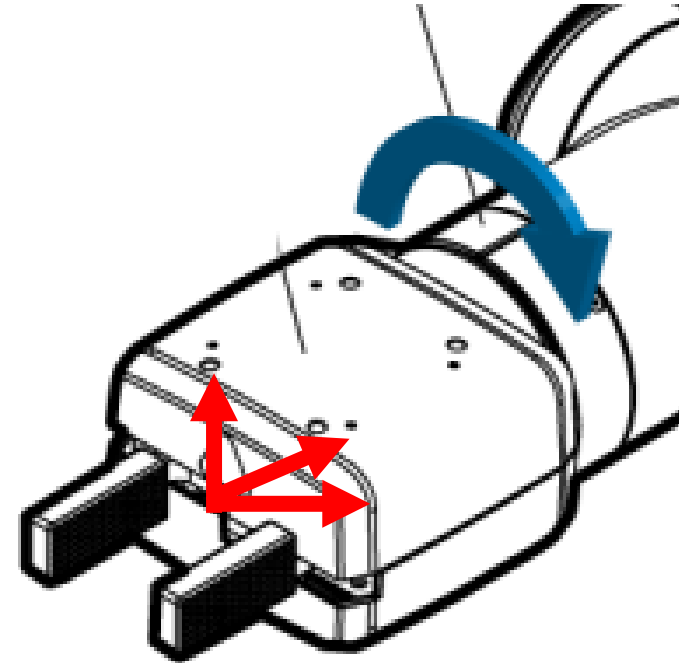
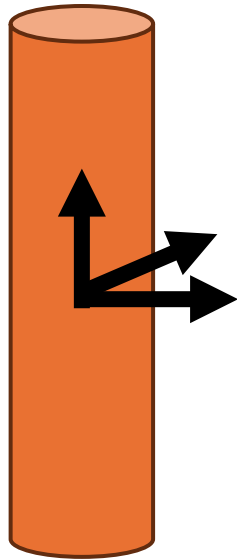
End-effector is the robot part making contact with a rigid object.
The selective contact is a grasp.



Where is the end-effector?

Grasping

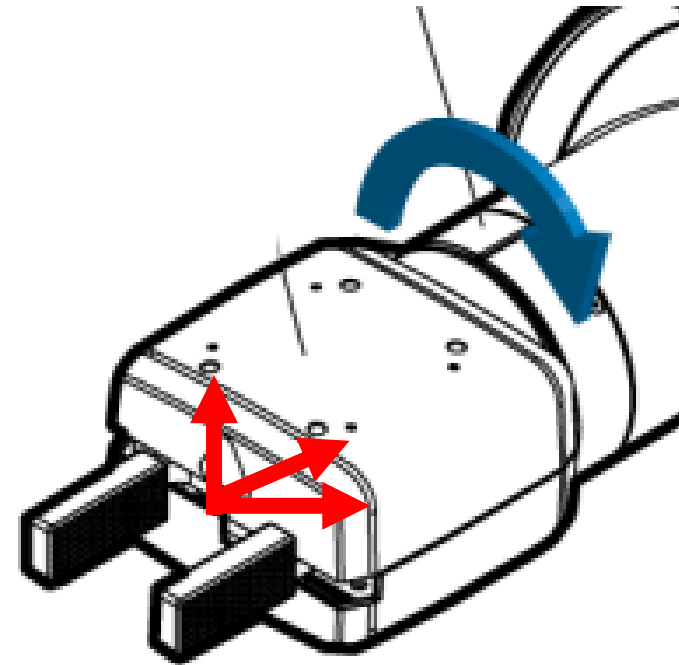
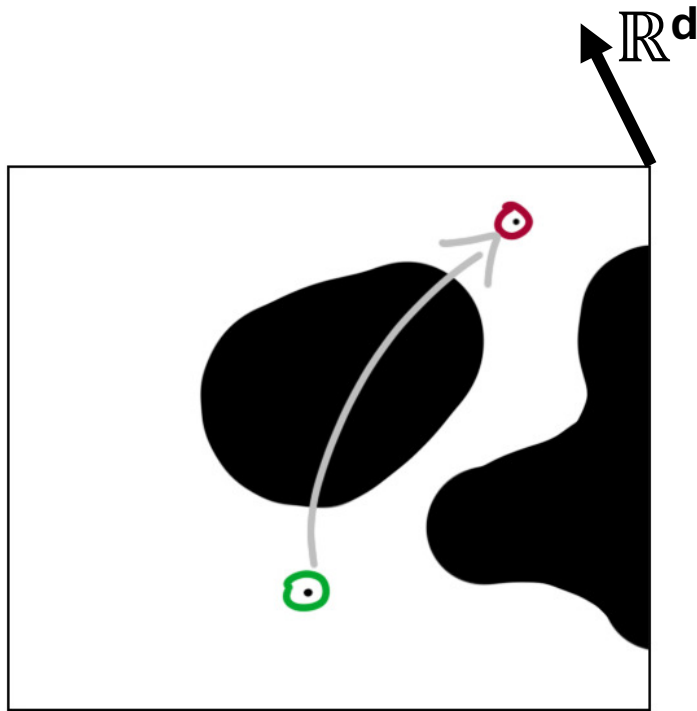
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Where is the end-effector?

Grasping

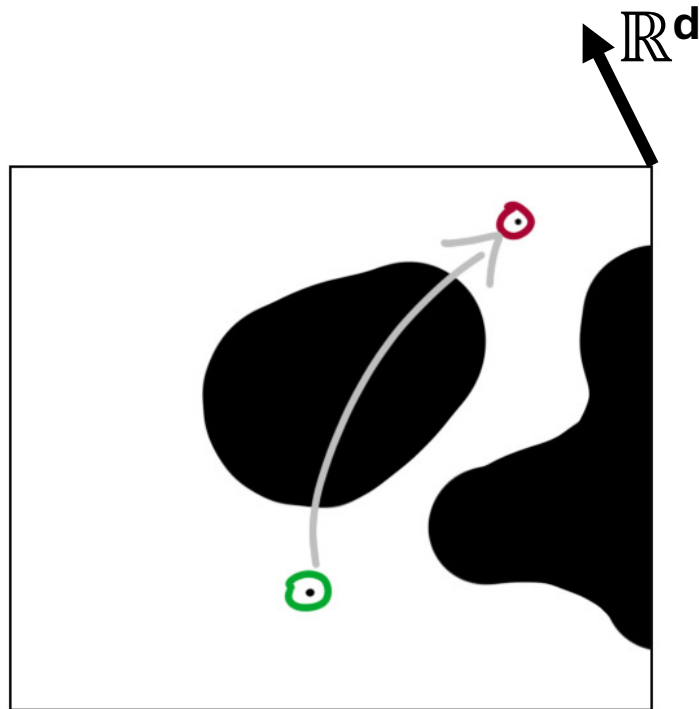
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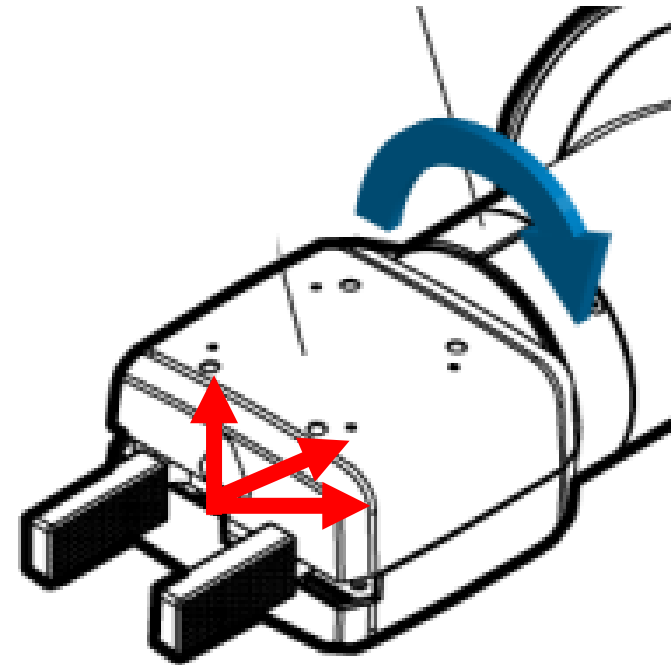
Where is the end-effector?

Grasping

End-effector is the robot part making contact with a rigid object.
The selective contact is a grasp.

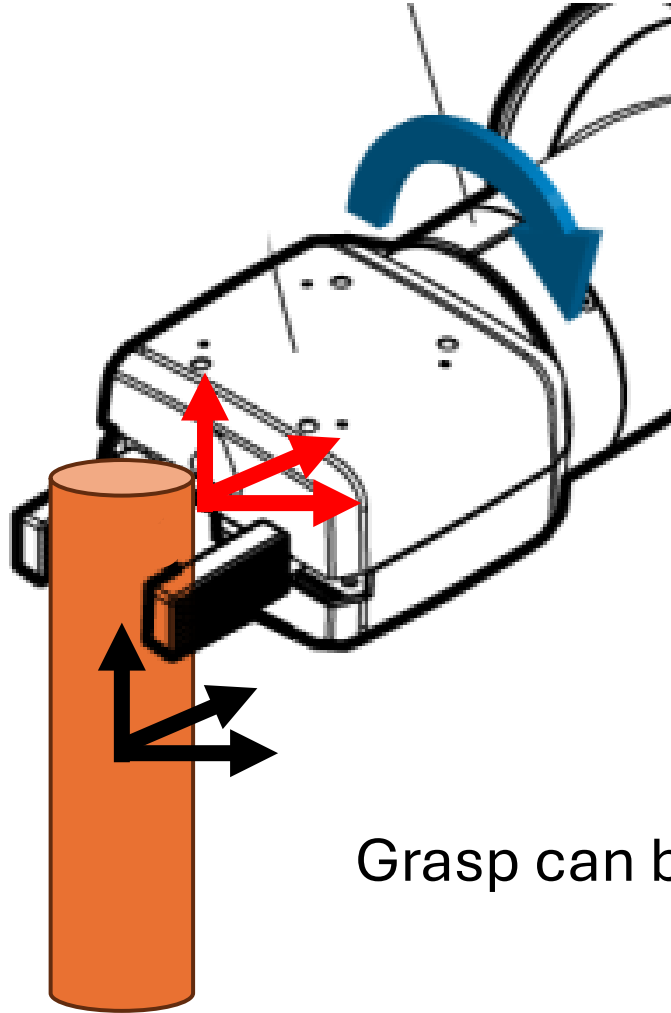


FK

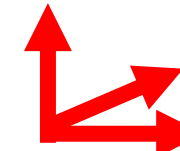


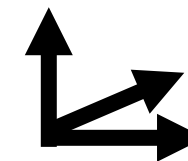
End-effector pose from FK.

Grasp



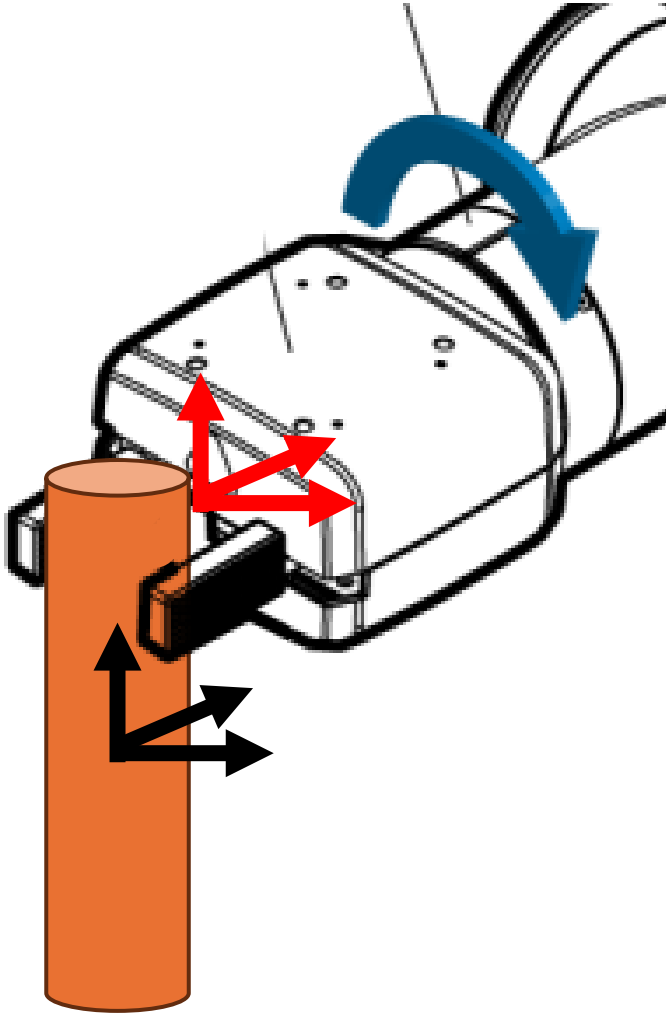
SE(3) Transformation or Relative Pose

 End-effector
pose at selective
contact

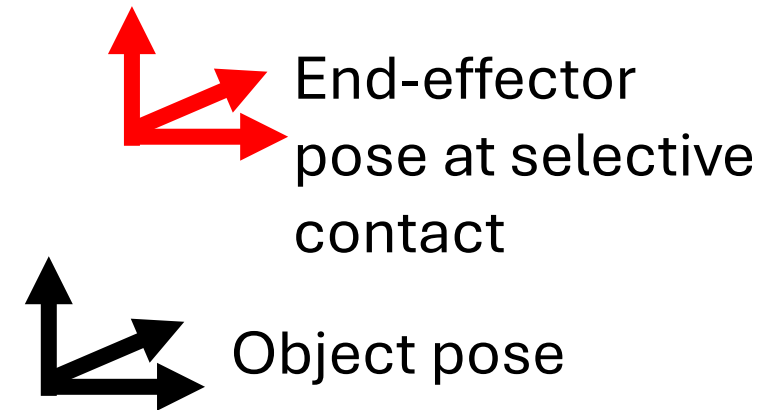
 Object pose

Grasp can be defined with the relative transformation
from the object to the end-effector.

Grasp

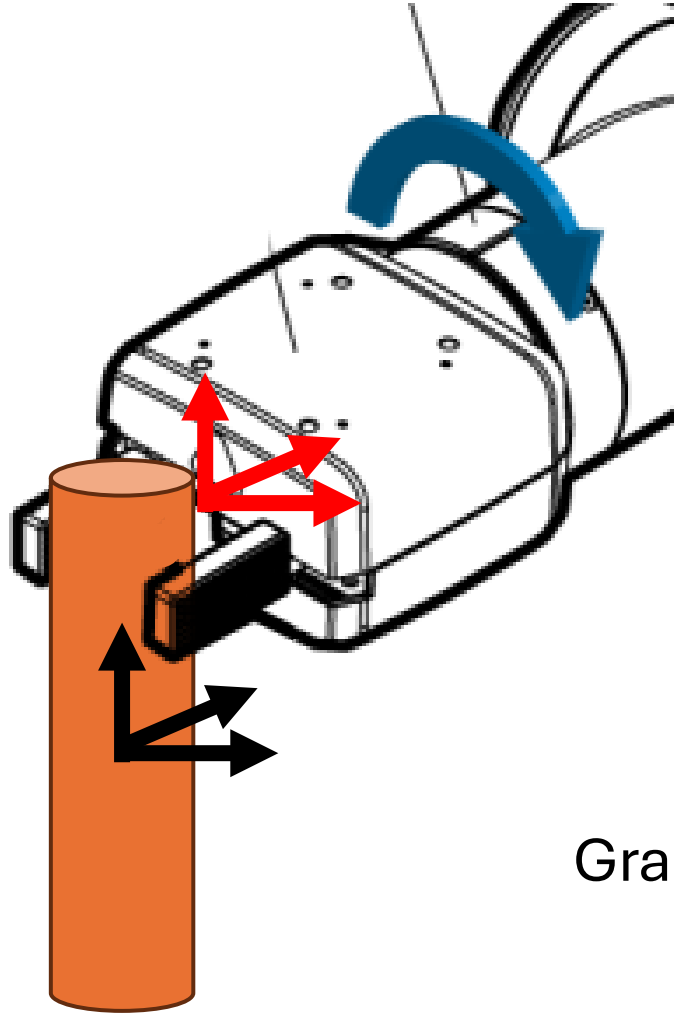


SE(3) Transformation or Relative Pose

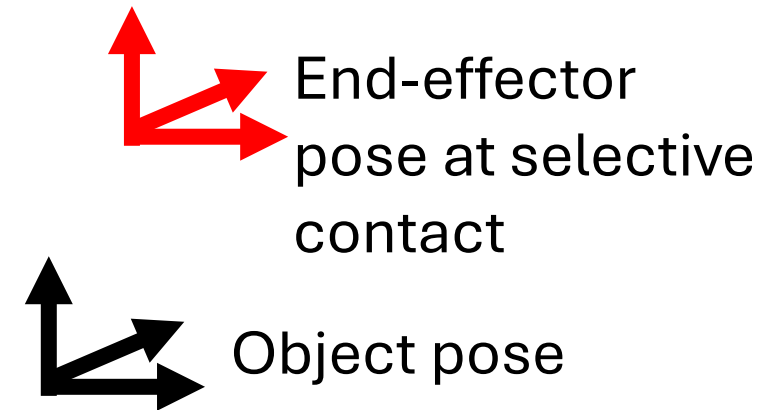


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Grasp

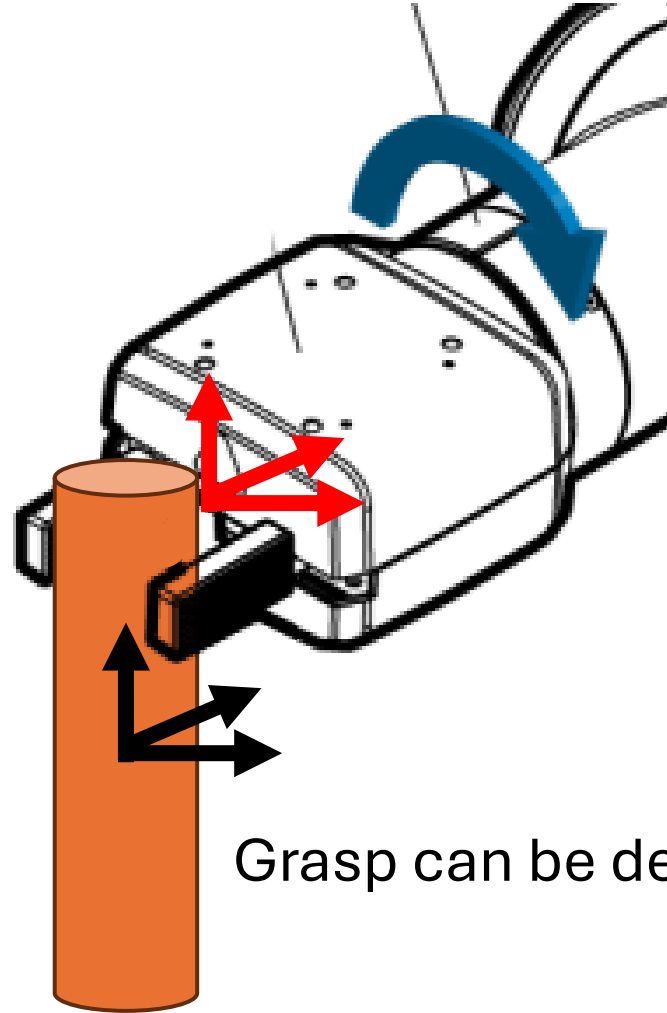


SE(3) Transformation or Relative Pose

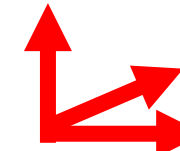


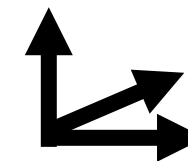
Grasp can be defined with the relative transformation from the object to the end-effector.

Grasp



SE(3) Transformation or Relative Pose

 End-effector
pose at selective
contact

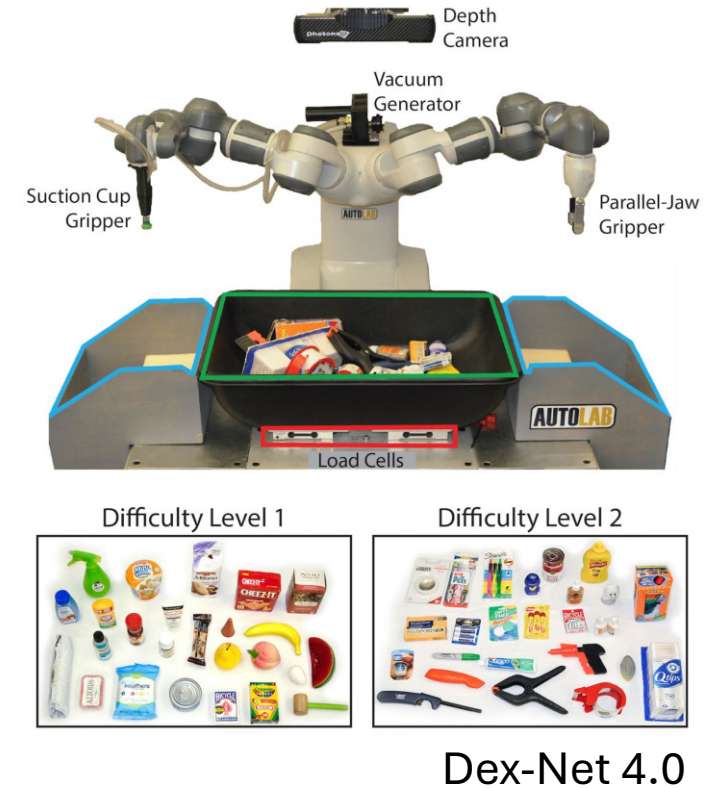
 Object pose

Grasp can be defined with the relative transformation
from the object to the end-effector.

Grasp Generation

Model-based methods use known geometric information about ee and objects (graspit!).

Model-free methods can operate directly on sensing information (dex-net, kpm)



Miller, Andrew T., and Peter K. Allen. "Graspit! a versatile simulator for robotic grasping." *IEEE Robotics & Automation Magazine* 11, no. 4 (2004): 110-122.

Mahler, Jeffrey, Matthew Matl, Xinyu Liu, Albert Li, David Gealy, and Ken Goldberg. "Dex-net 3.0: Computing robust vacuum suction grasp targets in point clouds using a new analytic model and deep learning." In *2018 IEEE International Conference on robotics and automation (ICRA)*, pp. 5620-5627. IEEE, 2018.

Manuelli, Lucas, Wei Gao, Peter Florence, and Russ Tedrake. "kpm: Keypoint affordances for category-level robotic manipulation." In *The International Symposium of Robotics Research*, pp. 132-157. Cham: Springer International Publishing, 2019.

Mahler, Jeffrey, Matthew Matl, Vishal Satish, Michael Danielczuk, Bill DeRose, Stephen McKinley, and Ken Goldberg. "Learning ambidextrous robot grasping policies." *Science Robotics* 4, no. 26 (2019): eaau4984.

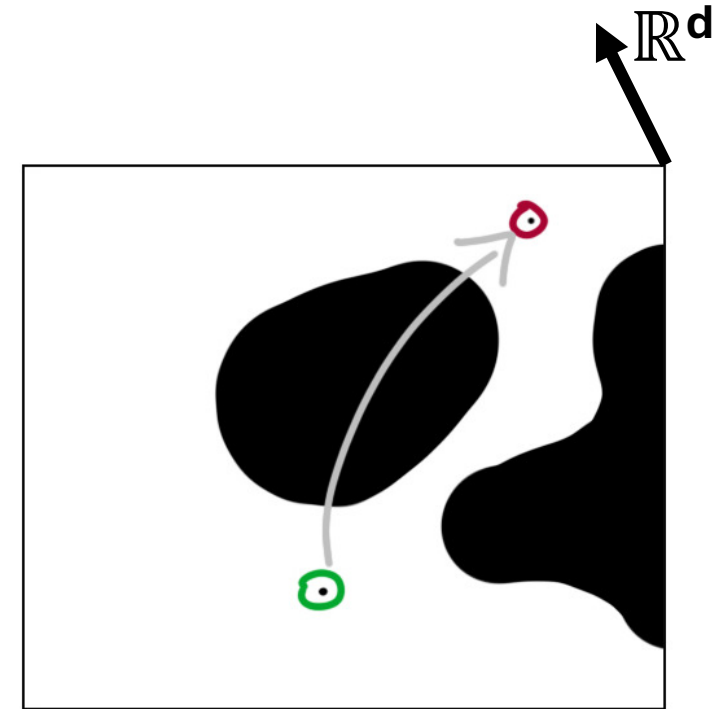
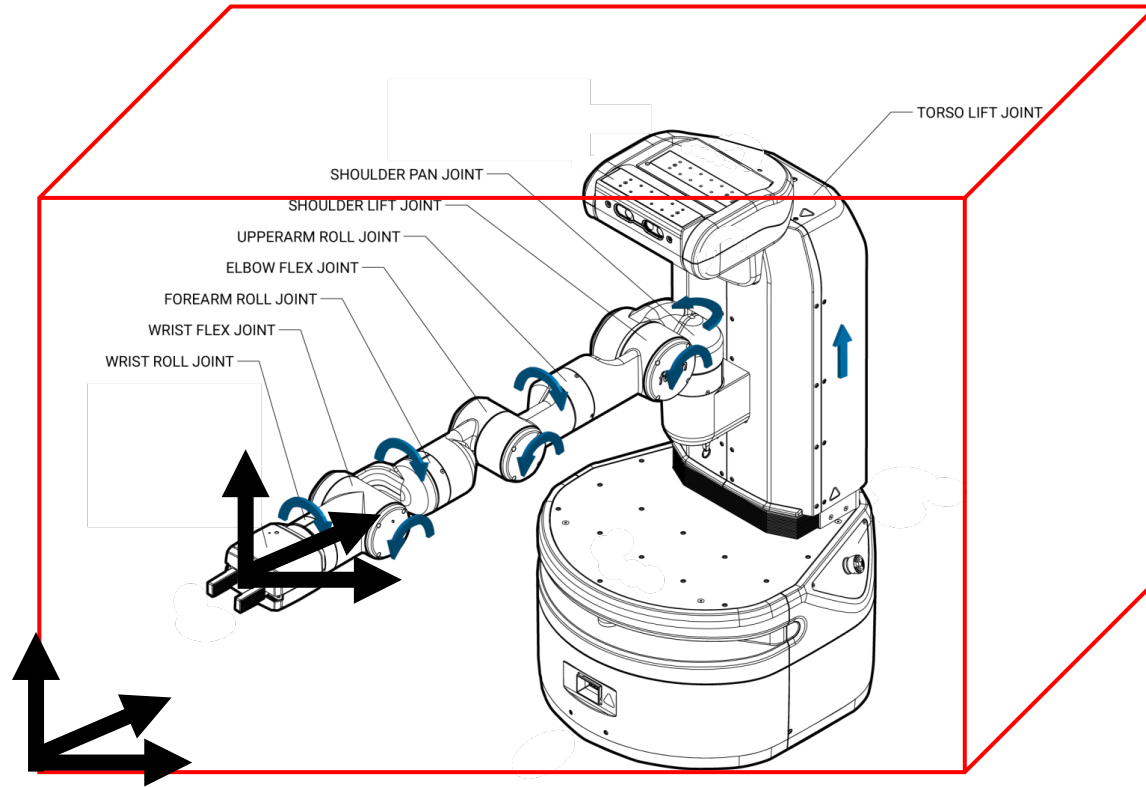
Grasp Generation



Park, Hyeonjun, and Donghan Kim. "An open-source anthropomorphic robot hand system: HRI hand." *HardwareX* 7 (2020): e00100.

<https://roboticsandautomationnews.com/2021/01/22/ati-launches-end-effector-kits-for-fanuc-collaborative-robots/39852/>

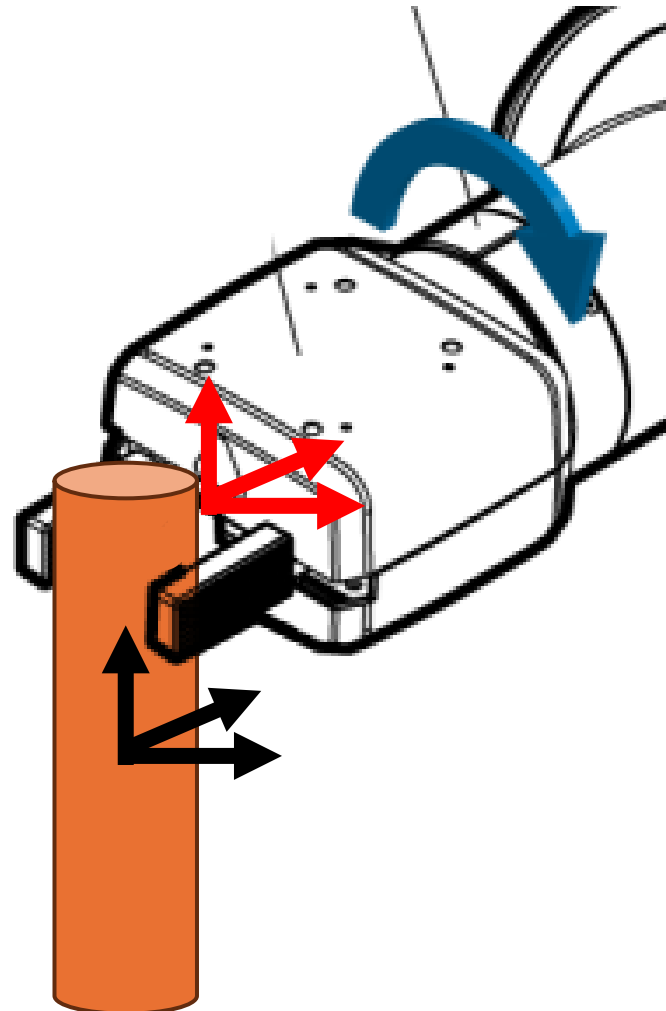
Task Space



The space where the task will be defined wrt an end-effector.
Here, $SE(3)$. Closely related to operation space.

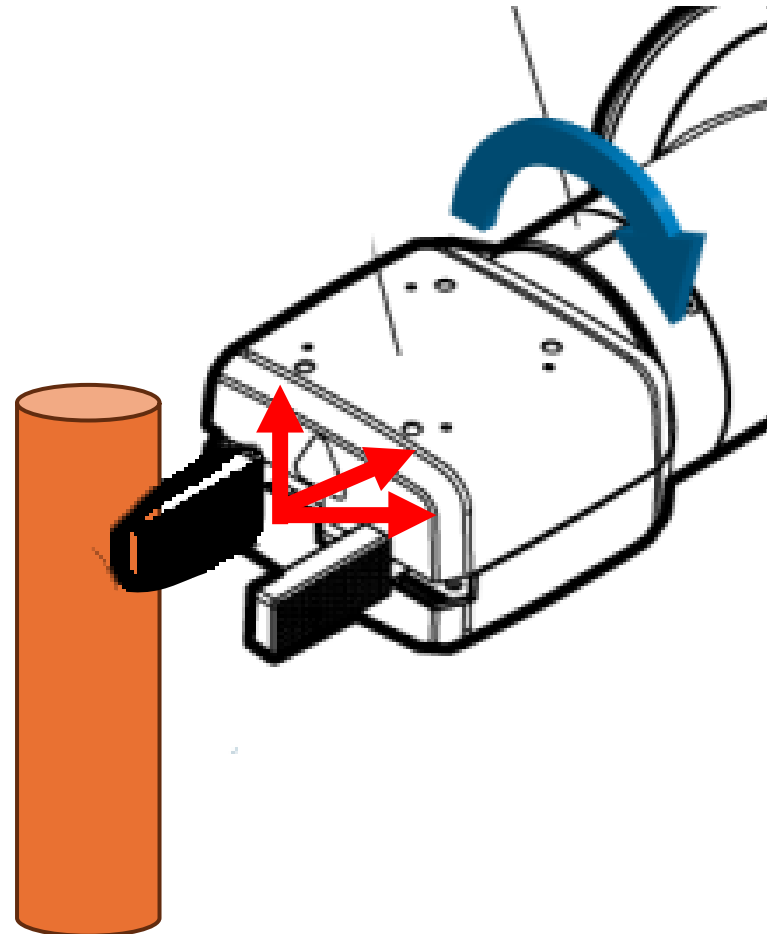
Prehensile Manipulation

The relative grasping pose does not change while the grasp is engaged.



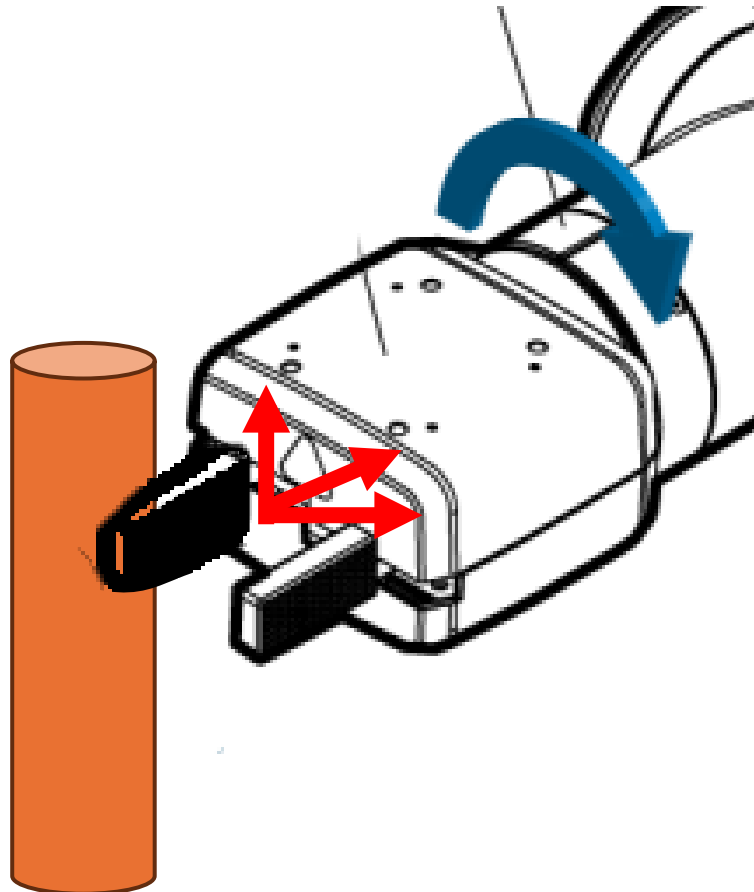
Non-prehensile Manipulation

The relative grasping pose can change while the contact is engaged.



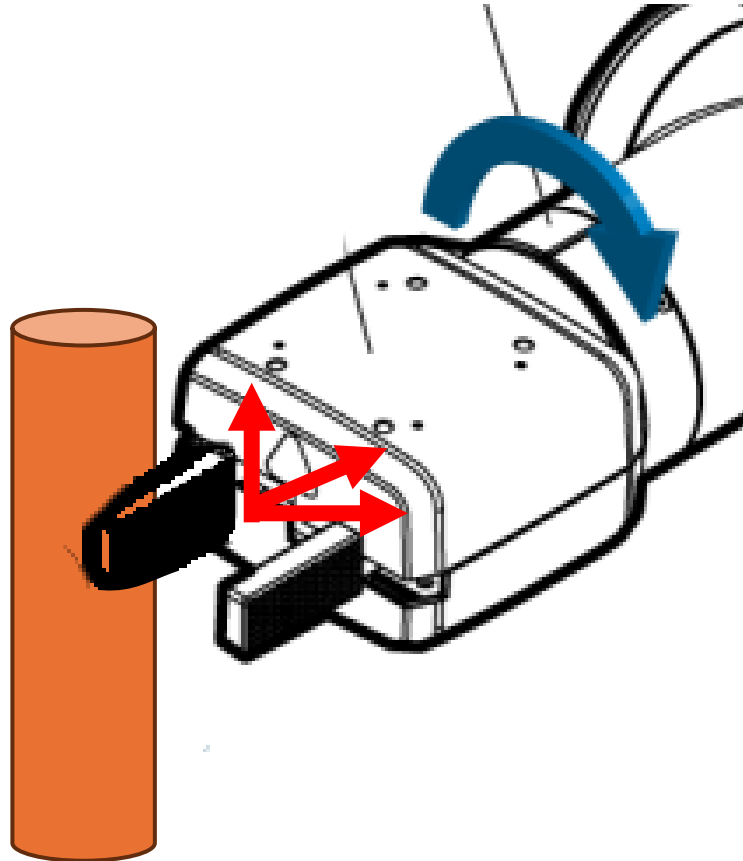
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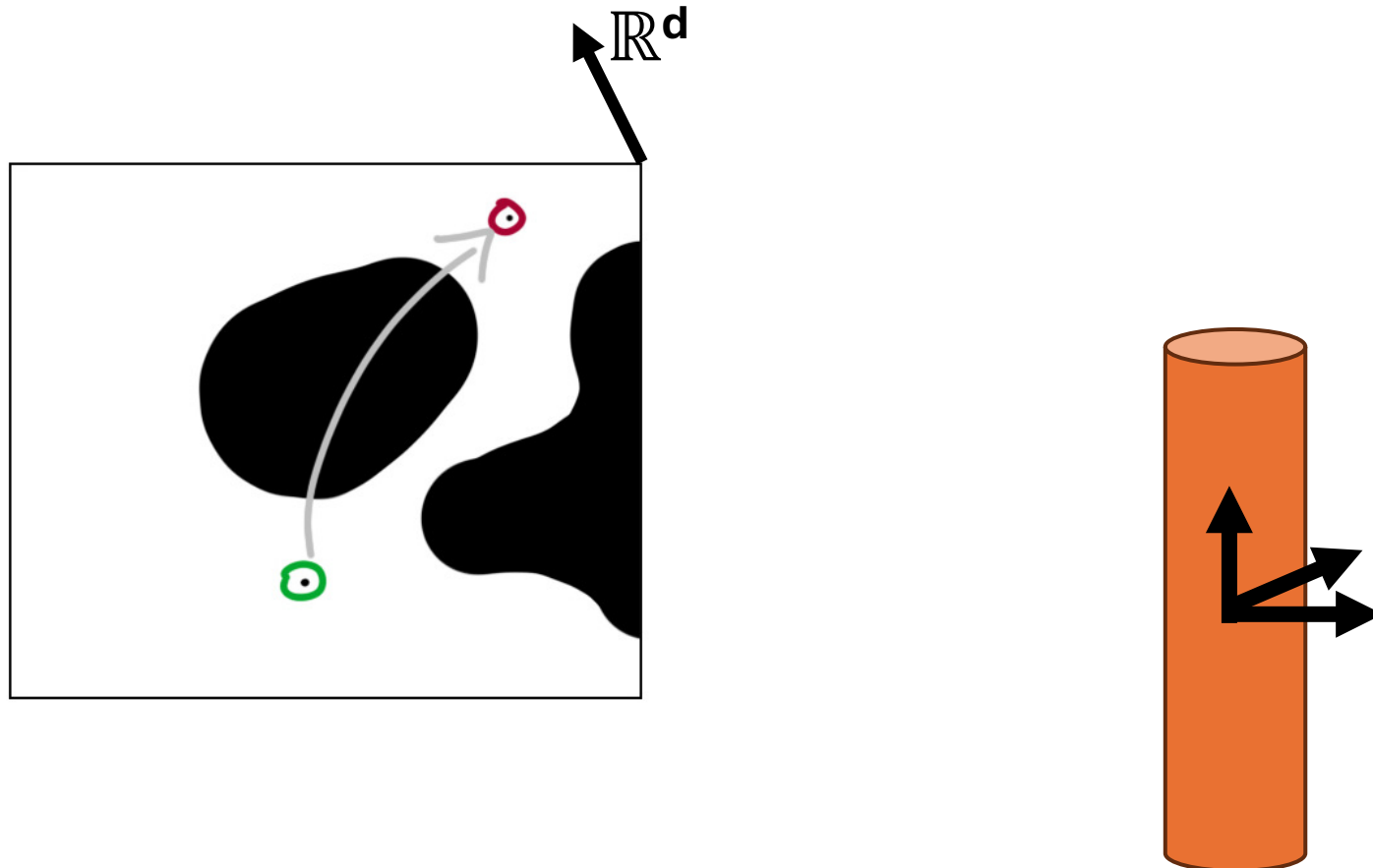
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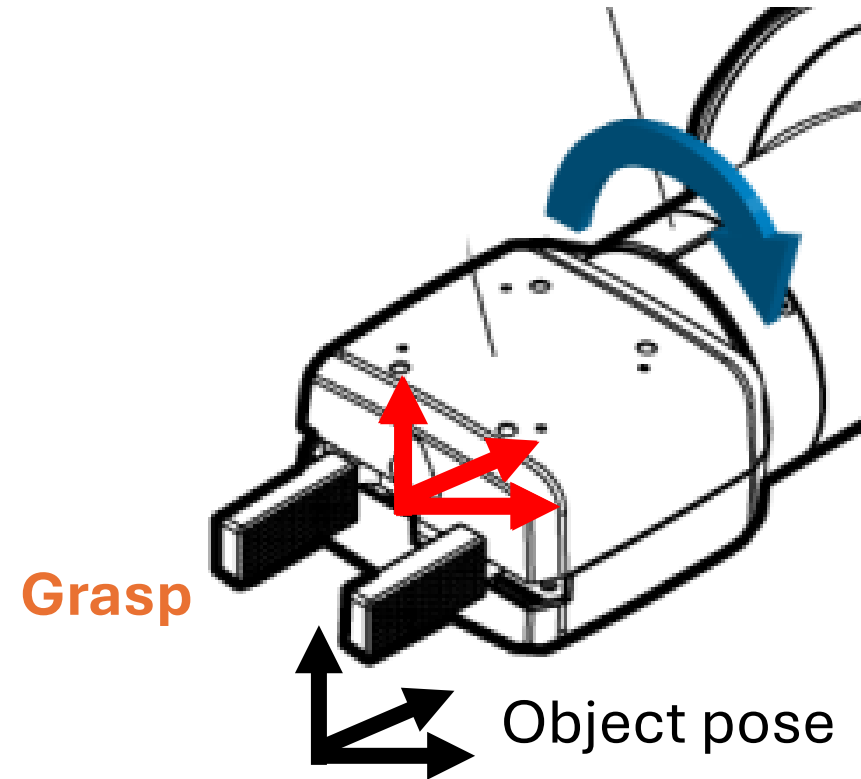
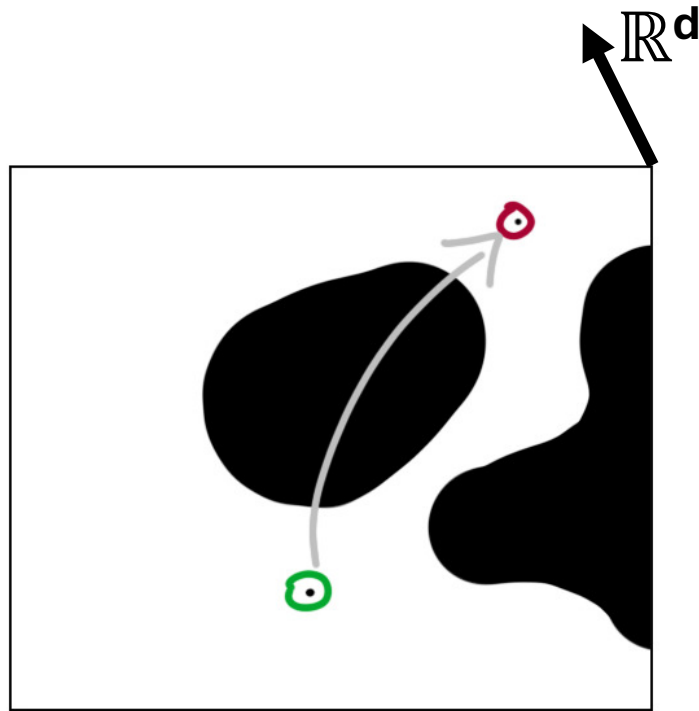
Grasping Configurations From Object Poses

For manipulation planning we need a configuration to plan to.



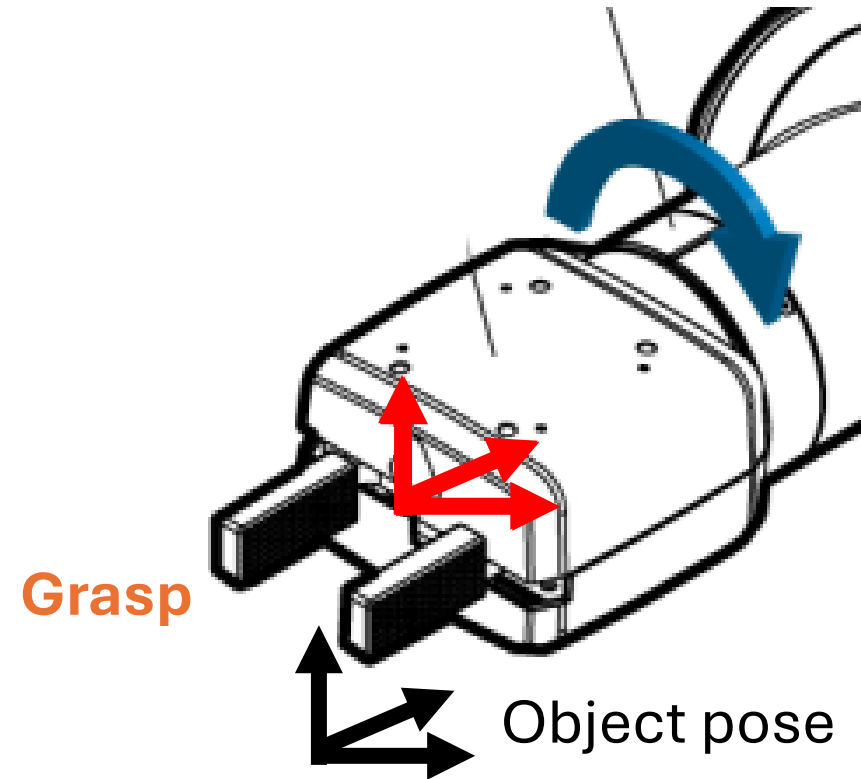
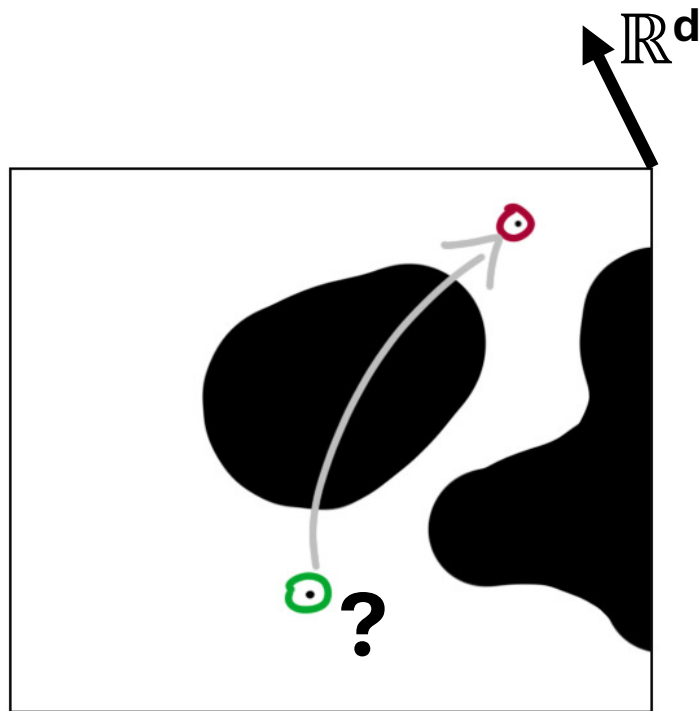
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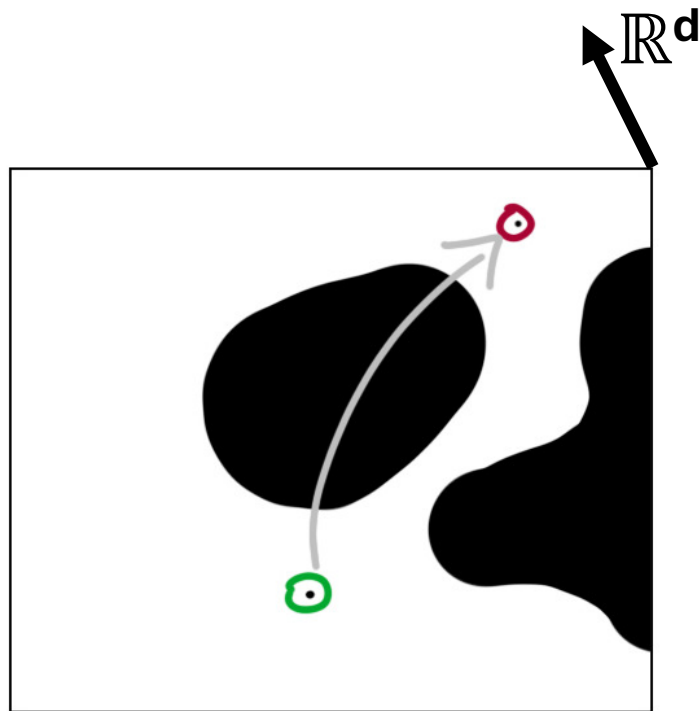
Grasping Configurations From Object Poses

For manipulation planning we need a configuration to plan to.

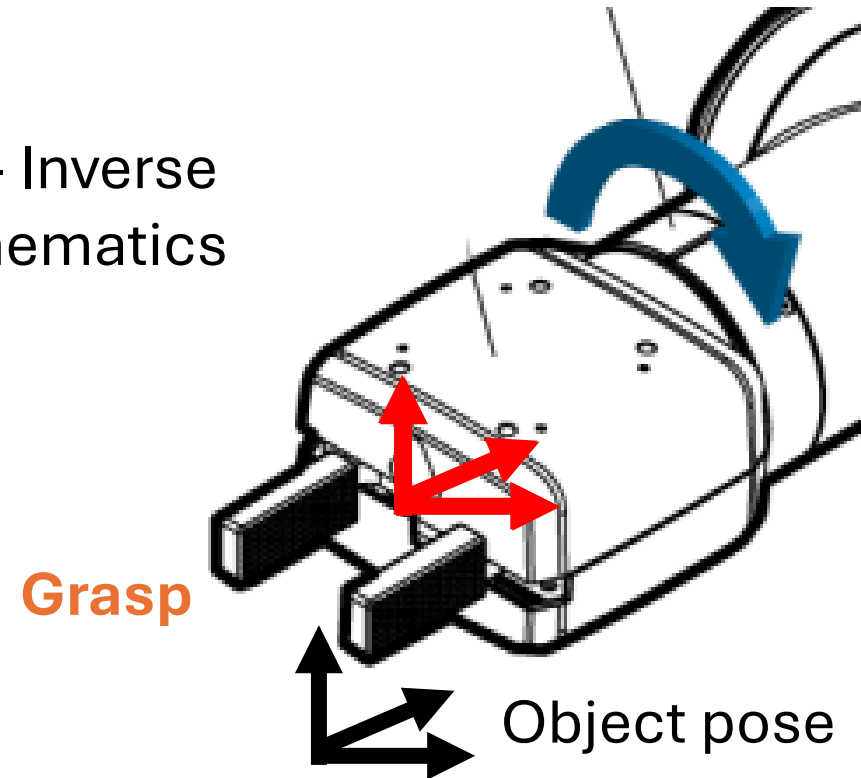


Grasping Configurations From Object Poses

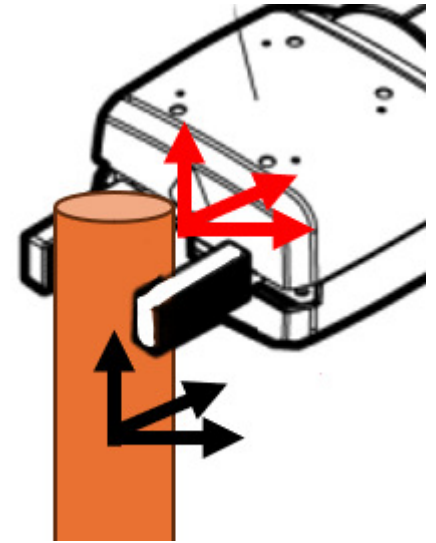
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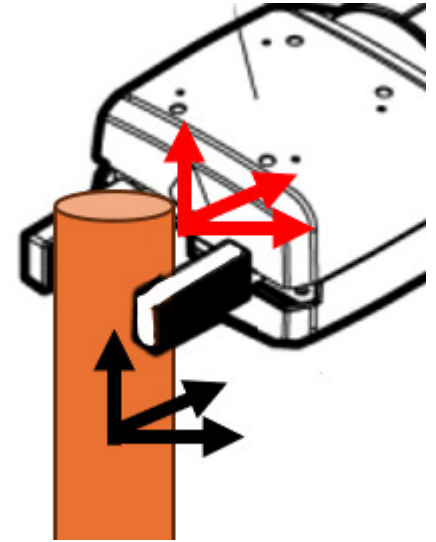
IK - Inverse
Kinematics



Does this configuration look correct?

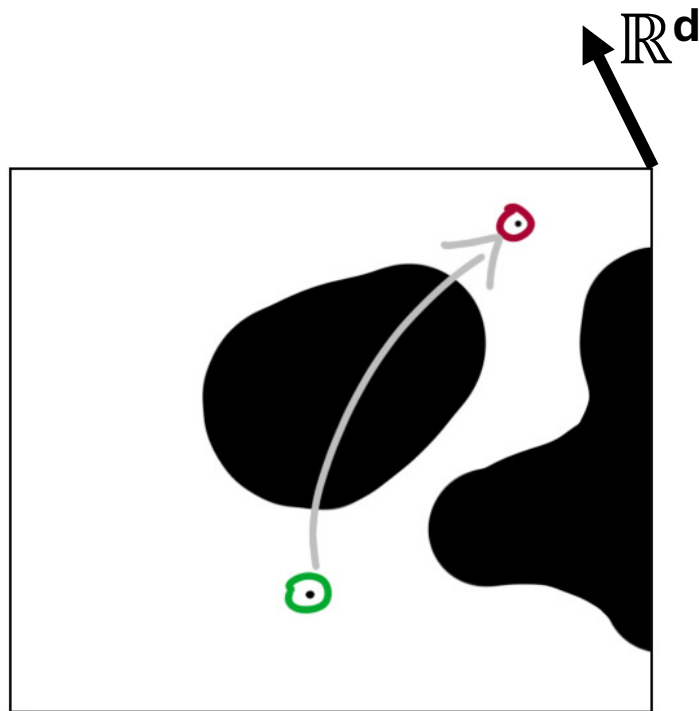


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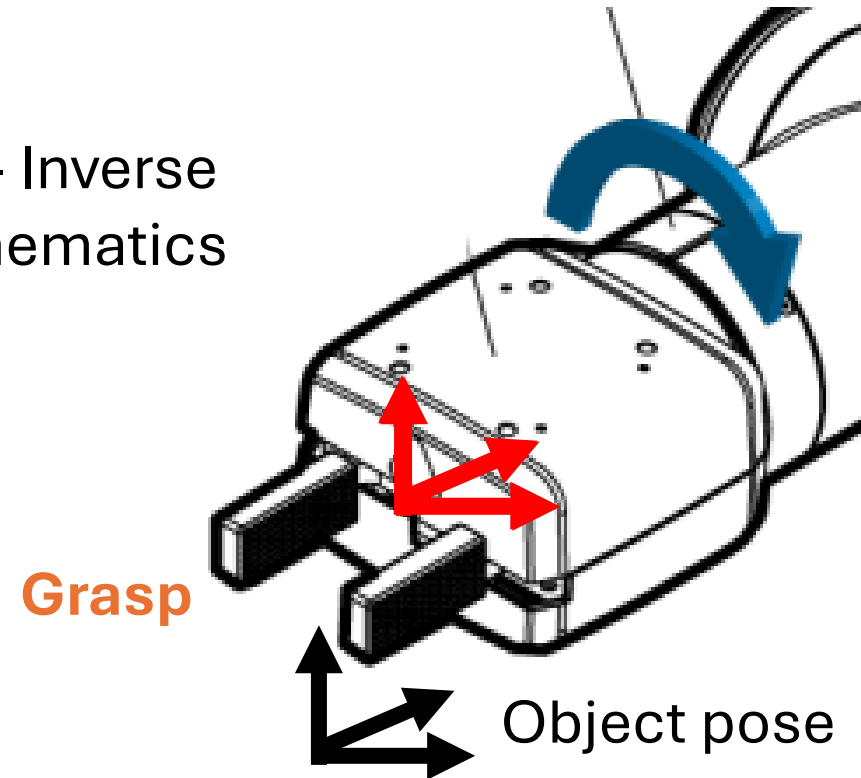


Grasping Configurations From Object Poses

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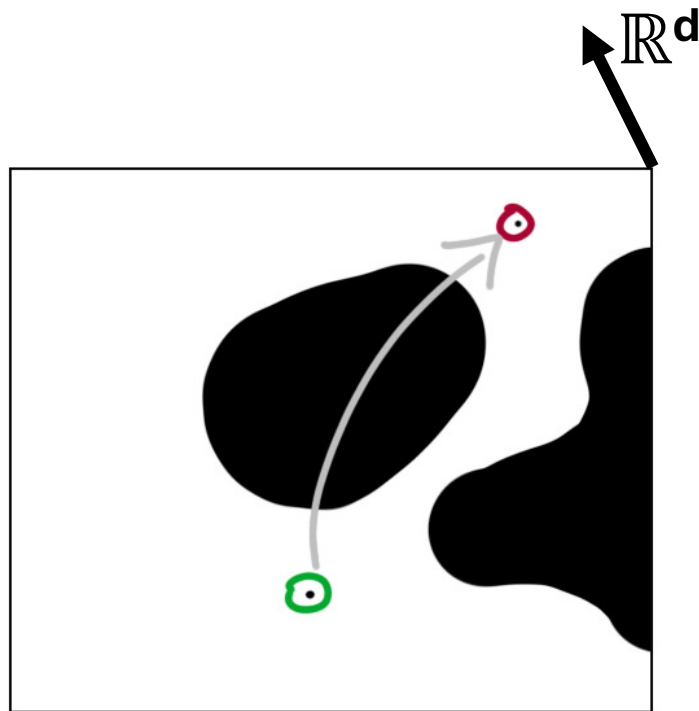


IK - Inverse
Kinematics

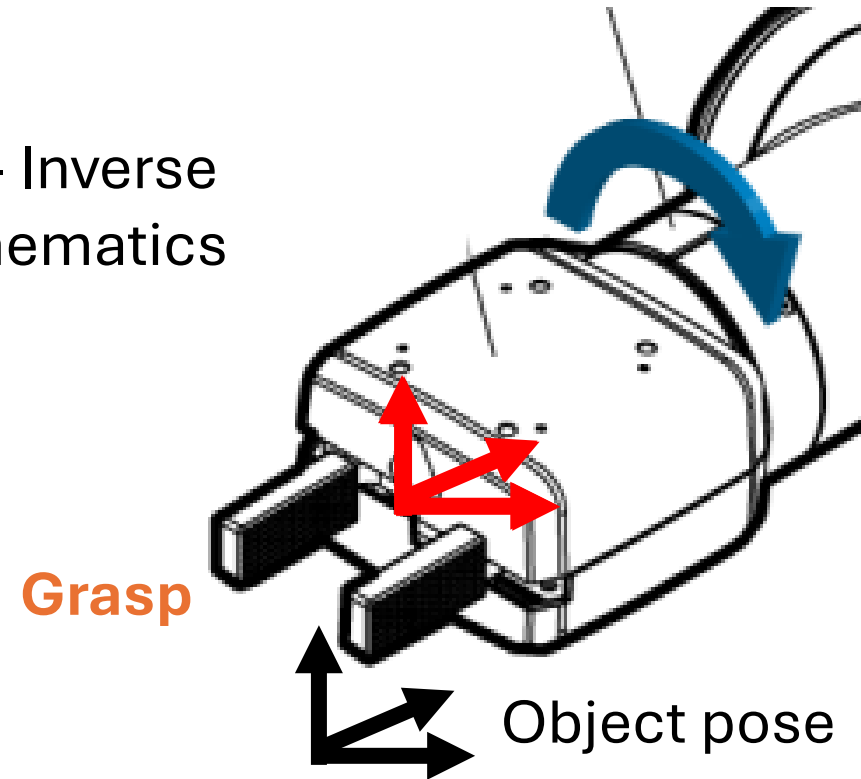


One-to-Many

For manipulation planning we need a configuration to plan to.

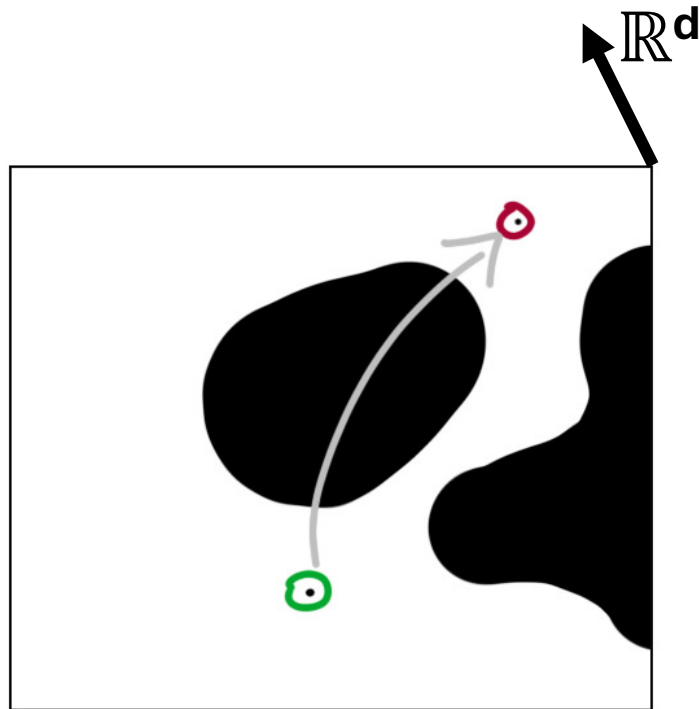


IK - Inverse
Kinematics

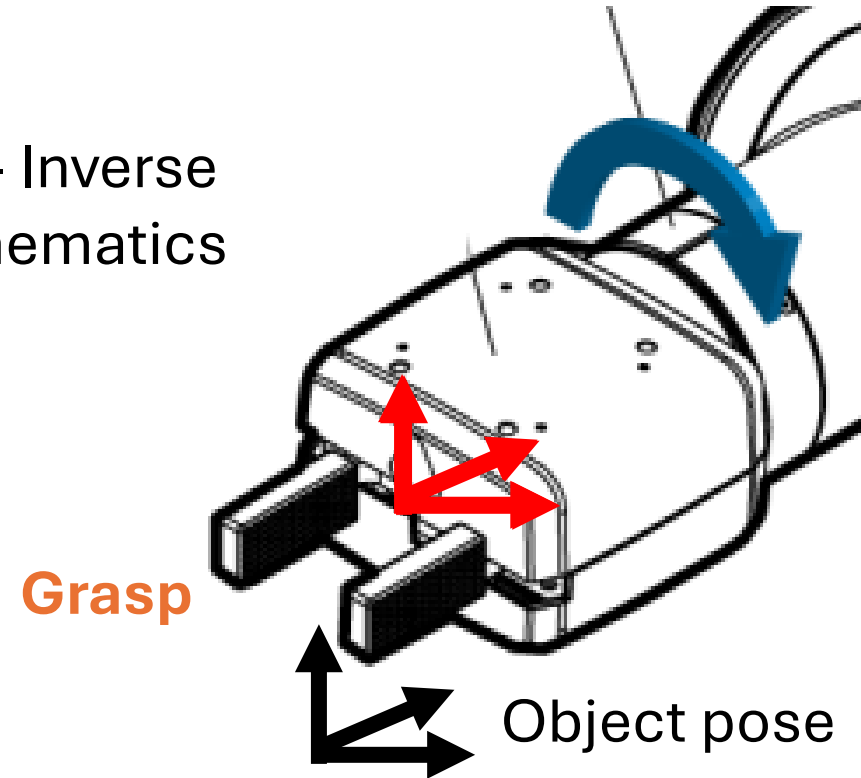


One-to-Many - Continuous

For manipulation planning we need a configuration to plan to.

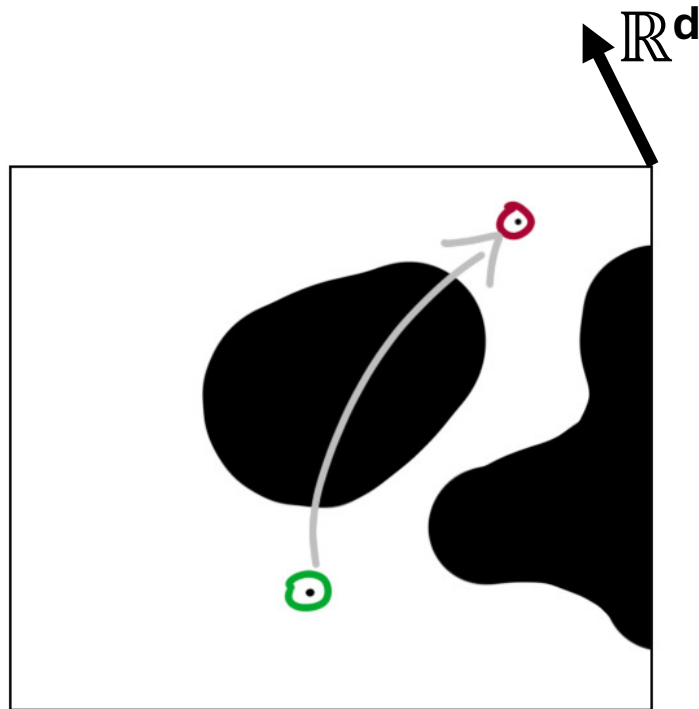


IK - Inverse
Kinematics

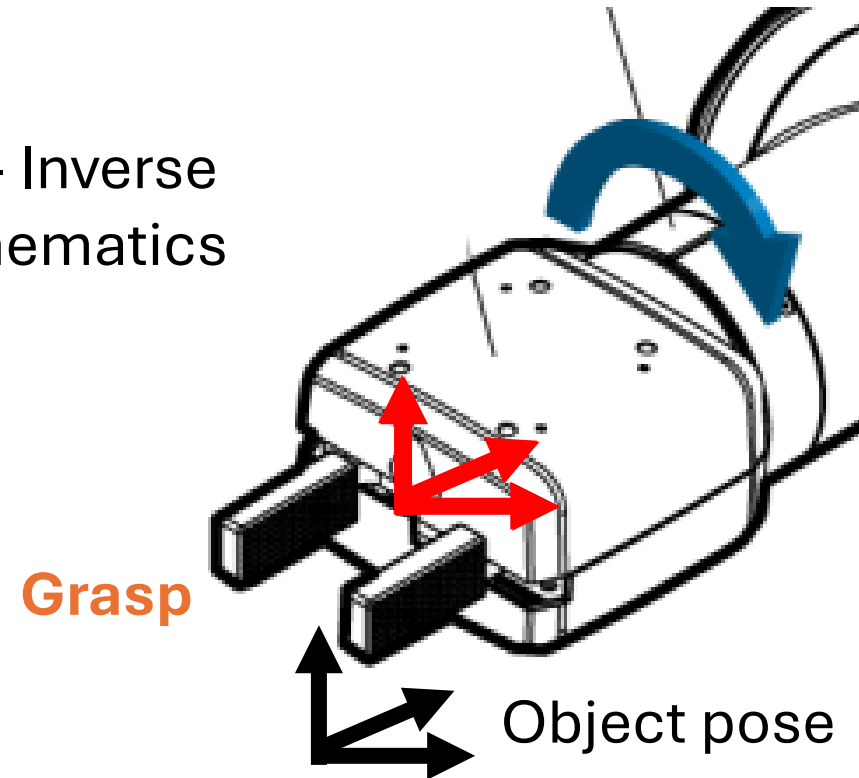


Pick and Place Manipulation Planning

Motion plan between manipulation configurations
- grasping and stable placements

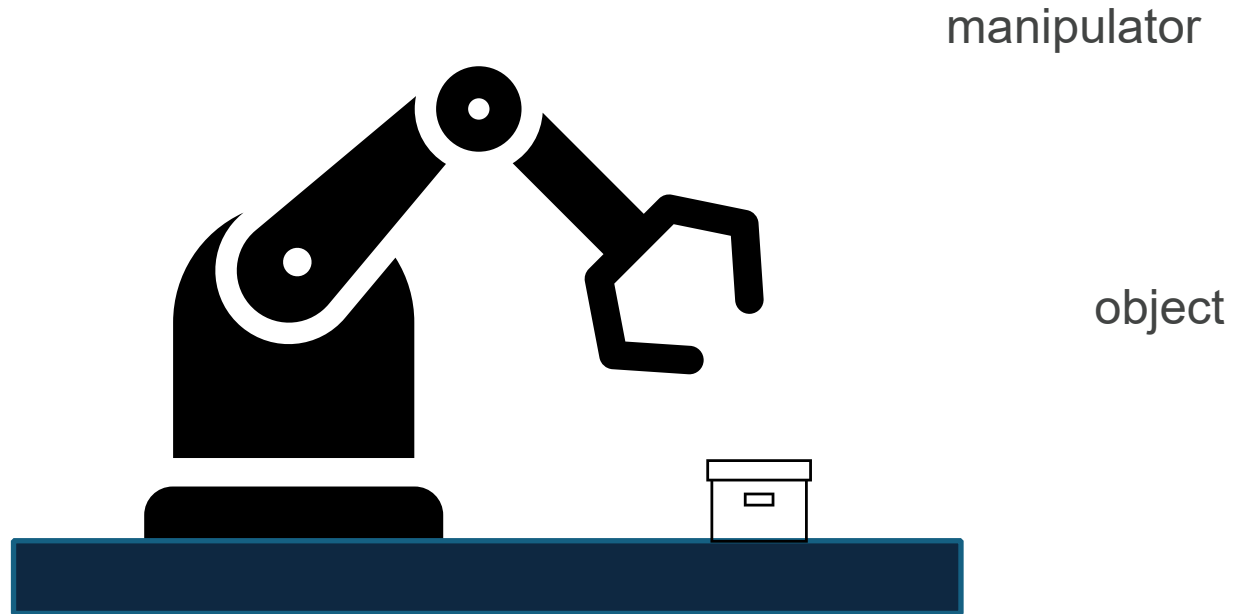


IK - Inverse
Kinematics

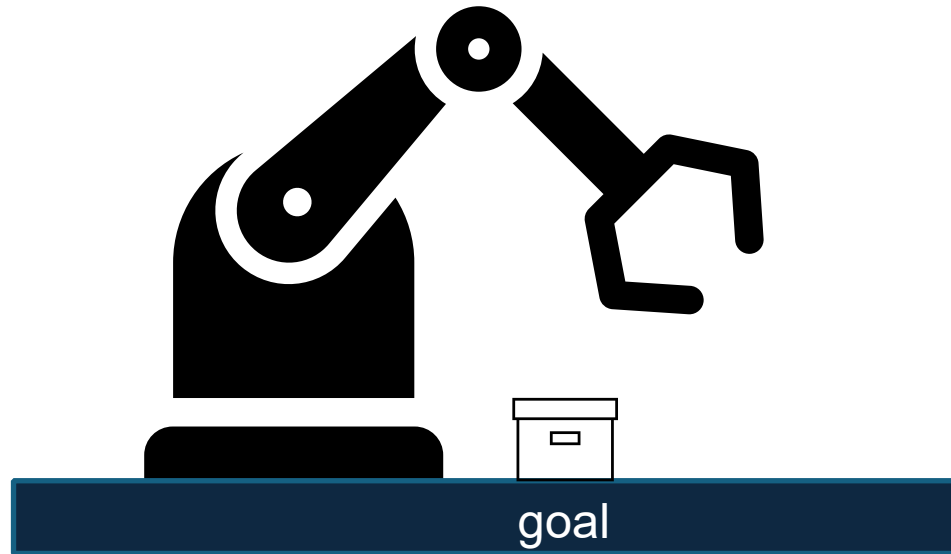


Planning Perspective

Modes (Manipulation)

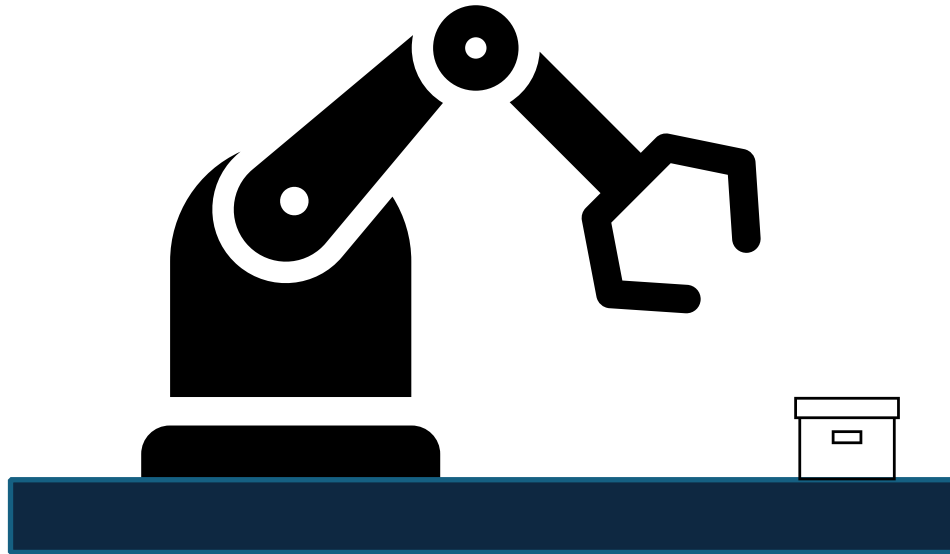


Modes (Manipulation)

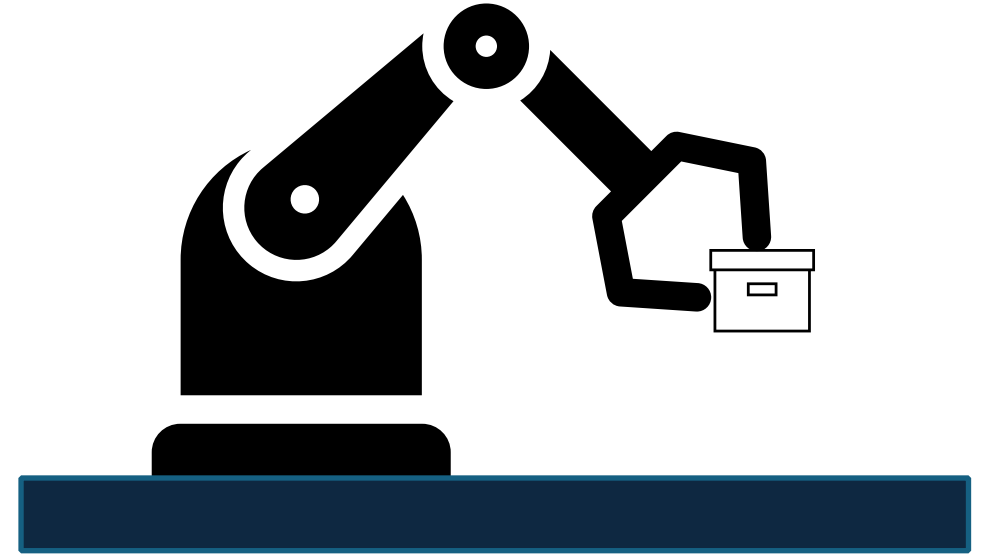


Brock O., Kuffner J., Xiao J. (2008) Motion for Manipulation Tasks. In: Siciliano B., Khatib O. (eds) Springer Handbook of Robotics. Springer, Berlin, Heidelberg.
https://doi.org/10.1007/978-3-540-30301-5_27

Modes (Manipulation)



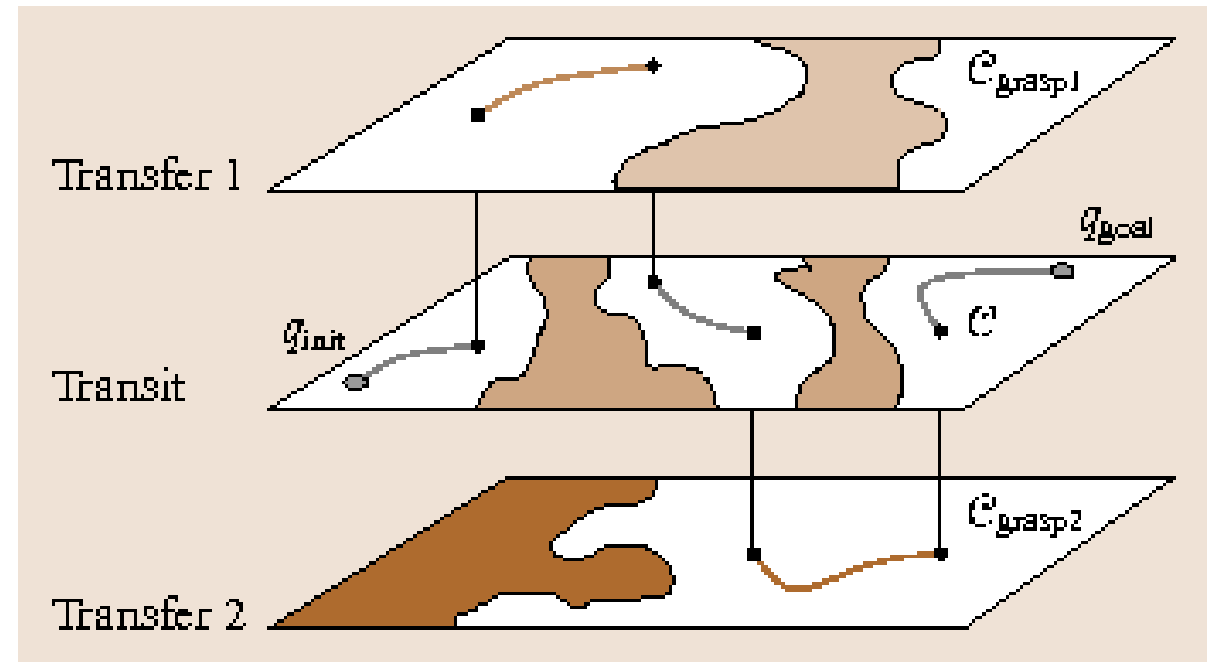
the manipulator moves alone
(transit mode)



the manipulator moves with grasped object
(transfer mode)

Modes (Manipulation)

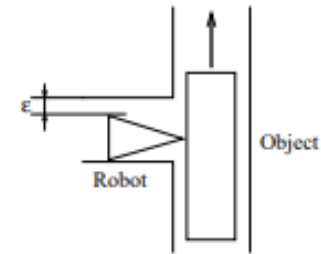
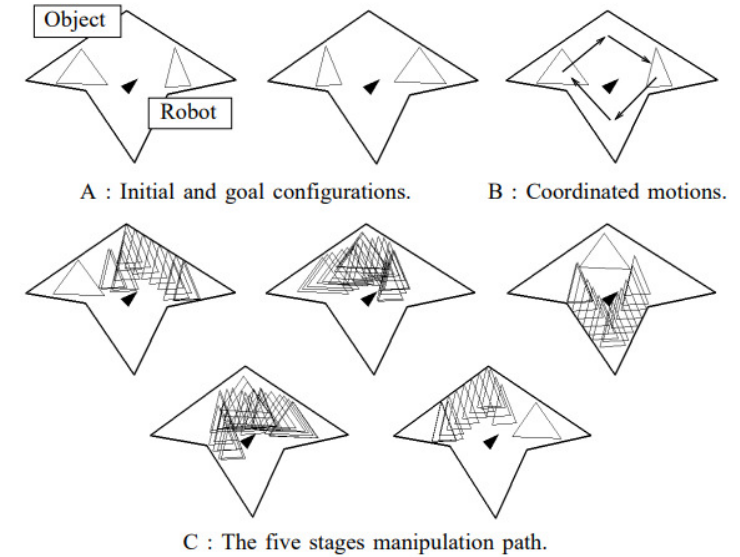
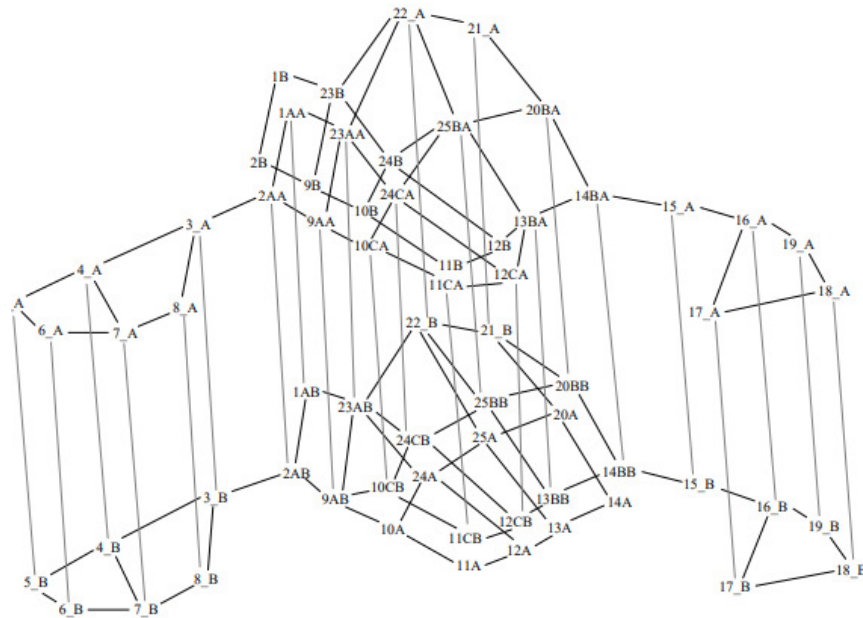
🔗 Transit and transfer modes create a set of hybrid continuous configuration spaces. In this example, transfer paths in different $\mathcal{C}$ -spaces arise from different ways of rigidly grasping the object



Brock O., Kuffner J., Xiao J. (2008) Motion for Manipulation Tasks. In: Siciliano B., Khatib O. (eds) Springer Handbook of Robotics. Springer, Berlin, Heidelberg.
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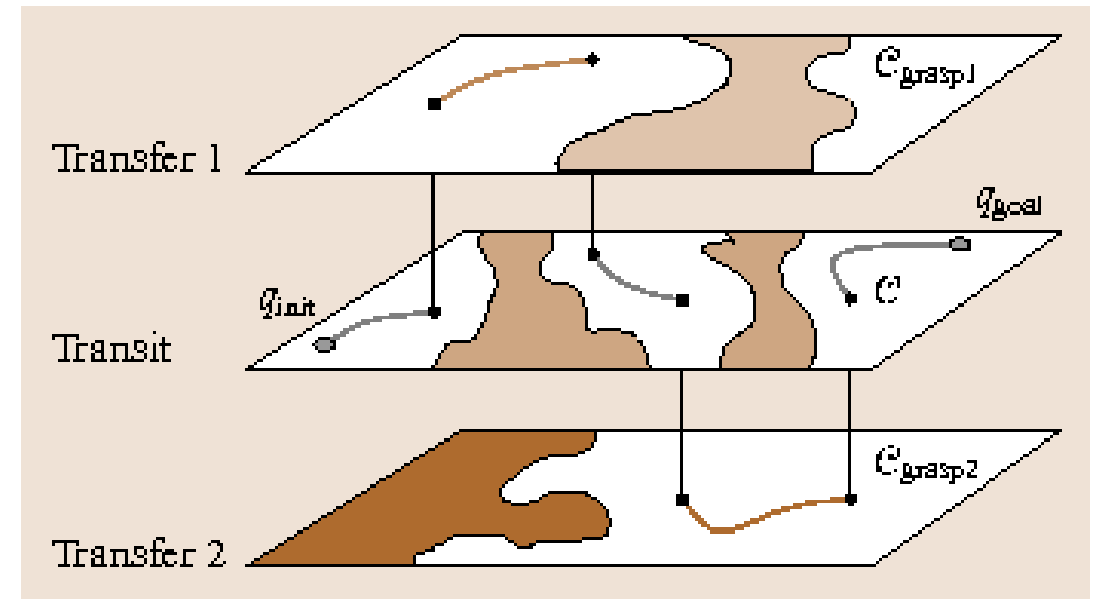
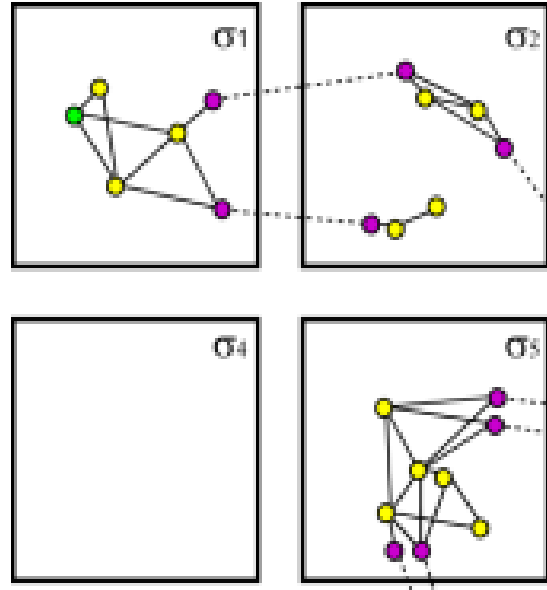
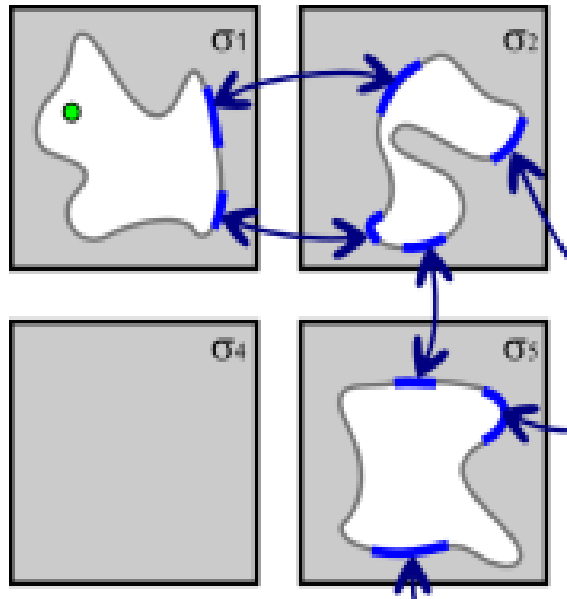
Manipulation Graph

 **Manipulation planning problem:** Find a finite sequence of transit-paths and transfer-paths to reach the final configuration.



 Alami, Rachid, Jean-Paul Laumond, and Thierry Siméon. "Two manipulation planning algorithms." In WAFR Proceedings of the workshop on Algorithmic foundations of robotics, pp. 109-125. AK Peters, Ltd. Natick, MA, USA, 1994.

Abstractions for discrete search space



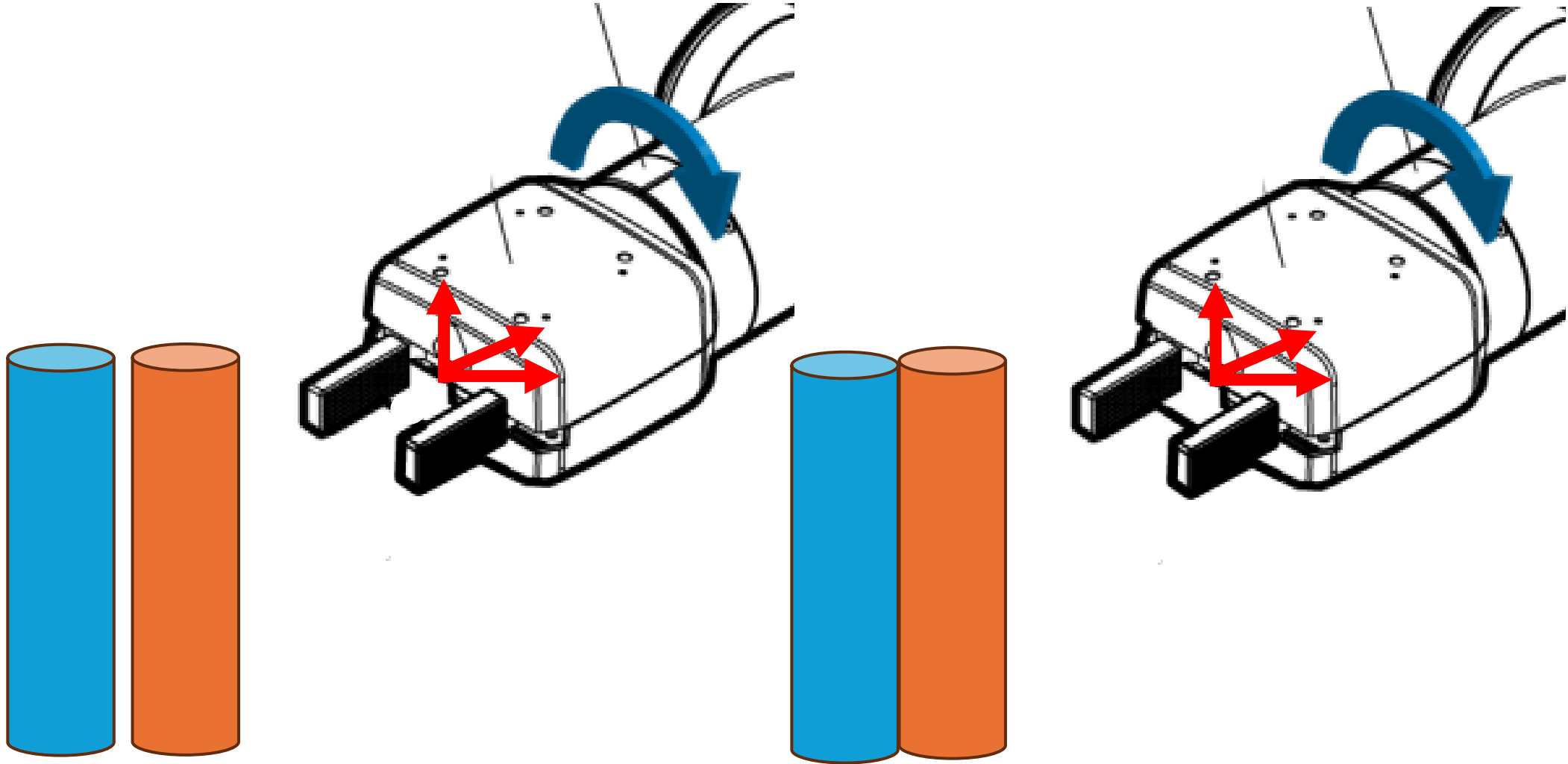
Transit Transfer PICK PLACE

Hauser, Kris, and Jean-Claude Latombe. "Multi-modal motion planning in non-expansive spaces." *The International Journal of Robotics Research* 29, no. 7 (2010): 897-915.

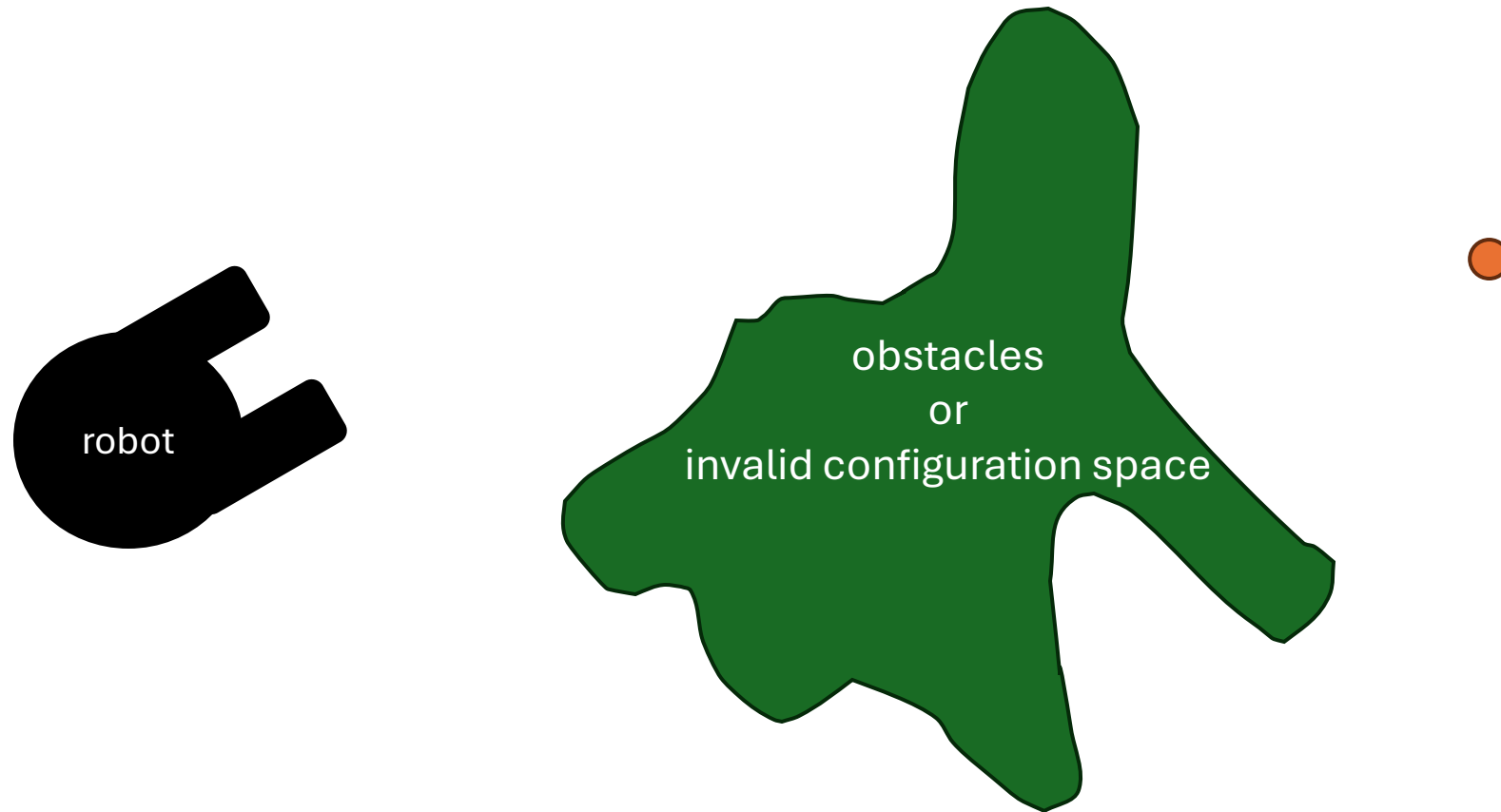
Brock O., Kuffner J., Xiao J. (2008) Motion for Manipulation Tasks. In: Siciliano B., Khatib O. (eds) Springer Handbook of Robotics. Springer, Berlin, Heidelberg.

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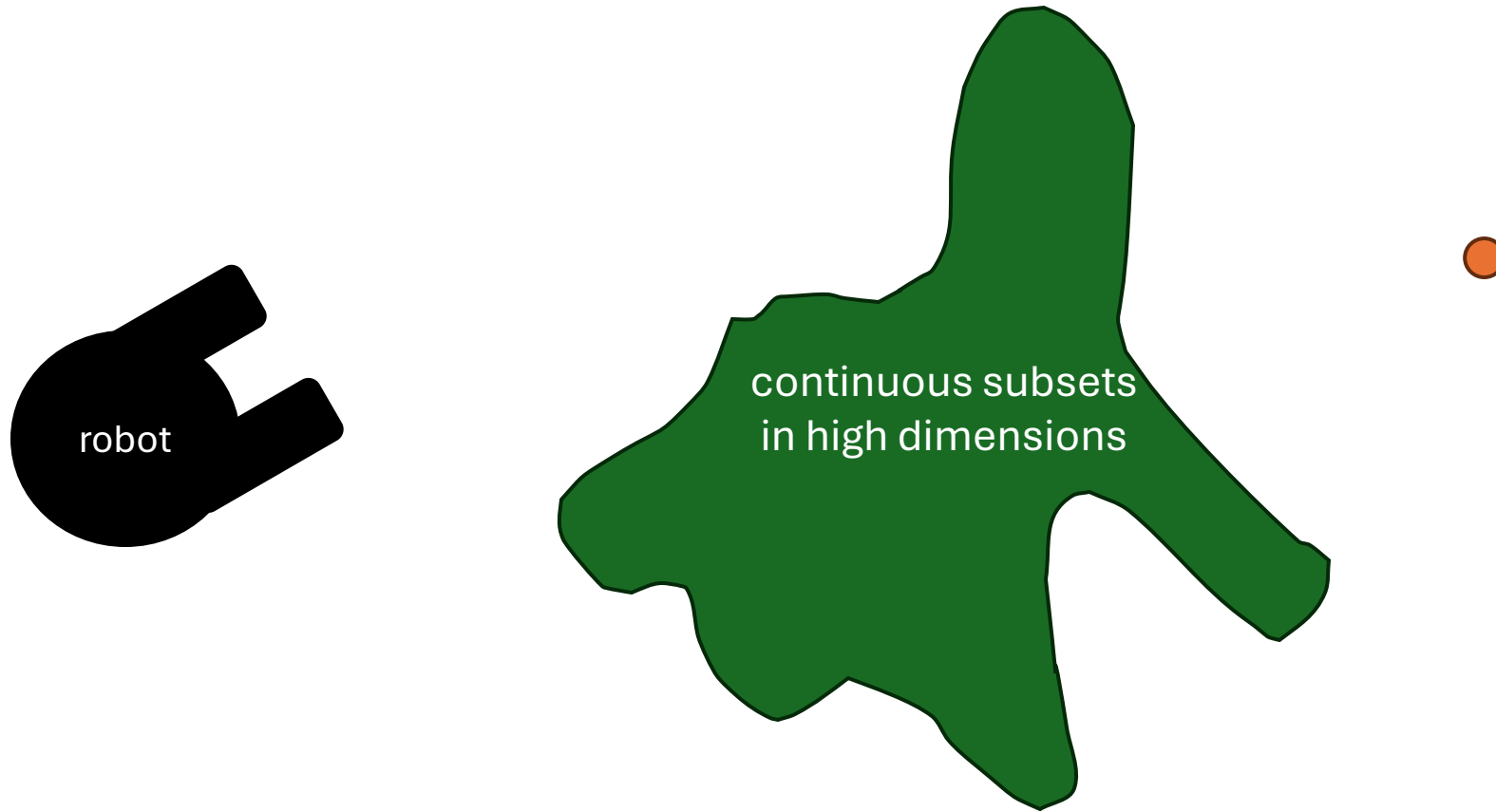
Distances: How Close is it Anyway?



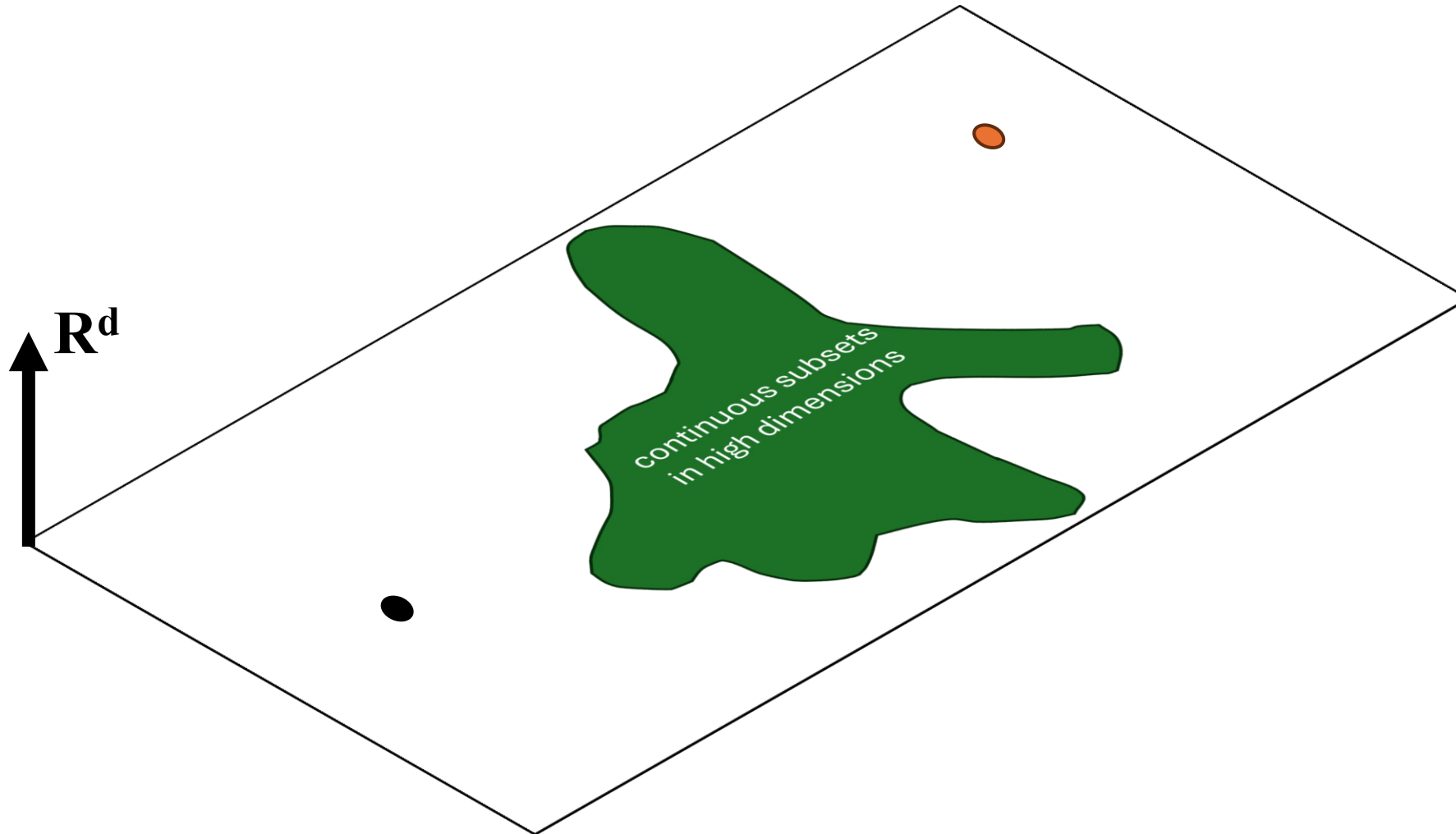
Recap: Robot Motion Planning



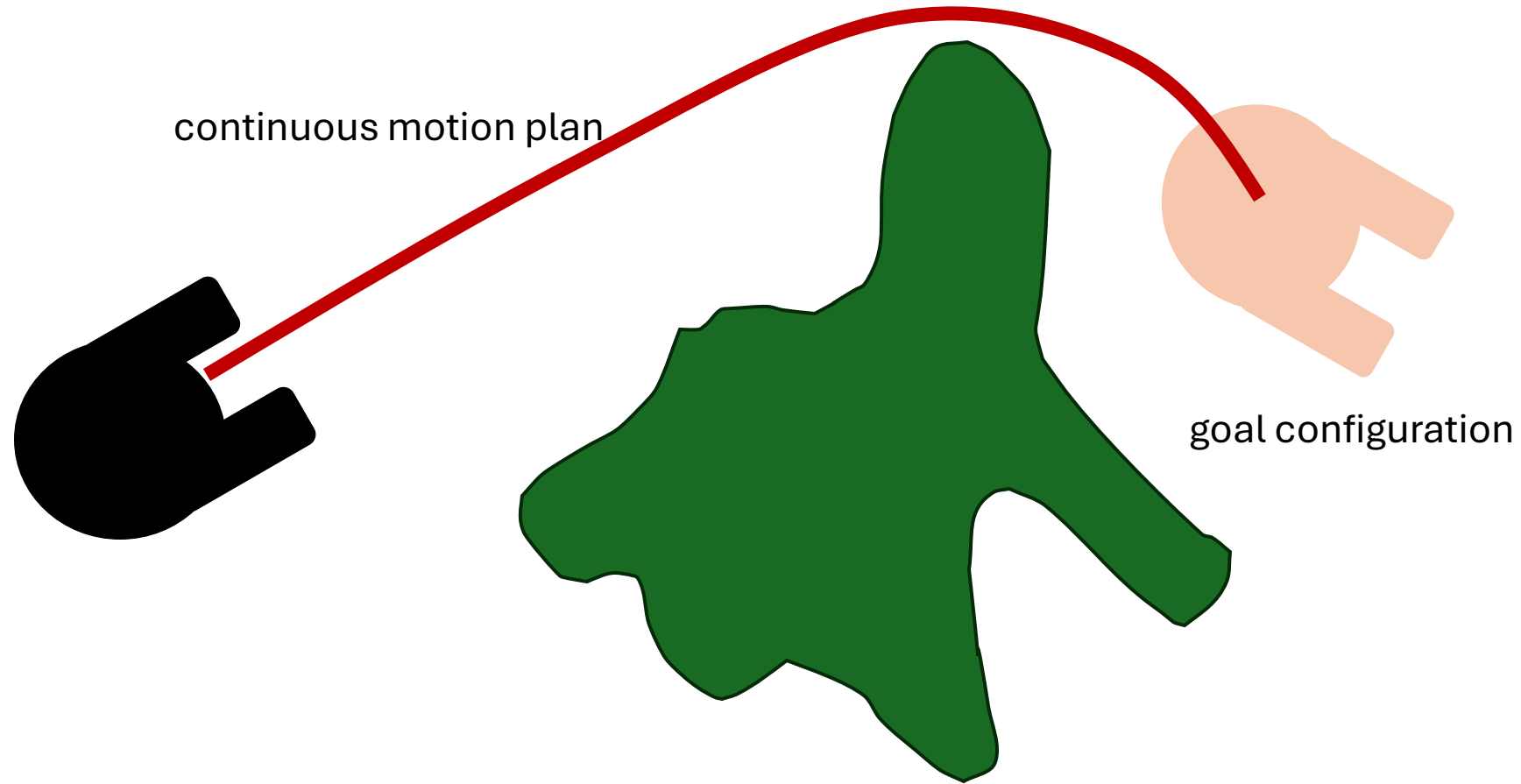
Robot Motion Planning



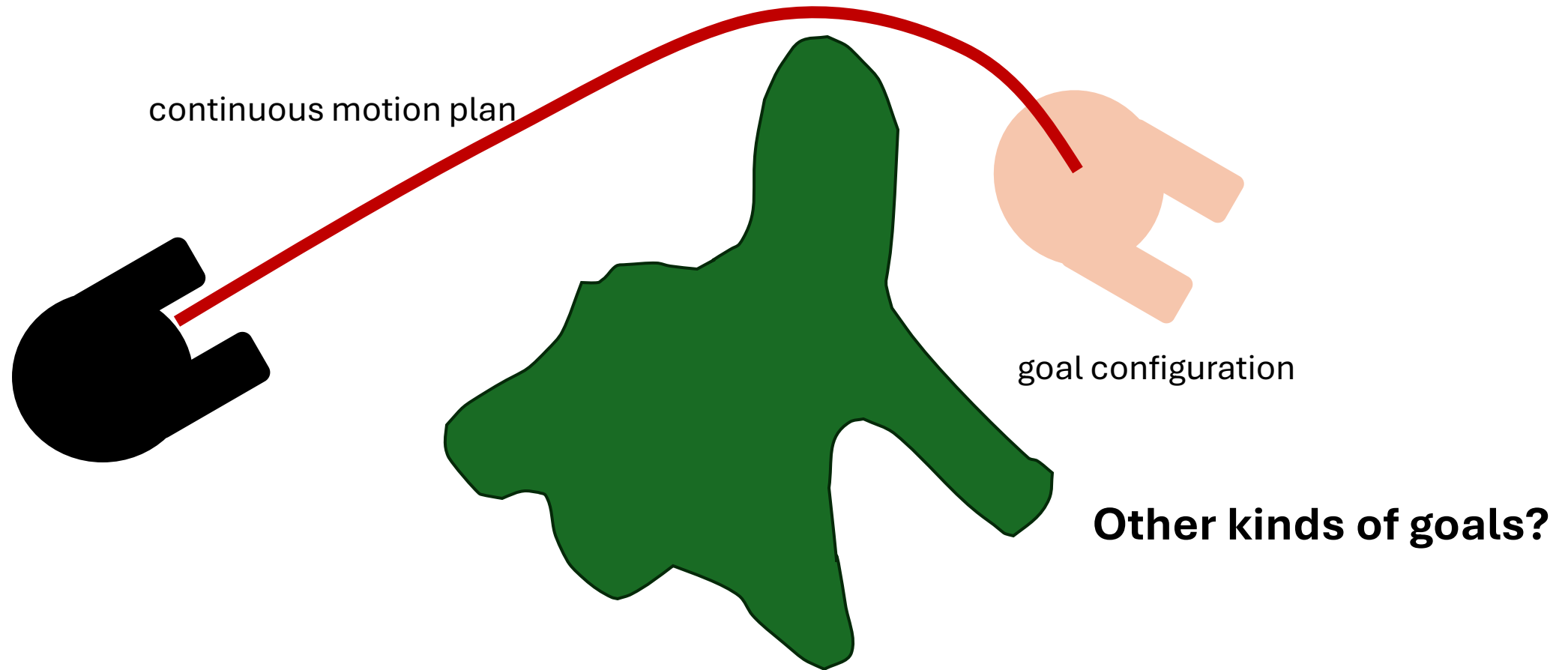
Robot Motion Planning



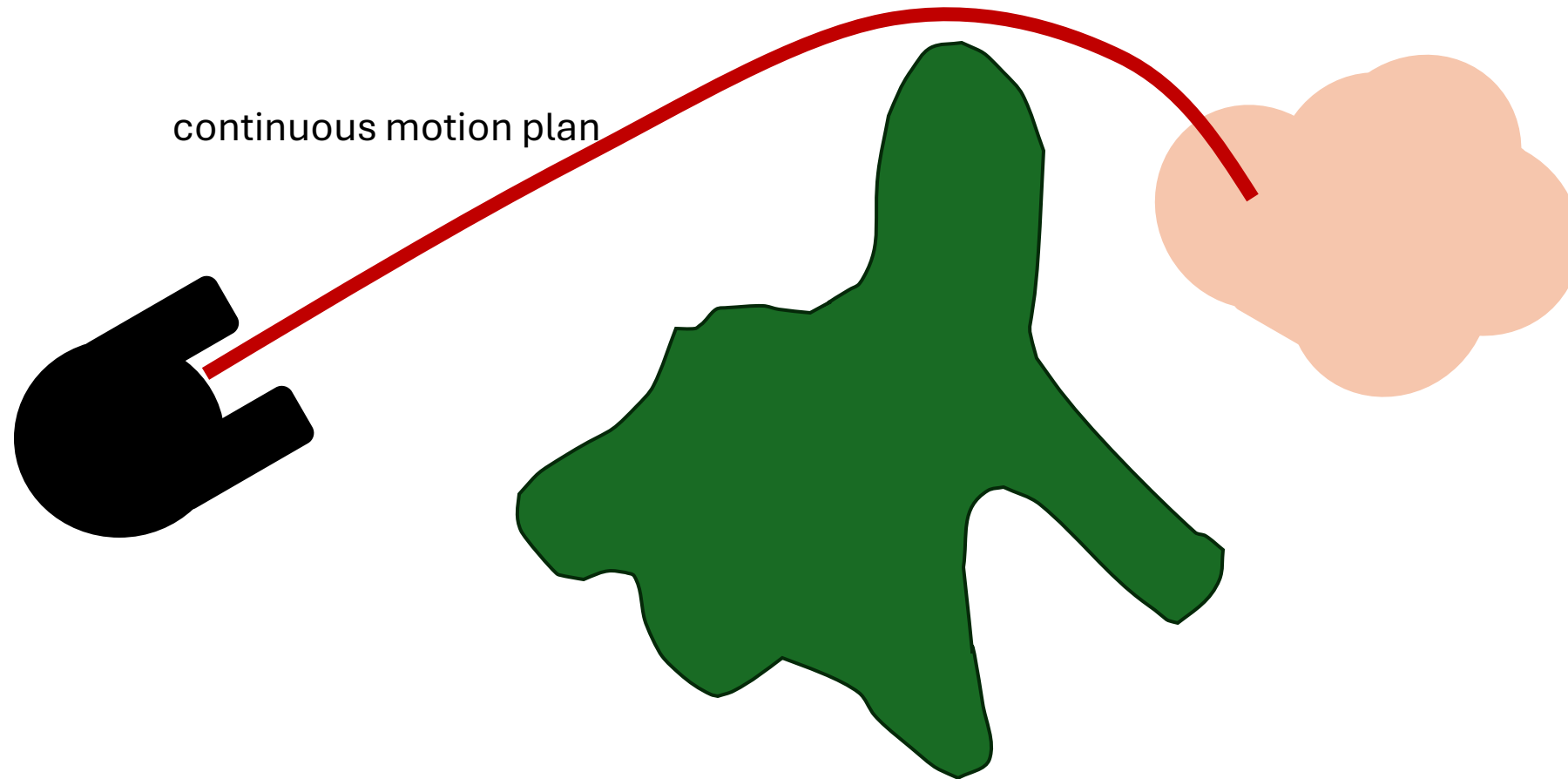
Robot Motion Planning



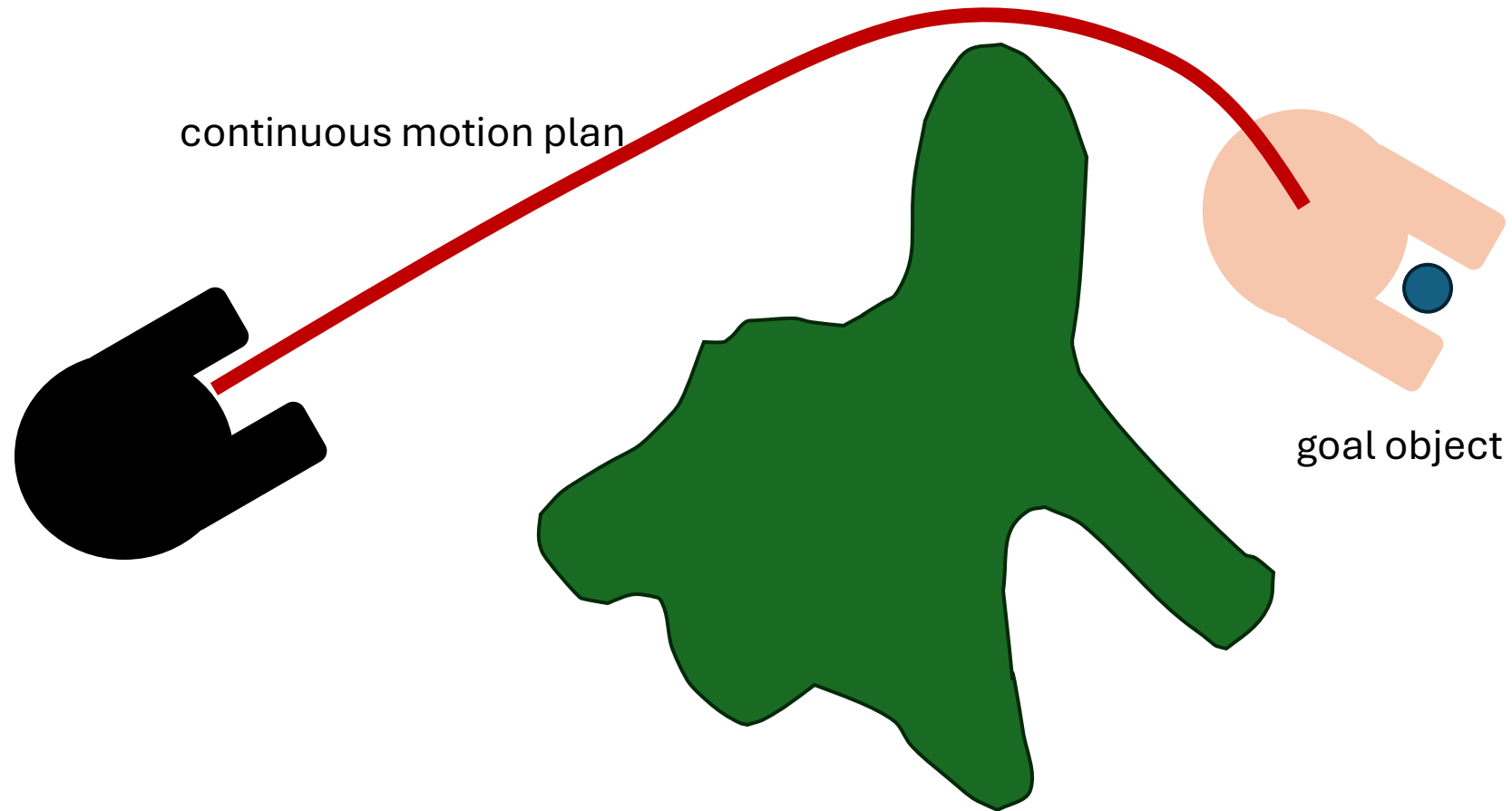
Robot Motion Planning



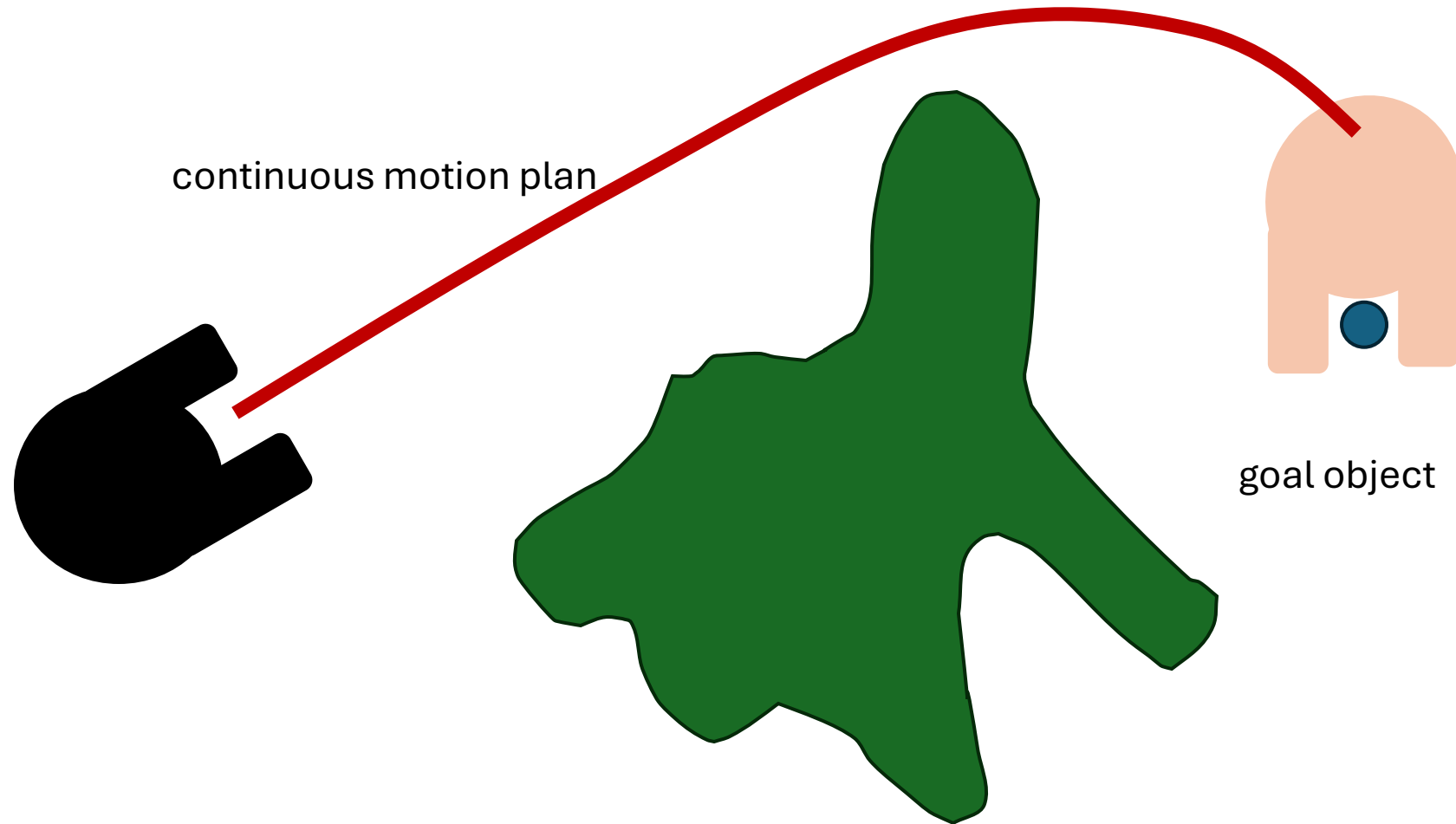
Goal Sets



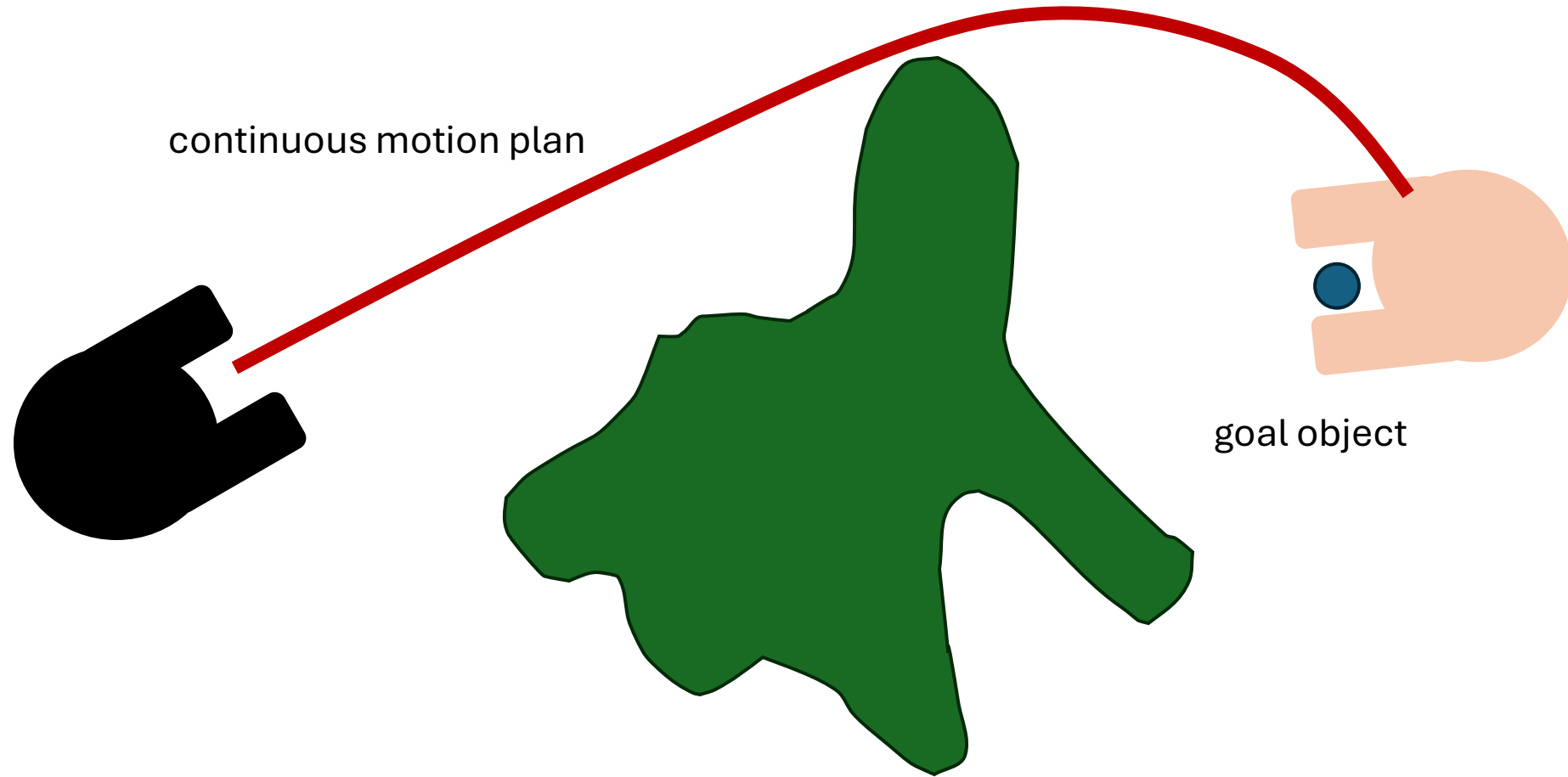
Goal: PICK the object



Goal: PICK the object

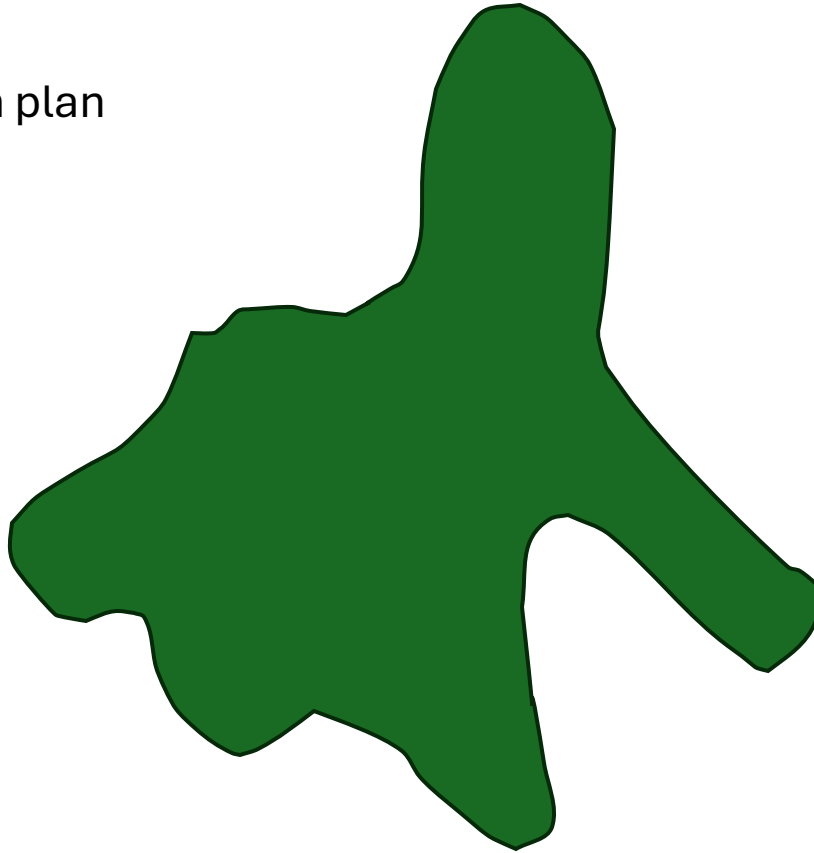
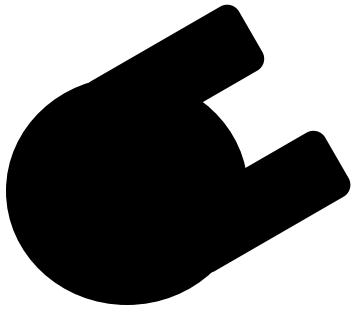


Goal: PICK the object



Goal set: PICK the object

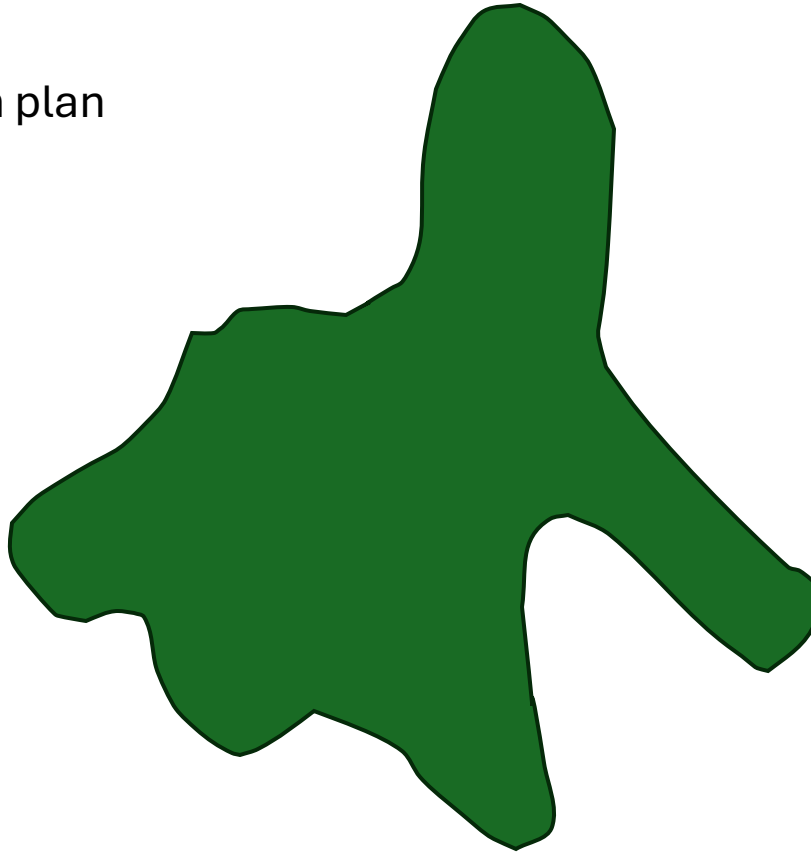
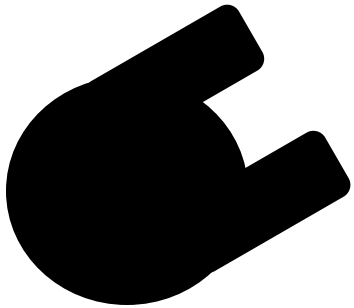
continuous motion plan



Goal set: PICK the object

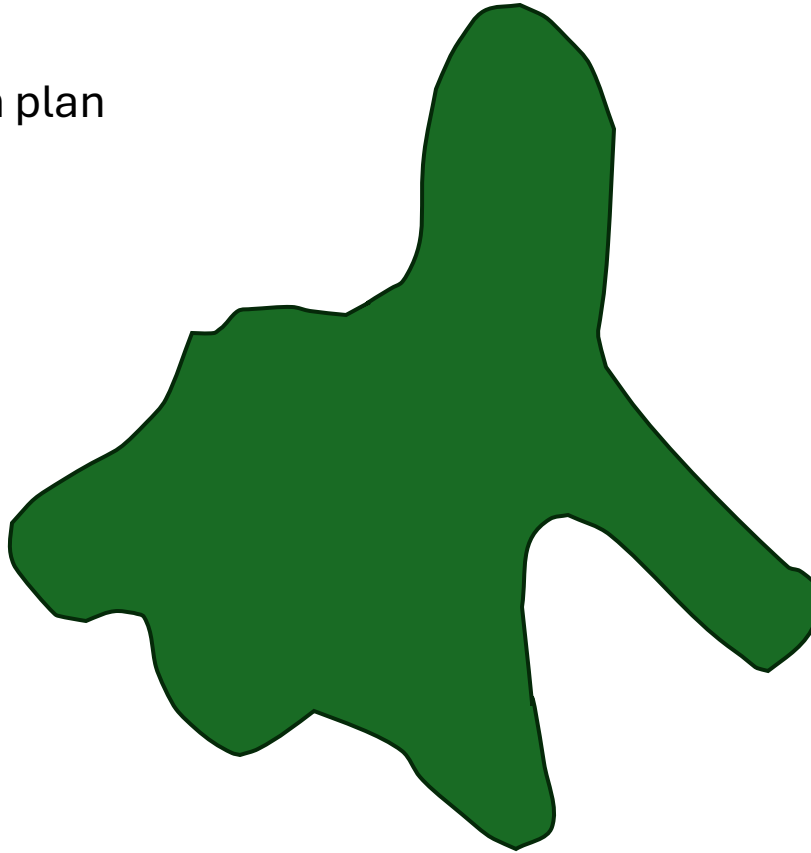
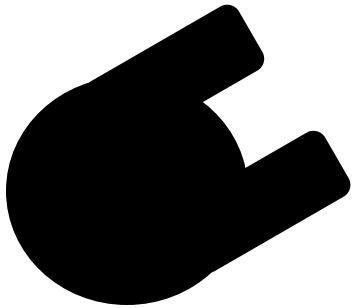
🤖 What is the volume of this set?

continuous motion plan



Goal set: PICK the object

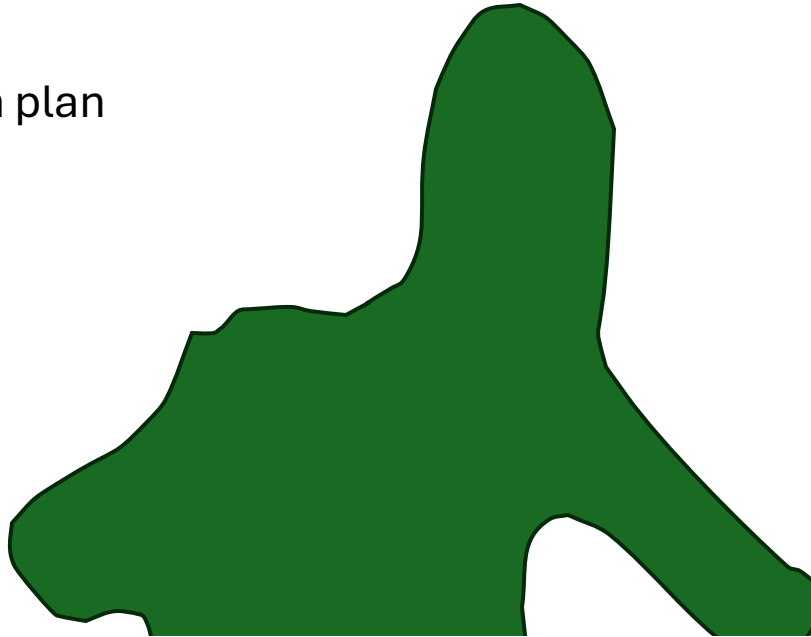
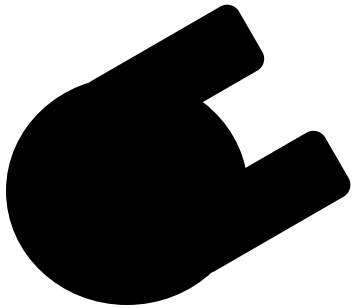
continuous motion plan



The ring is a lower-dimensional manifold

Actions as Constraint Manifolds

continuous motion plan



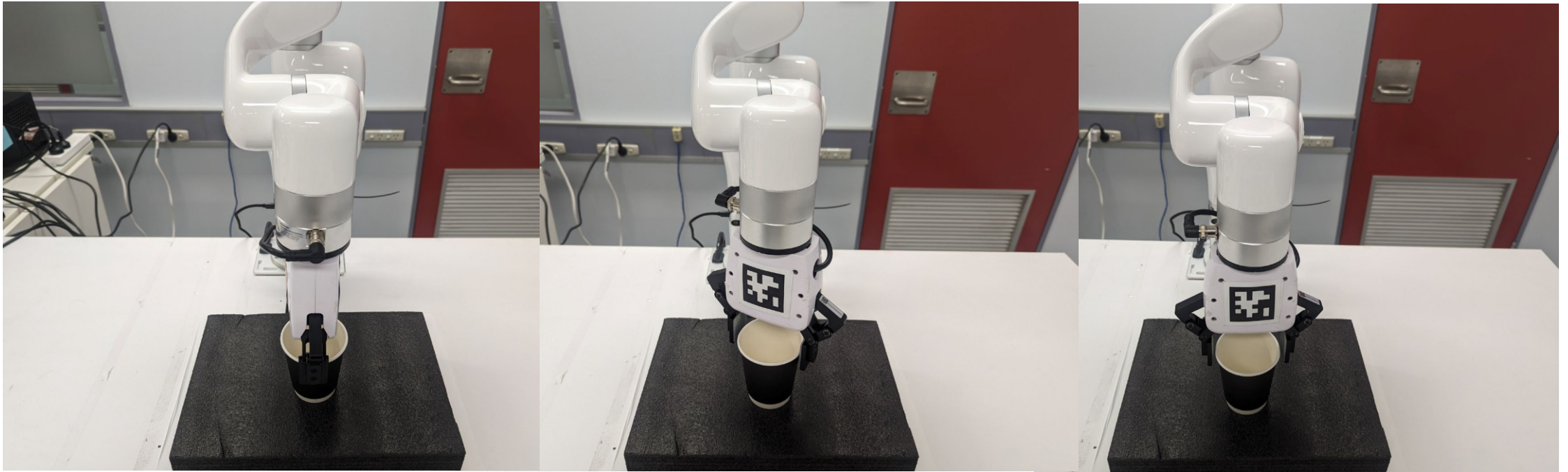
The ring is a lower-dimensional manifold

Actions as Constraints

PICK()



Actions as Constraint Manifolds



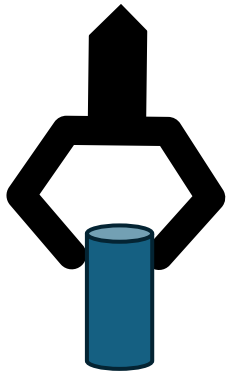
Actions as Constraints

PICK()

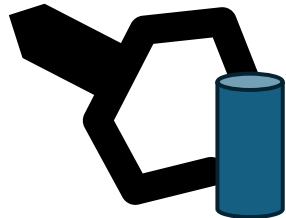


Actions as Constraint Manifolds

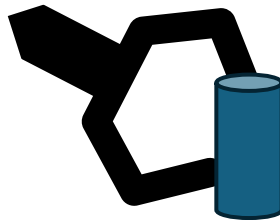
- 🔗 Configuration Space – all the ways the robot can move
- 🔗 Constraint manifold – subset in space which can be zero volume and lower dimensional



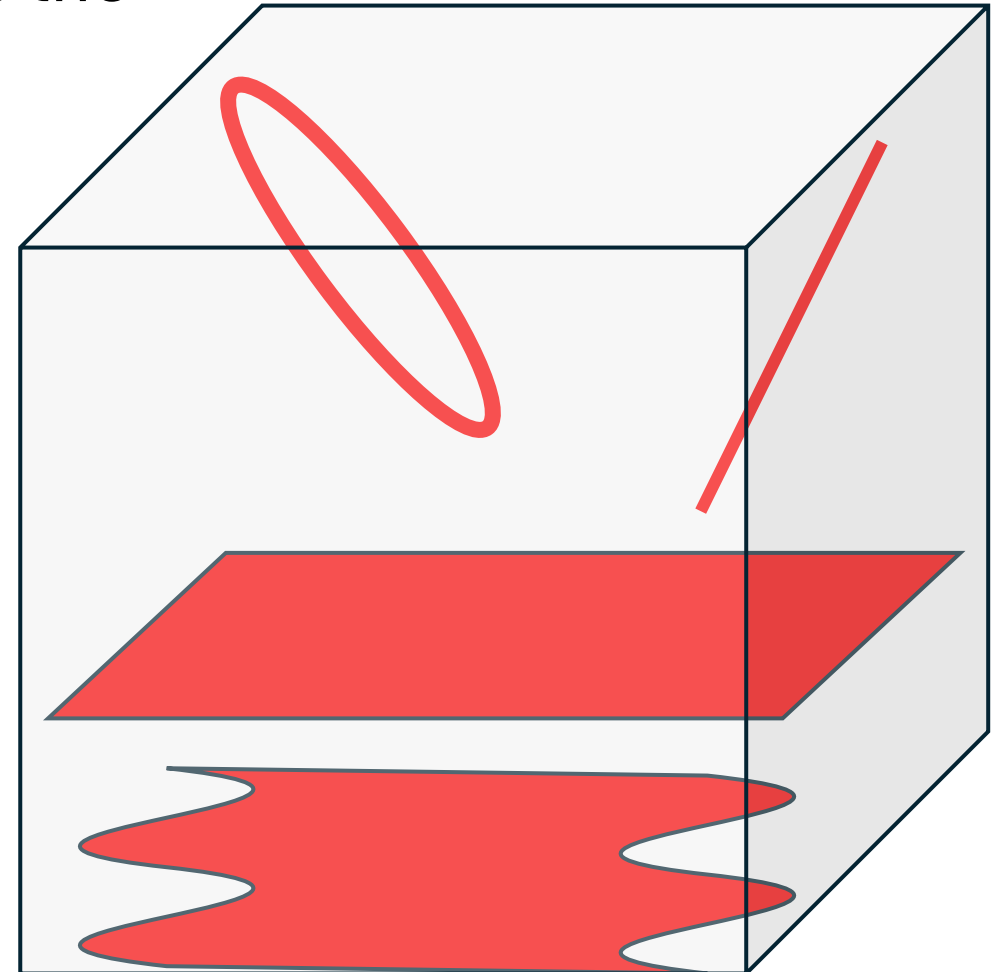
Pick up



Keep upright

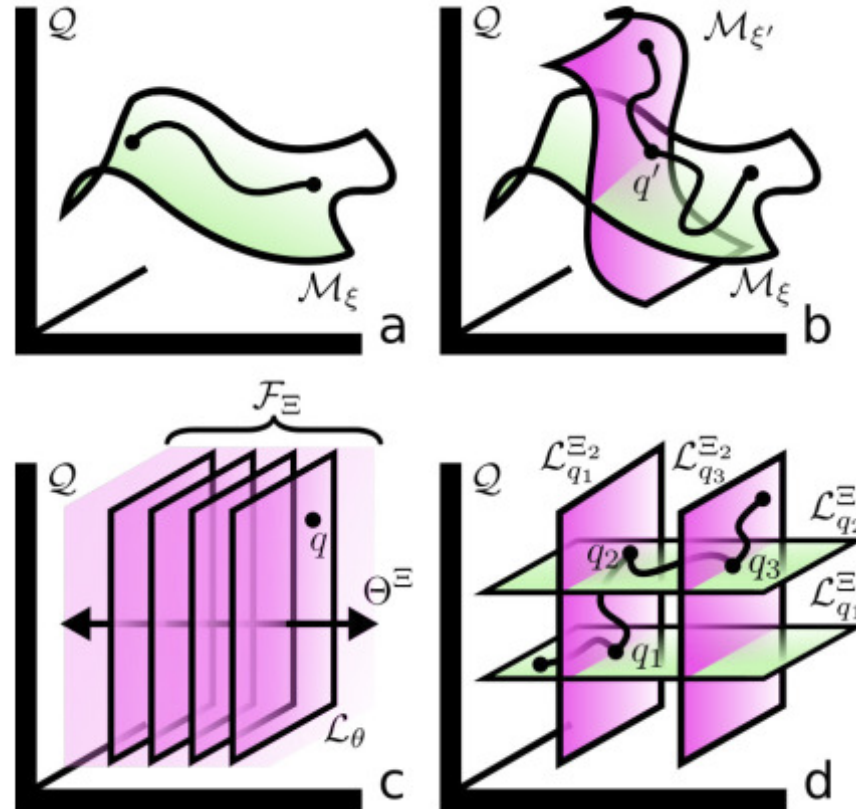


Place



Planning through Sequences of Constraints

🔗 Multi-modal Planning

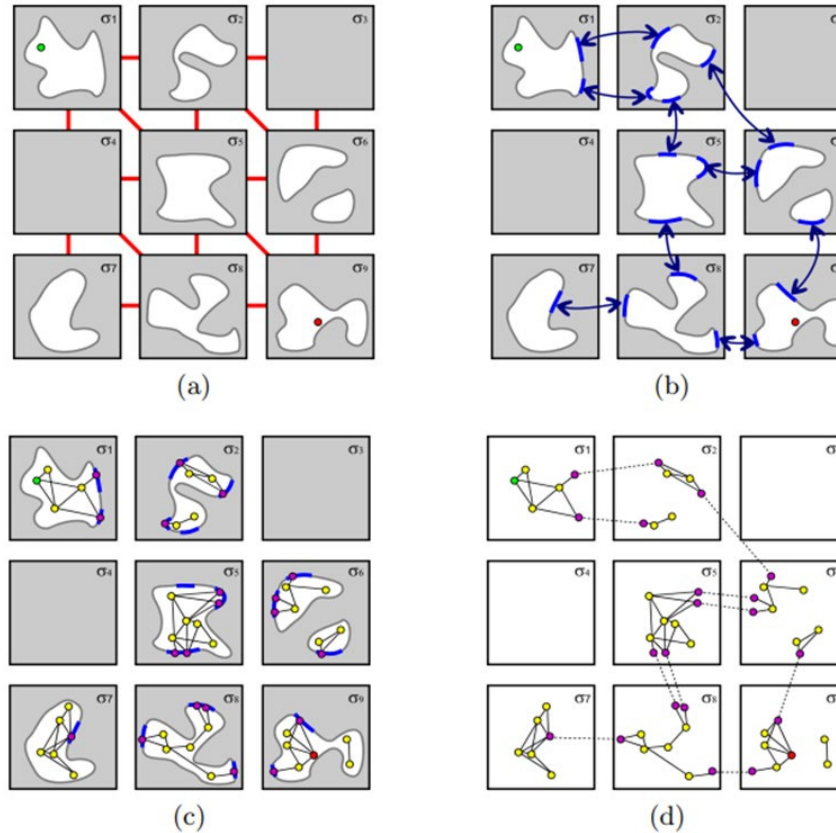


Kingston, Zachary, Andrew M. Wells, Mark Moll, and Lydia E. Kavraki. "Informing multi-modal planning with synergistic discrete leads." In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3199-3205. IEEE, 2020.

Kingston, Zachary, Mark Moll, and Lydia E. Kavraki. "Decoupling constraints from sampling-based planners." In *Robotics Research*, pp. 913-928. Springer, Cham, 2020.

Planning through Sequences of Constraints

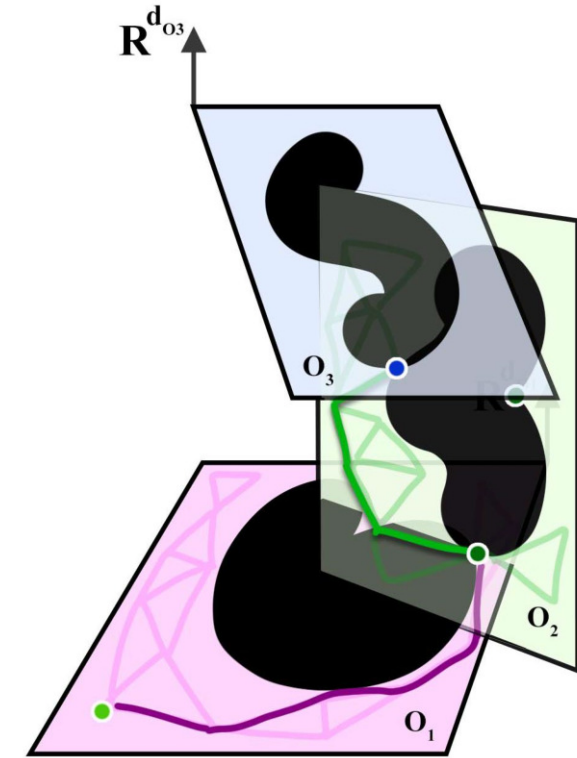
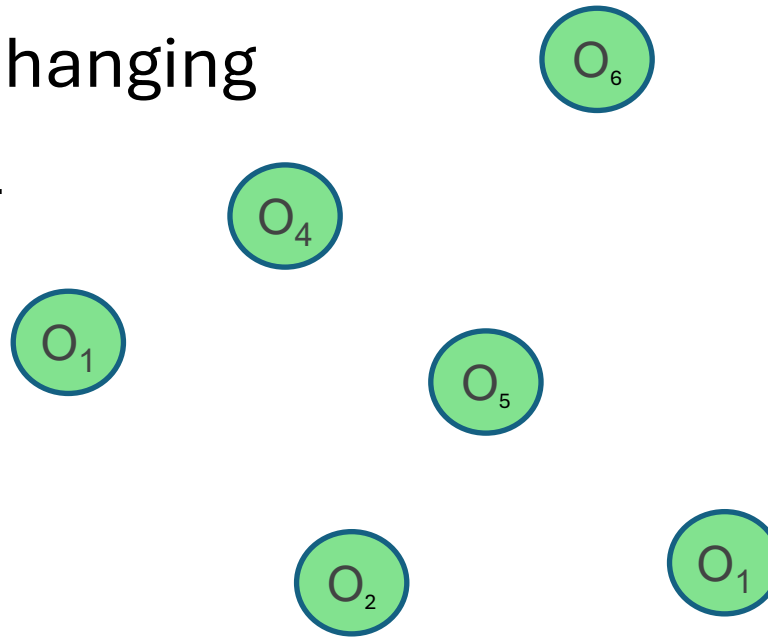
🔍 Multi-modal Planning



Hauser, Kris, and Jean-Claude Latombe. "Multi-modal motion planning in non-expansive spaces." *The International Journal of Robotics Research* 29, no. 7 (2010): 897-915.

Planning through Sequences of Constraints

- 🤖 These search spaces are hybrid
- 🤖 These search spaces are massive
- 🤖 Dimensions keep changing
- 🤖 Lower dimensional components cannot be sampled naively
- 🤖 Need for guidance



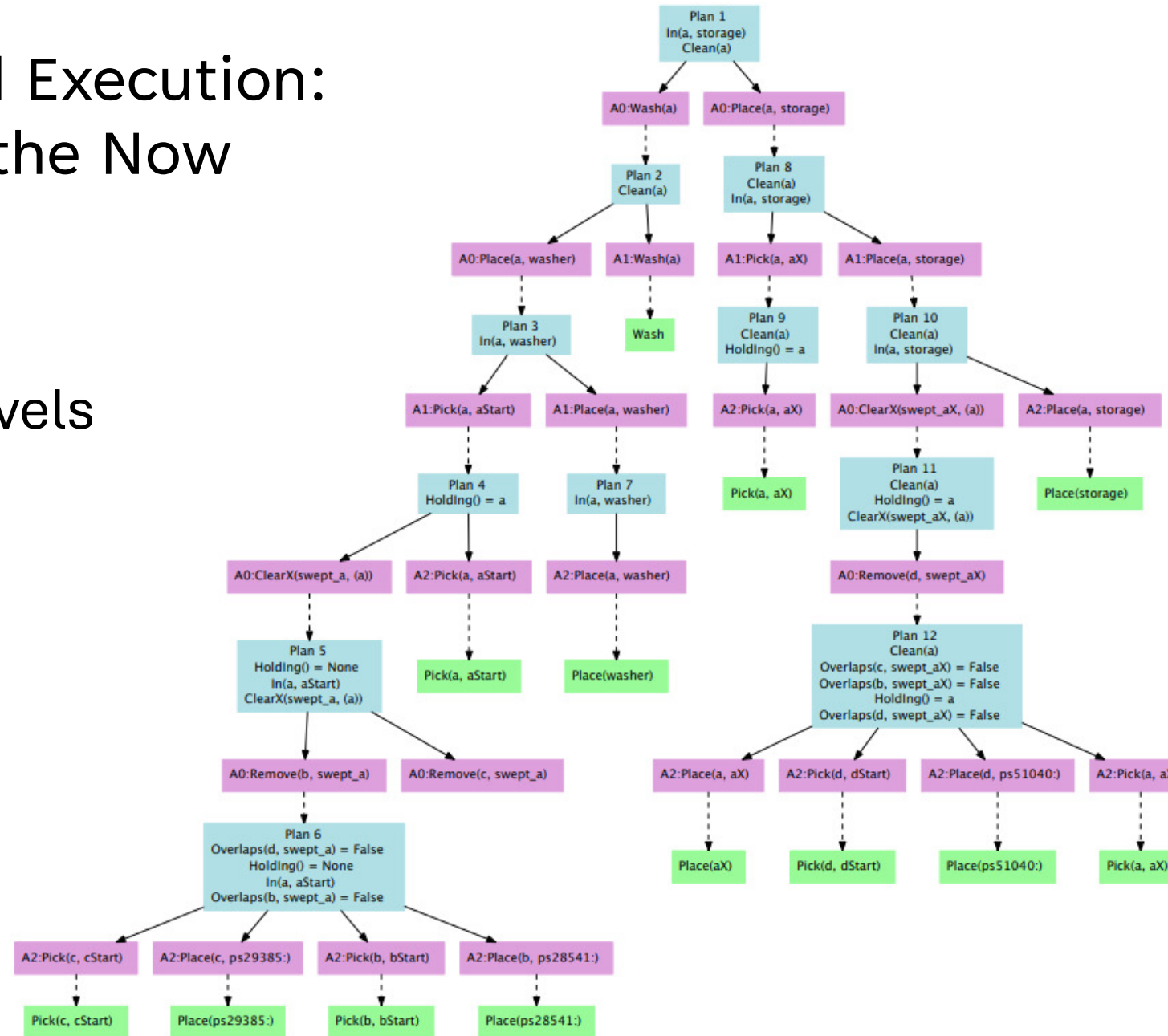
Vega-Brown, William, and Nicholas Roy. "Asymptotically optimal planning under piecewise-analytic constraints." In *Algorithmic Foundations of Robotics XII*, pp. 528-543. Springer, Cham, 2020.

Shome, Rahul, Daniel Nakhimovich, and Kostas E. Bekris. "Pushing the boundaries of asymptotic optimality in integrated task and motion planning." In *International Workshop on the Algorithmic Foundations of Robotics*, pp. 467-484. Springer, Cham, 2020.

Brief History of TAMP

Interleaved Planning and Execution: Hierarchical Planning in the Now

🔧 Recursively plan across
refinement (abstraction) levels
of actions
-- ***then motion plan then
execute***



Kaelbling, Leslie Pack, and Tomás Lozano-Pérez. "Hierarchical planning in the now." In *Workshops at the Twenty-Fourth AAAI Conference on Artificial Intelligence*. 2010.

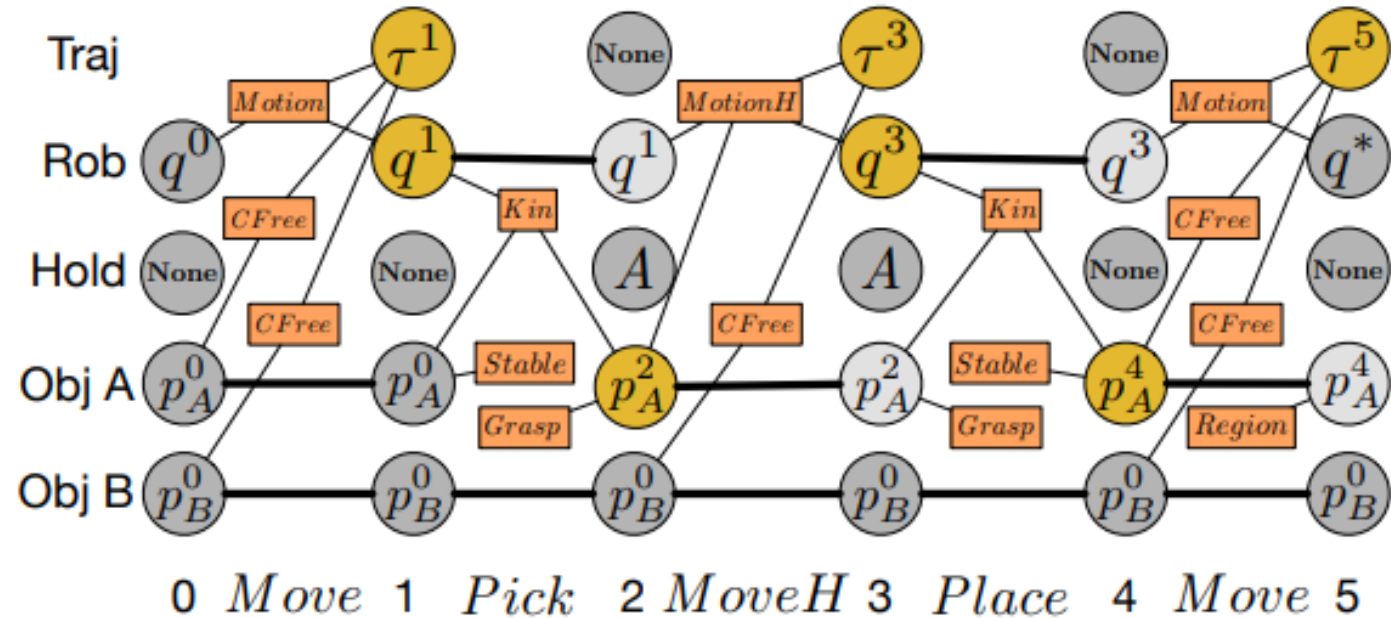
Composing Samplers

Factored transition system

- factored search space

arm	$C_{\text{manipulator}}$	X
objects	P_{objA}	X
grasped?	$\{\phi, \text{objA}, \text{objB}\}$	X

Pick-and-place sampler and constraint network



Garrett, Caelan Reed, Tomás Lozano-Pérez, and Leslie Pack Kaelbling. "Ffrob: An efficient heuristic for task and motion planning." In *Algorithmic Foundations of Robotics XI*, pp. 179-195. Springer, Cham, 2015.

Garrett, Caelan Reed, Tomás Lozano-Pérez, and Leslie Pack Kaelbling. "Sample-Based Methods for Factored Task and Motion Planning." In *Robotics: Science and Systems*, vol. 13. 2017.

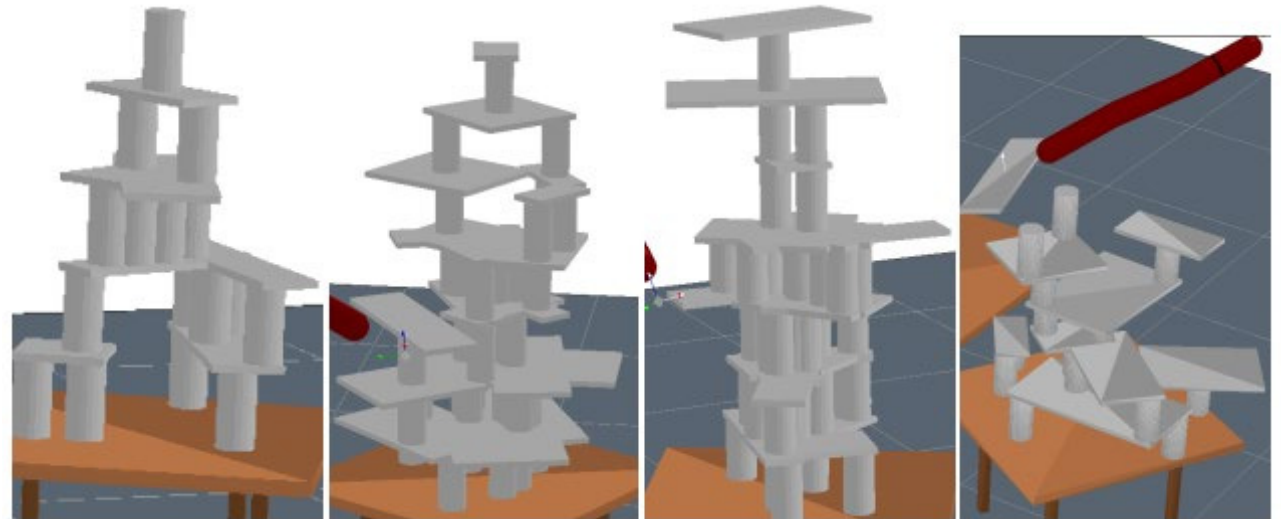
Garrett, Caelan Reed, Tomas Lozano-Perez, and Leslie Pack Kaelbling. "FFRob: Leveraging symbolic planning for efficient task and motion planning." *The International Journal of Robotics Research* 37, no. 1 (2018): 104-136.

Garrett, Caelan Reed, Tomás Lozano-Pérez, and Leslie Pack Kaelbling. "PDDLStream: Integrating symbolic planners and blackbox samplers via optimistic adaptive planning." In *Proceedings of the International Conference on Automated Planning and Scheduling*, vol. 30, pp. 440-448. 2020.

Garrett, Caelan Reed, Chris Paxton, Tomás Lozano-Pérez, Leslie Pack Kaelbling, and Dieter Fox. "Online replanning in belief space for partially observable task and motion problems." In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5678-5684. IEEE, 2020.

Optimization: Logic-geometric Programming

- 🤖 Find plan skeleton (sequence of actions)
- 🤖 Then optimize
- 🤖 While not violating constraints
- 🤖 With some objective function



Build the tallest structure

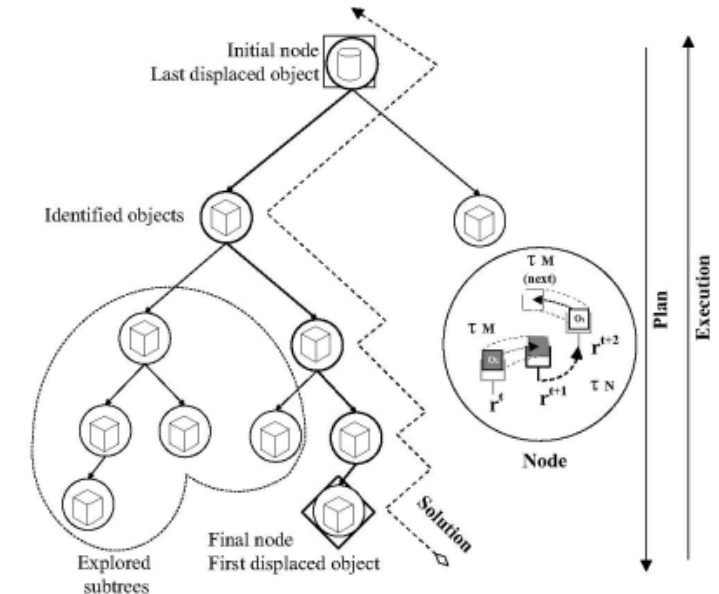
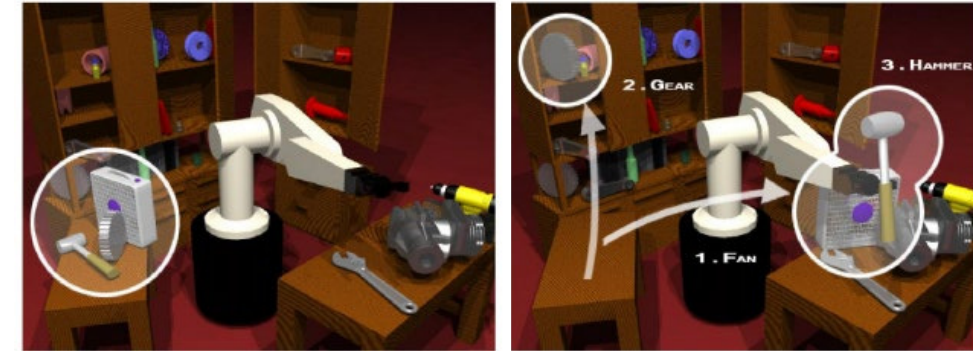
Toussaint, Marc. "Logic-geometric programming: An optimization-based approach to combined task and motion planning." In *Twenty-Fourth International Joint Conference on Artificial Intelligence*. 2015.

Manipulation Among Movable Obstacles

🤖 Movable object:

🤖 Monotone: each object moved **once**

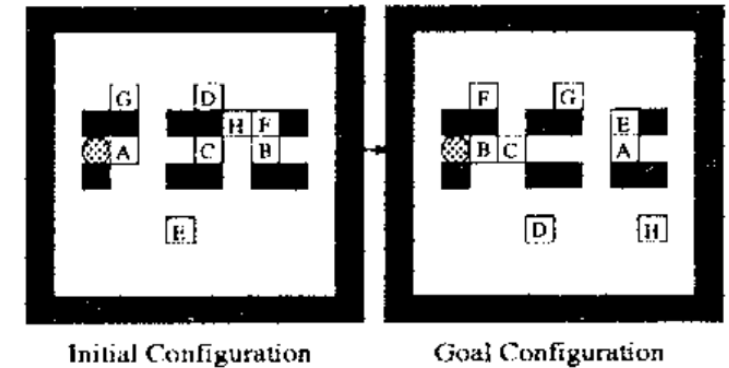
Geometry One closed triangular mesh.
Center of Gravity A given point.
Motion Constraints One vector specifying allowable motion in generalized coordinates.
Grasps A set of workspace transforms for the end effector in the local object frame.



🤖 Stilman, Mike, Jan-Ullrich Schamburek, James Kuffner, and Tamim Asfour. "Manipulation planning among movable obstacles." In Proceedings 2007 IEEE international conference on robotics and automation, pp. 3327-3332. IEEE, 2007.

Pushing Advances

- 🤖 Non-prehensile actions: pushing
- 🤖 Rearranging objects to accomplish a task

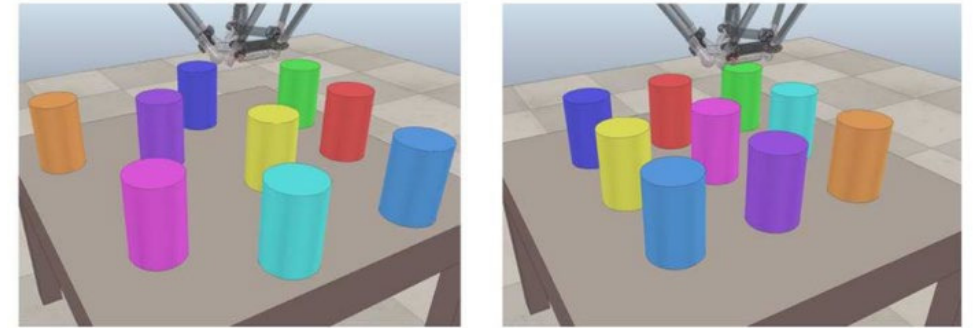


- 🤖 Dogar, Mehmet R., and Siddhartha S. Srinivasa. "A planning framework for non-prehensile manipulation under clutter and uncertainty." *Autonomous Robots* 33, no. 3 (2012): 217-236..

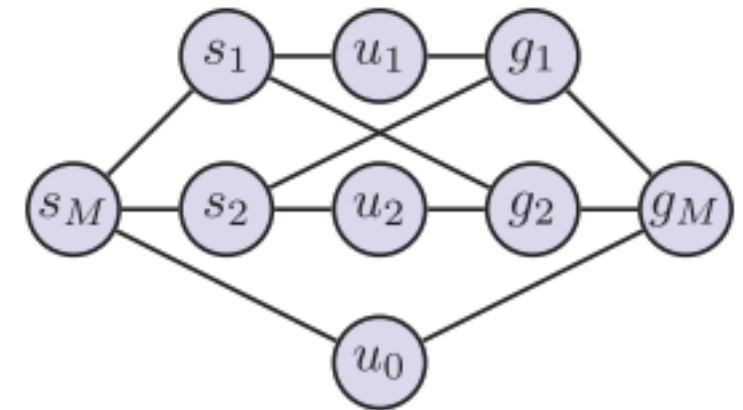
Optimal Tabletop Rearrangement

🔧 **Monotone**
(single grasp per object -> find the best ordering)

🔧 **The optimal solution is a Travelling Salesperson Problem**



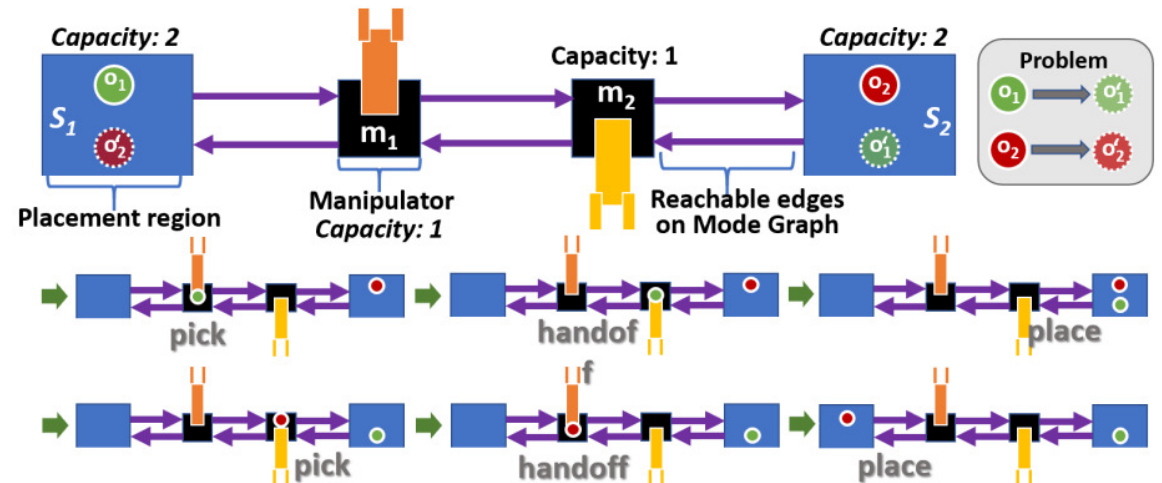
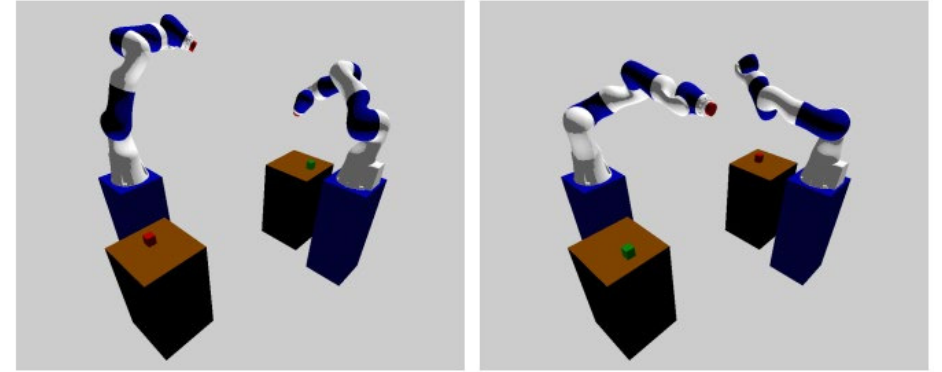
Costs here are closely related to object locations



🔧 Han, Shuai D., Nicholas M. Stiffler, Athanasios Krontiris, Kostas E. Bekris, and Jingjin Yu. "Complexity results and fast methods for optimal tabletop rearrangement with overhand grasps." The International Journal of Robotics Research 37, no. 13-14 (2018): 1775-1795.

Many-arm Tabletop Rearrangement

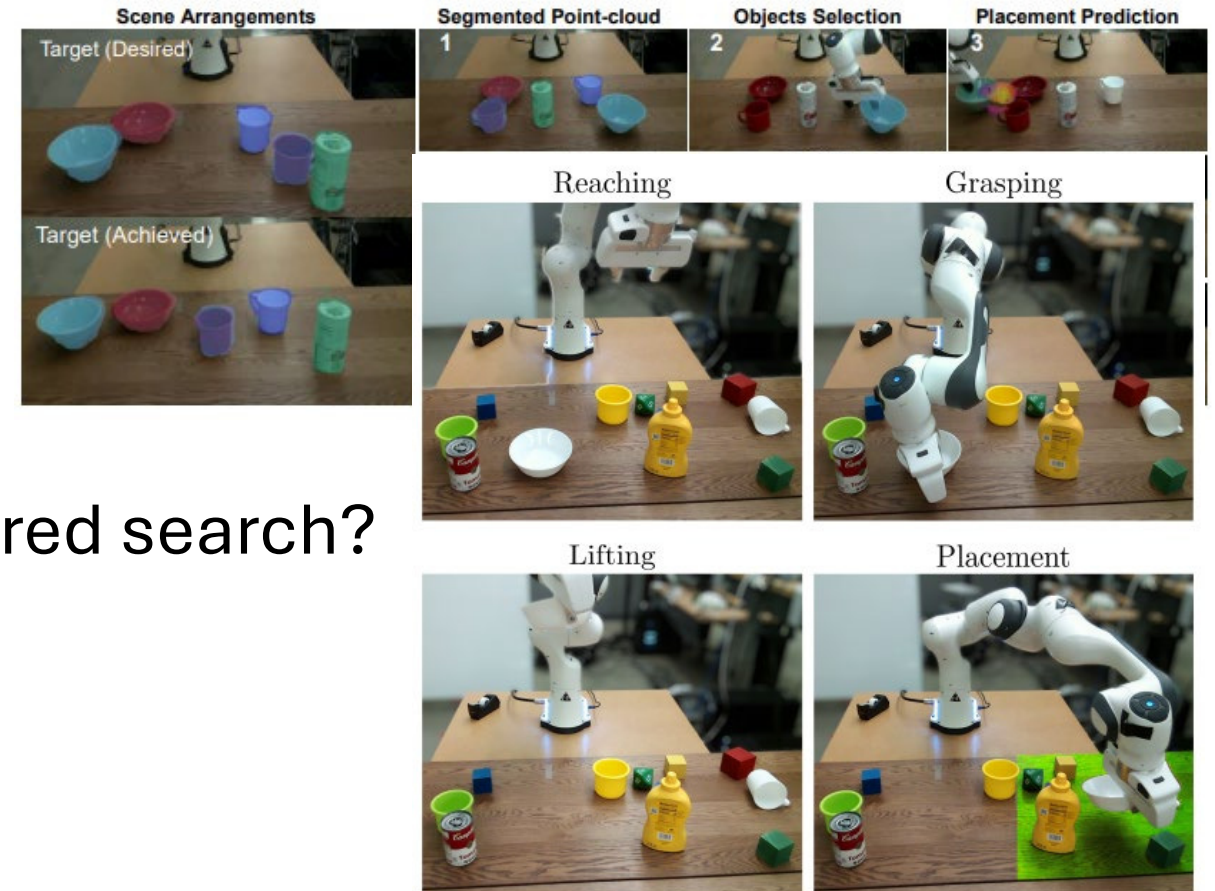
- High-level TAMP search
- Arm x Object-centric guidance reducible to **network flow problem**



- Shome, Rahul, and Kostas E. Bekris. "Synchronized multi-arm rearrangement guided by mode graphs with capacity constraints." WAFR (2020).

What Abstraction is the Best Abstraction?

- 🔧 Continuous robot reasoning?
- 🔧 Object-centric reasoning?
- 🔧 Manipulation mode reasoning?
- 🔧 Mode x grasp x IK x pose – factored search?
- 🔧 Learn it all?



- 🔧 Qureshi, Ahmed H., Arsalan Mousavian, Chris Paxton, Michael C. Yip, and Dieter Fox. "NeRP: Neural Rearrangement Planning for Unknown Objects." arXiv preprint arXiv:2106.01352 (2021).
- 🔧 Danielczuk, Michael, Arsalan Mousavian, Clemens Eppner, and Dieter Fox. "Object rearrangement using learned implicit collision functions." In 2021 IEEE International Conference on Robotics and Automation (ICRA), pp. 6010-6017. IEEE, 2021.

Learning

🔧 Learn abstractions of the discrete space

-- states

-- actions

-- transitions

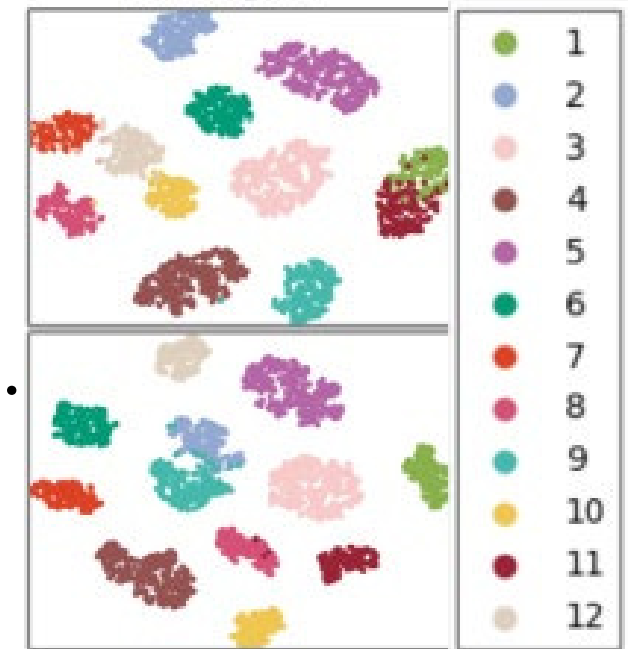
🔧 Learn symbolic states from sensing

🔧 Learn action parametrizations

🔧 Learn heuristics, objectives, rewards, constraints...



Siam-pair



Chamzas, Constantinos, Martina Lippi, Michael C. Welle, Anastasiia Varava, Alessandro Marino, Lydia E. Kavraki, and Danica Kragic. "State Representations in Robotics: Identifying Relevant Factors of Variation using Weak Supervision." In *Robot Learning Workshop, Neurips*. 2020.

Curtis, Aidan, Xiaolin Fang, Leslie Pack Kaelbling, Tomás Lozano-Pérez, and Caelan Reed Garrett. "Long-Horizon Manipulation of Unknown Objects via Task and Motion Planning with Estimated Affordances." *arXiv preprint arXiv:2108.04145* (2021).

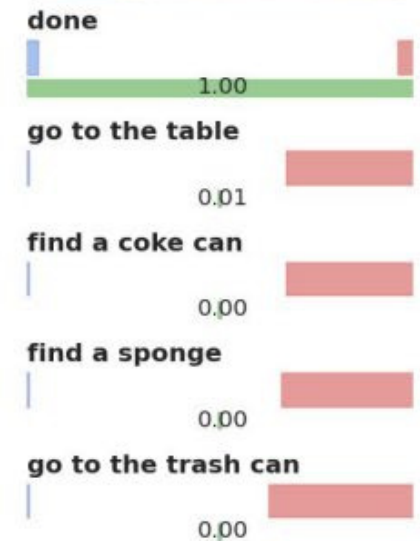
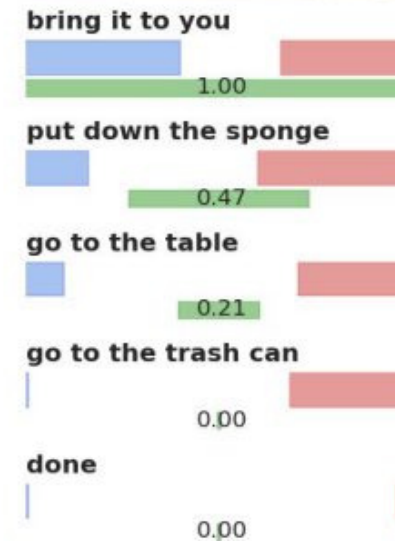
Wang, Zi, Caelan Reed Garrett, Leslie Pack Kaelbling, and Tomás Lozano-Pérez. "Learning compositional models of robot skills for task and motion planning." *The International Journal of Robotics Research* 40, no. 6-7 (2021): 866-894.

Generative Models for Actions and Motions

Human: I spilled my coke, can you bring me something to clean it up?

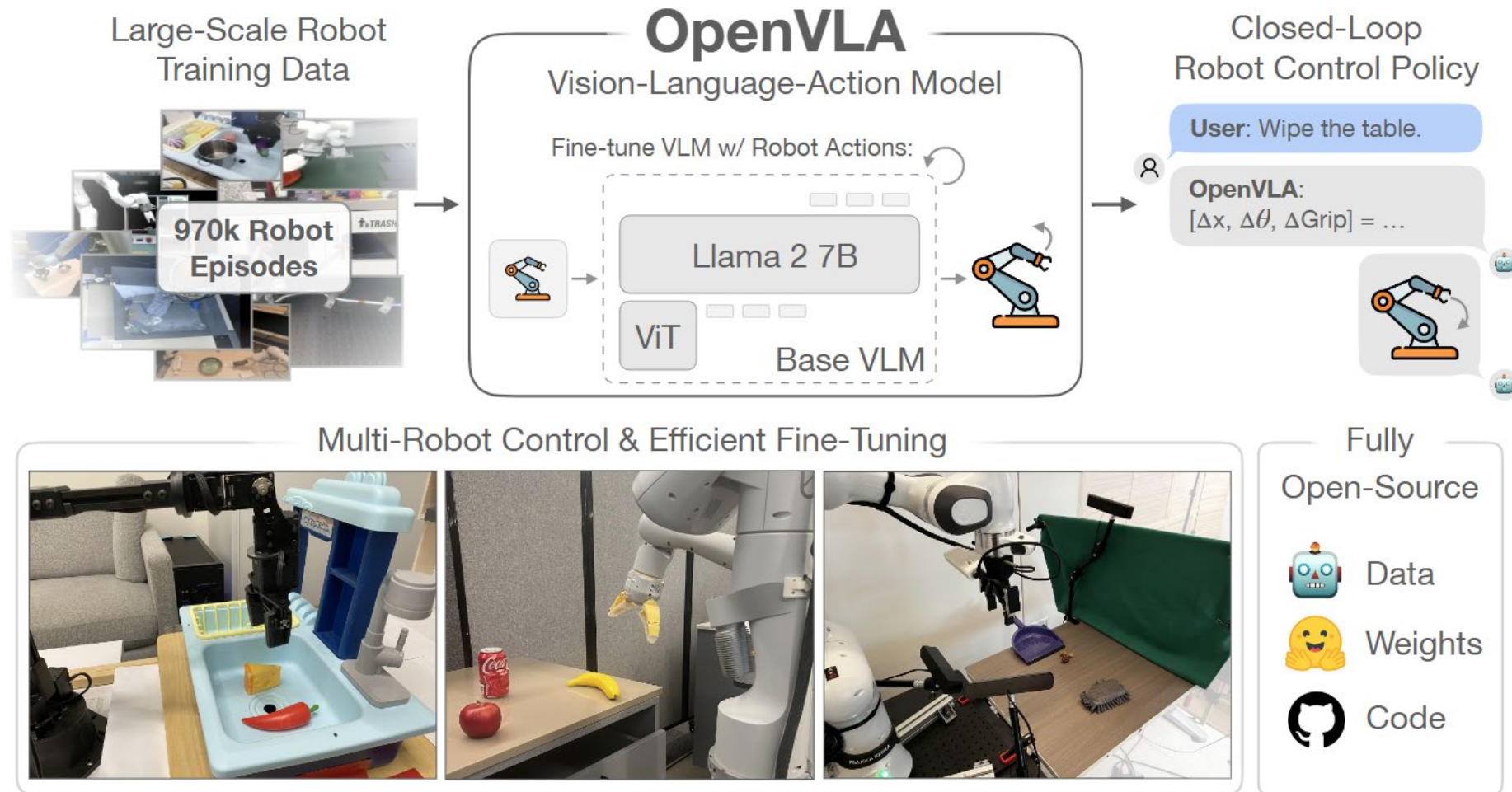
Robot: I would
1. Find a sponge
2. Pick up the sponge
3. Bring it to you
4. Done

Language × Affordance
Combined Score



Ahn, Michael, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn et al.
"Do as i can, not as i say: Grounding language in robotic affordances." *arXiv preprint arXiv:2204.01691* (2022).

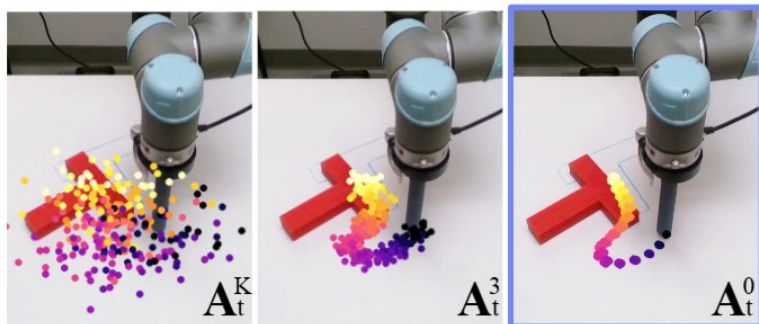
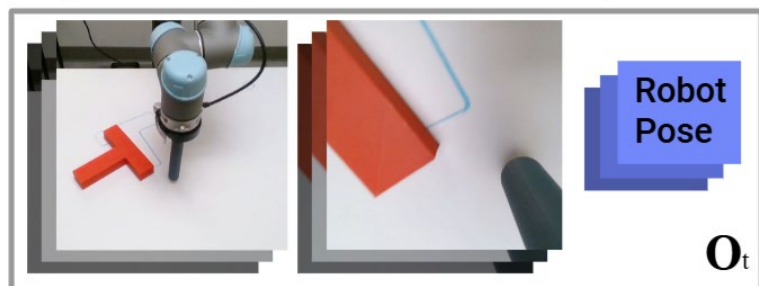
Generative Models for Actions and Motions



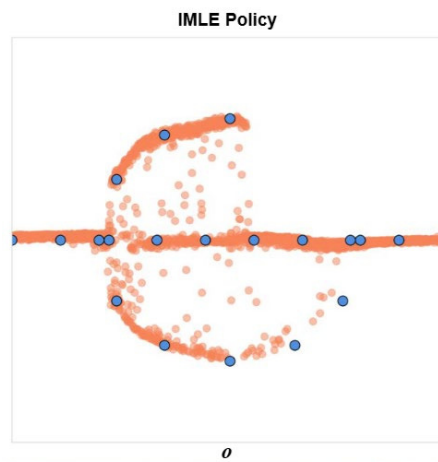
Kim, Moo Jin, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov et al.
"OpenVLA: An Open-Source Vision-Language-Action Model." In 8th Annual Conference on Robot Learning, 2024

Generative Models for Actions and Motions

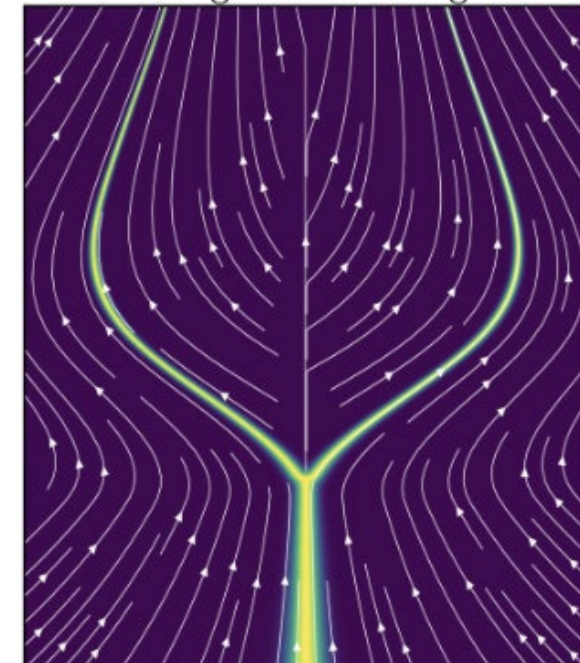
Input: Image Observation Sequence



Output: Action Sequence



Multimodal marginal flow *learned*
using flow matching



Chi, Cheng, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. "Diffusion policy: Visuomotor policy learning via action diffusion." The International Journal of Robotics Research (2023): 02783649241273668.

Rana, Krishan, Robert Lee, David Pershouse, and Niko Suenderhauf. "Imle policy: Fast and sample efficient visuomotor policy learning via implicit maximum likelihood estimation." arXiv preprint arXiv:2502.12371 (2025).

Jiang, Sunshine, Xiaolin Fang, Nicholas Roy, Tomás Lozano-Pérez, Leslie Pack Kaelbling, and Siddharth Ancha.

"Streaming Flow Policy: Simplifying diffusion flow-matching policies by treating action trajectories as flow trajectories." arXiv preprint arXiv:2505.21851 (2025).

<https://rahulsho.me>

5X

Task and Motion Planning for Execution in the Real, Pan Tianyang and Shome, Rahul and Kavraki, Lydia E. 2024 IEEE Transactions on Robotics. Honorable Mention for Best Paper 2024.



📌 Logical constraints and motion, geometric, physical constraints remain fundamental features of task and motion planning.